# Packet Interface of SCT3258

Version:1.13

Date: 2014/12/16

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# Change History

| Version | Date | Change Descriptions | Author |
|---|---|---|---|
| 1.0 | 2013/08/15 | Initial version | Hao Ye |
| 1.1 | 2013/10/21 | Add DMR Tier 2 support | Hao Ye |
| 1.2 | 2013/11/25 | DPMR_CALL_OPTION report | Hao Ye |
| 1.3 | 2013/12/16 | Update NEW_EVENT command<br>Update ANALOG_CONFIG command | Hao Ye |
| 1.4 | 2013/12/25 | Update DPMR_CALL_OPTION for report 1B 09 and 1B 0A<br>RSSI_LEVEL command change to SQ_LEVL command, add fields | Hao Ye |
| 1.5 | 2014/02/24 | Update 3700 calibration procedure<br>Update PROTOCOL_OPTION for 12.5kHz DPMR<br>Update ANALOG_CONFIG for fm 25kHz | Hao Ye |
| 1.6 | 2014/03/07 | Change the command ID of COMPANDER_CONFIG from 0x37 to 0x06<br>Update PROTOCOL_OPTION for new way of all call mapping<br>Update description for VOCODER_CMODE and ANALOG_CONFIG | Hao Ye |

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| | | | |
|---|---|---|---|
| 1.7 | 5/14/2014 | Add command CTC_PARAM<br>Add TX equalizer<br>Add DCS high pass filter to improve DCS detection<br>Add phase adjustment for two point modulation<br>Add support for ES8338 codec | Hao Ye |
| 1.8 | 6/26/2014 | Fixing typos | Hao Ye |
| 1.9 | 10/15/2014 | New rx IQ calibration | Hou Dan |
| 1.10 | 11/13/2014 | Update DMR_CALL_SLOT and DMR_SLOT_FOUND for TDMA direct mode<br>Update CTC_PARAM | Hao Ye |
| 1.11 | 11/26/2014 | Update usage of DPMR_CALL_OPTION for sending power saving header | Fang Yao |
| 1.12 | 12/16/2014 | Update CTC_PARAM<br>Update SCRAMBLER_SEED<br>Add ANALOG_FEATURE command | Jinglong Fang |
| 1.13 | 2/13/2015 | Add DPMR_REPEAT_HEADER<br>Update DMR_CALL_SLOT<br>Update DMR_SLOT_FOUND | Chenxiabing |

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# 1 Packet Interface Descriptions

Packet interface is defined above the physical interface (HPI, Serial Port) for communications between SCT3258 and the host MCU. The packet format is identical regardless of physical interfaces.

As HPI in SCT3258 is preconfigured to use 16 bit format, the total number of bytes in a packet should be even. If the total number of byte is odd, a zero is appended at the end. This appended zero will not be counted in the LENGTH field.

## 1.1 Packet Format

The packet format is as shown in the table below. A packet consists of a packet header and number of packet fields and optional parity bytes. A packet header starts with 3 bytes of START_BYTES. The next two bytes contain the packet LENGTH and the next byte contains the TYPE. Each packet contains one or more fields which are shown as FIELD (0) through FILELD (N-1). Optionally, at the end of the packet, there are two parity bytes.

| Packet Header | | | Field | | | Parity | |
|---|---|---|---|---|---|---|---|
| START_BYTES | LENGTH | TYPE | FIELD(0) | ... | FIELD(N-1) | FBYTE | PARITY_BYTE |
| 3 byte | 2 byte | 1 byte | Variable | | Variable | 1 byte | 1 byte |

**Table 1-1 General Packet Format**

Packet details are described in the sections below.

### 1.1.1 START_BYTES

The three bytes of START_BYTES always have fixed values of 0x84, 0xA9, 0x61.

### 1.1.2 LENGTH

The PACKET LENGTH occupies two bytes of the packet. The MS byte of the packet length is the fourth byte of the packet and the LS byte of the packet length is the fifth byte of the packet. The PACKET LENGTH is the sum of each FIELD. Note that PACKET LENGTH excludes the first 6 bytes taken up by the Packet Header. If parity check is enabled, the length field includes the parity byte as well.

As HPI in SCT3258 is preconfigured to use 16 bit format, the total number of bytes in a packet should be even. If the total number of byte is odd, a zero is appended at the end. This appended zero will not be counted in the LENGTH field.

### 1.1.3 TYPE

TYPE field specifies the destination, read/write property and the types of the packet. It includes

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multiple fields, described in the table below:

| Bit Position | Field | Descriptions |
|---|---|---|
| 7 | AMBE bypass | External Vocoder message bypass<br>0: SCT3258 Control<br>1: AMBE control |
| 6 | Codec bypass | Codec message bypass<br>0: SCT3258 Control<br>1: Codec control, any bytes after packet type are send to codec directly |
| 5 | Read/Write | Read Write Control<br>0: Write. The MCU write fields to SCT3258<br>1: Read. SCT3258 report fields to the MCU. Not all parameters are readable. |
| 4 | Near/Far | Near end / Far end control<br>0: Near end<br>1: Far End |
| 3-0 | Packet Type | See Table Below |

**Table 1-2 Fields in TYPE Byte**

SCT3258 supports message bypass to external vocoder and to external codec chip. The MCU can send message to external vocoder (e.g. AMBE3000) or external codec (e.g. TI TMS320AIC3204) directly through SCT3258. This saves physical interface between the MCU and other chips.

Bypass for vocoder and codec is specified by bit 7 and bit 6 respectively.

Internally, the call related information is stored in registers in SCT3258. External MCU can write to and read from these registers through packet interface. The write/read control is through bit 5 of TYPE field. Not all registers are readable. Column 6 of Table 1-5 describes whether a register is readable or not.

Some fields, such as Own ID, Called ID, have two sets of internal registers, for far end and near end of calls. When writing a packet field, the content is always stored in the near end registers. When reading field information, the MCU can request the near end information or the far end information. The near end information is the information that is to be sent or already sent to the far end. The far end information is the information that is received from the far end. The near/far end control is through bit 4 of the TYPE byte.

For example, station A has an Own ID 1234567, station B has an Own ID of 1234563. The user (MCU) can set the Own ID for the two stations with field DPMR_OWN_ID_BCD, and with write flag. The user can then read back the Own ID of each station with read flag, and near end flag at the two stations. After a call is made from A to B, the user can read the Own ID of station A by issuing read command to station B with far end flag. Column 6 of Table 1-5 also describes

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whether far end register is available for reading.

The details of Packet Type are described in Table 1-3 below.

| Type Value | Packet Name | Descriptions |
|---|---|---|
| 0 | Control Packets | Used to set up chip operation mode, configure hardware, set up or end a call. When a control packet is received, the chip returns a control packet with response fields that contain response data or indication of errors in the control packet. |
| 3 | DPMR Packet | Used for DPMR protocol specific information. User can use these messages to change the elements of DPMR protocol. |
| 4 | Program packet | Used for host MCU to program SCT3258. |
| 5 | DMR | Used for DMR protocol specific information. User can use these messages to change the elements of DMR protocol. |
| Others | Reserved | Reserved |

**Table 1-3 Packet Types**

Note that the boot loader for DPMR mode and DMR mode is different. Depending on the boot loader types, SCT3258 enters DPMR mode or DMR mode. In DPMR mode, commands for DMR only (type 5) are not recognized. In DMR mode, command for DPMR only (type 3) are not recognized.

Also note that for program packet only (type 4), the packet header and parity bytes can be omitted. In that case, the boot loader searches for a match of the field and command code.

## 1.1.4 Packet Fields

The packet fields contain useful packet information. Various different packet fields each with their own format are defined in the next section, however, the general format of a field is shown in the table below.

A field consists of a field identifier followed by field data. The length of field data is dependent upon the field identifier. Many fields have fixed lengths. Some fields are variable in length; and in such cases the length of the field data is embedded inside field data explicitly or implicitly.

| Field Identifier | Field Data |
|---|---|
| 1 byte | L(n)-1 bytes |

**Table 1-4 General Field Format**

## 1.1.5 Parity Field

The parity field is a 2-byte field at the end of a packet. The first byte of the parity field is the parity

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field identifier and is always equal to 0x2f. The second byte of the parity field is the parity byte. It is obtained by "Exclusive-oring" every byte in the packet, except for the START_BYTES and the parity byte (last byte), together. SCT3258 checks the parity byte for all received packets, informs the MCU of any parity error, and discards any packet that has an incorrect parity byte. When parity check fails, the receiver replies with a packet (with field 0x17) indicating parity failure. Parity check is disabled at reset. It can be enabled when SCT3258 receives a control packet with PARITY_ENABLE field.

### 1.1.6 Time Out

If a valid message is received, SCT3258 responds to MCU in 40 milliseconds (except some especial commands in the table ). If a response packet is not received within 40 milliseconds, the MCU can consider the message lost and resend the packet if necessary.
When a response packet is received, the MCU or SCT3258 does not send an ACK to the response packet.

| Commands | Response time |
|---|---|
| CHAN_D | 80ms |
| VOCOER_SEL | 1-3s |
| CODEC_SELECT | 1 s |
| DIGC_DATA_FRAME | 80ms.<br>The response time of the first data is header frame(80ms) + Preamble + Silence |
| ACK_MESSAGE | The response time of receiving 17 0A after finishing sending is about 400ms. We have to wait the 17 0A to do the subsequent operation. |

## 1.2 Field Formats

A packet must contain one or more fields. The field formats are different for different types of packets. Each packet requires a response packet.

Control packets can be used to configure the chip prior to operation and also to query for information from the chip. The response packet for must fields just echoes back the control field identifier followed by a 0x00 byte to indicate the control field was received successfully. If multiple fields are present in the packets, the response packet only needs to echo the last field of the packet.

DPMR / DMR packets contain the call information. They can be originated either from host or from SCT3258. The response packets to DPMR / DMR packet echoes back the control field identifier followed by a 0x00 byte to indicate the field was received successfully. If multiple fields are present in the packets, the response packet only needs to echo the last field of the packet.

The Program Packet is used by the host to down load application program to SCT3258. The response packet to Program Packet echoes back the control field identifier followed by a 0x00

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byte to indicate the field was received successfully.

Either the MCU or SCT3258 can initiate a packet exchange. The packet can be used to write information to the other end (SCT3258 or MCU), or it can be used to request information from the other end. The Read/Write control is by bit 5 of the TYPE byte.

When requesting field information, the MCU can request near the end information or the far end information. The near end information is the information that is to be sent or already sent to the far end. The far end information is the information that is received from the far end. The near/far end control is through bit 4 of the TYPE byte.

The table below summarizes various fields used in SCT3258.

| Field Name | Field Identifier | Packet Type | Control Field Length (Bytes) | Response Field Length (Bytes) | Note | Descriptions |
|---|---|---|---|---|---|---|
| CHAN_D | 0x01 | 0 | Varies | 2 | YP | Encoded voice data |
| VOCODER_CMODE | 0x02 | 0 | 2 | 2 | N | Vocoder mode flags for current vocoder |
| AFC_CONFIG | 0x05 | 0 | 3 | 2 | YR | |
| COMPANDER_CONFIG | 0x06 | 0 | 8 | 2 | N | Compander Configuration |
| VOCODE_TONE | 0x08 | 0 | Varies | 2 | Y | Force current encoder/decoder to generate tone frames |
| ANALOG_FEATURE | 0x0A | 0 | 3 | 2 | N | Analog Feature configuration |
| VOCOER_SEL | 0x10 | 0 | 2 | 2 | N | Vocoder Selection |
| VOCODER_INI | 0x12 | 0 | 2 | 2 | N | Vocoder initialization |
| CHIP_RESET | 0x14 | 0 | 1 | 2 | N | Chip reset |
| CHIP_LOWPWR | 0x15 | 0 | 2 | 2 | N | Set chip to low power mode |
| PARITY_ENABLE | 0x16 | 0 | 2 | 2 | N | Enable parity check |
| ACK_MESSAGE | 0x17 | 0 | None | 2 | N | Message error report |
| WORK_MODE | 0x18 | 0 | 4 | 2 | Y | Set up SCT3258 work mode |
| CARRIER_READY | 0x19 | 0 | 2 | 2 | N | Carrier Ready Message |
| PROCESS_MODE | 0x1A | 0 | 2 | 2 | N | Process Mode |
| NEW_EVENT | 0x1B | 0 | None | 2 | YX | New event indication |
| DEMOD_GAIN | 0x1C | 0 | 3 | 2 | Y | Demodulator Gain |
| REPORT_FIELD | 0x1D | 0 | 2 | 2 | NX | Fields to report when receiving calls. |
| BER_REPORT | 0x1E | 0 | 2 | 10 | Y | Bit error rate test report |
| ANALOG_TONE | 0x1F | 0 | 18 | 2 | N | Generate Local Tone |
| CALL_START | 0x20 | 0 | 2 | 2 | N | Start call send |
| CALL_STOP | 0x21 | 0 | 1 | 2 | N | Stop call send |

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|---|---|---|---|---|---|---|
| CONNECT | 0x22 | 0 | 2 | 2 | NP | Send connection |
| DISCONNECT | 0x23 | 0 | 2 | 2 | NP | Send disconnection |
| DPMR_ACK | 0x24 | 0 | 2 | 2 | NP | Send ack |
| SYSTEM_INFO | 0x25 | 0 | 2 | 2 | NP | Send system or status information |
| ANALOG_SUBAUDIO | 0x26 | 0 | Varies | 2 | N | analog subaudio |
| CALL_MATCH | 0x27 | 0 | None | 2 | N | Call match report |
| ADDRESSING_MODE | 0x28 | 3 | 2 | 2 | NP | Dialing mask length |
| GROUP_ID | 0x29 | 3 | 5 | 2 | YP | Group ID number |
| SCRAMBLER_SEED | 0x2A | 0 | Varies | 2 | NP | Scrambler seed |
| MOD_GAIN | 0x2C | 0 | 3 | 2 | N | Modulator Gain |
| CODEC_SELECT | 0x2D | 0 | 2 | 2 | N | Codec Selection |
| SET_I2C_ADDRESS | 0x2E | 0 | 2 | 2 | N | Set the I2C address for the codec |
| SYNC_MISS_CNT | 0x2F | 0 | 2 | 2 | N | Set the number of Sync miss before reporting |
| Q_PRO_STR | 0x30 | 0 | 1 | Varies | Y | Query for product code |
| Q_HW_VER | 0x31 | 0 | 1 | Varies | Y | Query for hardware version |
| Q_SW_VER | 0x32 | 0 | 1 | Varies | Y | Query for software version |
| Q_CID_SN | 0x33 | 0 | 1 | 9 | Y | Query for customer ID |
| DMR_GROUP_ID | 0x36 | 5 | 6 | 2 | YD | Group ID number |
| DC_OFFSET | 0x39 | 0 | 13 | 2 | N | DC Offset |
| ANALOG_CONFIG | 0x3c | 0 | 12 | 2 | N | Analog Configuration |
| VOCODER_IO_SET | 0x3e | 0 | 2 | 2 | NP | Vocoder IO settings |
| CTC_PARAM | 0x3d | 0 | 23 | 2 | N | CTC detection parameters |
| MODOUT_CONFIG | 0x41 | 0 | 6 | 2 | N | Two point modulation delay |
| IQCOMP_COEF | 0x42 | 0 | 10 | 2 | YR | IQ compensation configuration |
| DIGC_DATA_FRAME | 0x43 | 0,3,5 | Varies | 2 | YFX | Data frames |
| DPMR_PREAMBLE_LENGTH | 0x46 | 0,3 | 3 | N | N | Preamble and silence length |
| DPMR_CALL_OPTION | 0x47 | 3 | 6 | 2 | N | DPMR Call Option |
| DPMR_END_UE | 0x4B | 3 | 4 | 2 | YF | 17-bit un-encoded END |
| DIGC_FS1 | 0x4C | 3 | 7 | 2 | YF | 48-bit of FS1 |
| DIGC_CALLED_ID_BIN | 0x50 | 3, 5 | 4 | 2 | YF | 24-bit called ID in binary |
| DIGC_OWN_ID_BIN | 0x51 | 3, 5 | 4 | 2 | YF | 24-bit own ID in binary |
| DPMR_CALLED_ID_BCD | 0x52 | 3 | 5 | 2 | YFP | 7 digit called ID in BCD |
| DPMR_OWN_ID_BCD | 0x53 | 3 | 5 | 2 | YFP | 7 digit own ID in BCD |

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| | | | | | | |
|---|---|---|---|---|---|---|
| DPMR_M_V_F_E | 0x54 | 3 | 2 | 2 | YFP | 3-bit communication mode and 4-bit communication format |
| DPMR_PROTOCOL_OPTION | 0x55 | 3 | 2 | 2 | NP | DPMR protocol option |
| DPMR_SLD | 0x56 | 3 | 4 | 2 | YFP | 18-bit slow data |
| DPMR_HT_CI_PM | 0x57 | 3 | 3 | 2 | YFP | 4-bit header type and 11-bit call information |
| DPMR_CI | 0x5A | 3 | 3 | 2 | FP | 11-bit call information |
| DPMR_CC | 0x5C | 3 | 4 | 2 | YFP | 24-bit of color code (Di-bit) |
| DPMR_SEND_SF | 0x5F | 3 | 1 | 2 | NP | Send super frame command. Upon receiving this command, SCT3258 assemble a super frame and send to the far end |
| DPMR_SEND_HEADER | 0x60 | 3 | 1 | 2 | NP | Send header frame command. Upon receiving this command, SCT3258 assemble a header frame and send to the far end |
| DPMR_SEND_END | 0x61 | 3 | 1 | 2 | NP | Send end frame command. Upon receiving this command, SCT3258 assemble a end frame and send to the far end |
| DPMR_SEND_AD | 0x62 | 3 | 1 | 2 | NP | Send appended data |
| DIGI_MIC_GAIN | 0x68 | 0 | 3 | 2 | N | Digital Microphone Gain |
| DIGI_SPEAKER_GAIN | 0x69 | 0 | 3 | 2 | N | Digital Speaker Gain |
| I2C_OPERATION | 0x6A | 0 | Varies | 5 | Y | I2C Read and Write |
| MISC_GAIN | 0x6B | 0 | 13 | 2 | N | Misc. Gains |
| SQ_LEVEL | 0x6C | 0 | 13 | 2 | YR | RSSI and OOBE levels |
| SPI_OPERATION | 0x6D | 0 | 3 | 2 | YR | SPI Write Operation |
| DMR_SLC | 0x6E | 5 | 5 | 2 | ND | DMR SLC report |
| DMR_CALL_SLOT | 0x6F | 5 | 2 | 2 | ND | DMR call slot information |
| EQUALIZER_FILTER | 0x70 | 0 | 12 | 2 | N | Set coefficients for audio equalizer |
| DMR_FLC | 0x76 | 5 | 4 | 2 | YD | DMR Full LC |
| DMR_CC | 0x77 | 5 | 2 | 2 | YD | DMR Color Code |
| DMR_CALL_START | 0x78 | 5 | 4 | 2 | ND | DMR Call Start |
| DMR_CALL_OPTION | 0x79 | 5 | 6 | 2 | ND | DMR Call option |
| DMR_OFFSET | 0x7A | 5 | 9 | 2 | ND | DMR Slot Timing Offset |
| DMR_SLOT_TYPE | 0x7B | 5 | 2 | 2 | ND | DMR slot type |
| DMR_EMB | 0x7C | 5 | 2 | 2 | ND | DMR EMB field |
| DMR_CALLED_ID_BCD | 0x7D | 5 | 5 | 2 | YFD | 8 digit called ID in BCD |
| DMR_OWN_ID_BCD | 0x7E | 5 | 5 | 2 | YFD | 8 digit called ID in BCD |

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|---|---|---|---|---|---|---|
| DMR_SLOT_FOUND | 0x7F | 5 | 2 | 2 | ND | DMR_slot found |

**Table 1-5 Overview of Packet Fields**

In the note field in the above table, Y means the command can be used as both WRITE and READ command; N means the command can only be used as a WRITE command. F means the command can be used to read both Far End information and Near End information. R means the command is used only when paired with SCT3700. D means command can be used for DMR only. P means command can be used for DPMR only. X means command is interpreted differently for DMR and DPMR.

As most fields are used as commands to SCT3258 for controlling and configuration purposes, we sometimes call field as command. As a result we use the terms field and command in-discriminatively throughout the document.

The detailed descriptions of each filed is described in the following sections. Only field format in Write Packet is shown. Read Packets usually only have the field ID byte.

## 1.2.1 CHAN_D

This field can be used to pass encoded voice data between the MCU and SCT3258. This command is only used in DPMR mode.

| Field Identifier | Data Length | Vocoder Configuration field |
|---|---|---|
| 1 Byte | 1 Byte | 1 Bytes |
| 0x01 | The length of the data in bytes | Encoded voice data. 36 bytes per 80 milliseconds. |
| Default: | 36 | |

**Table 1-6 CHAN_D Field Format**

## 1.2.2 VOCODER_CMODE

This field can be used to change the mode of the vocoder.

| Field Identifier | Vocoder Configuration field |
|---|---|
| 1 Byte | 1 Bytes |
| 0x02 | { AMBE_COMP, 0, 0, 0, NS, 0, TD, AGC} |
| Default: | 0x08 |

**Table 1-7 VOCODER_CMODE Field Format**

The meaning of each bit is given below:

| Bit Name | Descriptions | Default |
|---|---|---|

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| | | |
|---|---|---|
| AMBE_ COMP | AMBE vocoder compatibility bit | 0 |
| NS | Noise Suppression Enable<br>0: disable<br>1: enable | 1 |
| Reserved | Reserved | 0 |
| TD | Tone Detection Enable<br>0: disable<br>1: enable | 0 |
| AGC | AGC Enable<br>0: disable<br>1: enable | 0 |

**Table 1-8 VOCODER_CMODE Field Options**

## 1.2.3 AFC_CONFIG

This command is used to configure AFC (automatic frequency control), when used with SCT3700.

If the receiver carrier frequency is not matched to the transmitter carrier frequency, the performance degrade.

The use the AFC, the user first use this command to read the frequency offset between the transmitter and the receiver. To compensate for the frequency offset, the user has two options:

1. Change SCT3700 receiver LO, using SCT3700 frequency configuration command.
2. Change the internal mixer inside SCT3258, using this command.

The AFC_CONFIG write command has the following format:

| Field Identifier | Internal frequency offset |
|---|---|
| 1 Byte | 1 Bytes |
| 0x05 | Internal frequency offset applied at the SCT3258 mixer. The unit is Hz |
| Default: | 0 |

**Table 1-9 AFC_CONFIG Field Format**

AFC_CONFIG command is also used to read the frequency offset between the receiver and the transmitter.

To read the frequency offset, AFC_CONFIG command should be sent with “Read/Write” bit in the TYPE field set to 1. The AFC_CONFIG command used in “read” mode has the following format:

| Field Identifier | DC Time Constant |
|---|---|

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| 1 Byte | 1 Byte |
|---|---|
| 0x05 | Time constant for averaging AFC frequency offset |
| Default | 3 |

Time constant of 0 corresponds to average time of 31.25 ms, which is the time it takes to reach within 10% of the steady state energy value. If the constant is n, the average time is given by the following formula:

T = 31.25 * 2^n.

The Response to the read AFC_CONFIG command has the following format:

| Field Identifier | Frequency offset | Reserved |
|---|---|---|
| 1 Byte | 1 Byte | 1 Bytes |
| 0x05 | Frequency offset between the transmitter and the receiver, in Hz | Reserved |
| Default | - | - |

## 1.2.4 COMPANDER_CONFIG

This command is used to configure the compander in analog call mode. The configurable parameters include compander reference level for the transmitter and the receiver, the energy estimation timing constant, and the compander gain update interval.

The compander energy level is calculated as the running average of the absolute values of received audio signal, given by the equation below:

E(n) = (1-alpha) * E(n-1) + alpha * |x(n)|,

where |x(n)| is the absolute value of the received audio signal, alpha is the time constant. The smaller is the alpha, the larger is the time constant. Roughly, T = 4 / alpha, where T is the time it takes to reach within 10% of the steady state energy value. For example, with the default alpha = 68, the time constant is T = 4 / 68 = 60 ms.

The TX and RX compander reference levels are the signal levels that the input signal are compared against. At the TX side, if the input signal energy level is greater than the TX compander reference level, the input signal are reduced; if it is less than the TX compander reference level, it is enlarged. At the RX side, if the input signal energy level is greater than the

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RX compander reference level, the input signal are enlarged; if it is less than the RX compander reference level, it is reduced.

The compander gain update interval is the interval that the compander gains are re-calculated. Within the interval, the compander gain is unchanged. The compander gain update interval is in terms of 24 kHz samples.

| Field Identifier | Compander Update interval | Compander TX reference | Comander RX reference | Compander Energy Constant |
|---|---|---|---|---|
| 1 Byte | 1 Byte | 2 Bytes | 2 Bytes | 2 Bytes |
| 0x06 | Compander update interveral in terms of 24 samples | Compander TX reference, 0-32768 | Compander RX reference, 0-32768 | Compander Energy update constant, 0-16384 |
| Default | 60 | 2048 | 2048 | 68 |

**Table 1-10 COMPANDER_CONFIG Field Format**

## 1.2.5 VOCODE_TONE

This field is used to force the encoder to transmit a tone frame. The frequency (or frequencies) and amplitude of the tone are specified by this field. It is also used to report the detected DTMF tone or single tone to the MCU.

| Field Identifier | Tone Configuration | Tone Data | Amplitude Data | 16 bit Frequency |
|---|---|---|---|---|
| 1 Byte | 1 Byte | 1 Byte | 1 Byte | 2 Bytes (Optional) |
| 0x08 | See Table 1-10 for details | See Table 1-12 for details | Amplitude scale in dB<br>+3dB to -90 dB | Single tone Frequency in 16 bit decimal |
| Default | 0 | 0x80 | -10 dB (0xF6) | 0 |

**Table 1-11 VOCODE_TONE Field Format**

The details of Tone Configuration byte is as follows:

| Bit Position | Descriptions | Default |
|---|---|---|
| Bit 7-3 | Reserved | 0 |

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| | | |
|---|---|---|
| 2 | Optional 16 bit frequency field flag<br>0: No 16 bit frequency field<br>1: 16 bit frequency field present | 0 |
| 1 | Tone Direction<br>0: Send or detect near end Vocoder tone<br>1: Send or detect far end Vocoder tone | 0 |
| 0 | Tone on / off flag<br>0: Tone Off<br>1: Tone On | 0 |

**Table 1-12 Tone Configuration Details**

| DTMF Tone Data | Tone Type | Frequency 1 (Hz) | Frequency 2 (Hz) |
|---|---|---|---|
| 0-4 | Invalid Tone | NA | NA |
| 0x05-0x7A | Single Tone | Tone Data * 31.25 Hz | NA |
| 0x7B-0x7F | Invalid tone | NA | NA |
| 0x80 | DTMF “0” | 1336 | 941 |
| 0x81 | DTMF “1” | 1209 | 697 |
| 0x82 | DTMF “2” | 1336 | 697 |
| 0x83 | DTMF “3” | 1477 | 697 |
| 0x84 | DTMF “4” | 1209 | 770 |
| 0x85 | DTMF “5” | 1336 | 770 |
| 0x86 | DTMF “6” | 1477 | 770 |
| 0x87 | DTMF “7” | 1209 | 852 |
| 0x88 | DTMF “8” | 1336 | 852 |
| 0x89 | DTMF “9” | 1477 | 852 |
| 0x8a | DTMF “A” | 1633 | 697 |
| 0x8b | DTMF “B” | 1633 | 770 |
| 0x8c | DTMF “C” | 1633 | 852 |
| 0x8d | DTMF “D” | 1633 | 951 |
| 0x8e | DTMF “*” | 1209 | 941 |
| 0x8f | DTMF “#” | 1477 | 941 |
| 0x90 | KNOX “0” | 1162 | 820 |
| 0x91 | KNOX “1” | 1052 | 606 |
| 0x92 | KNOX “2” | 1162 | 606 |
| 0x93 | KNOX “3” | 1279 | 606 |
| 0x94 | KNOX “4” | 1052 | 672 |
| 0x95 | KNOX “5” | 1162 | 672 |
| 0x96 | KNOX “6” | 1279 | 672 |
| 0x97 | KNOX “7” | 1052 | 743 |
| 0x98 | KNOX “8” | 1162 | 743 |
| 0x99 | KNOX “9” | 1279 | 743 |
| 0x9a | KNOX “A” | 1430 | 606 |
| 0x9b | KNOX “B” | 1430 | 672 |

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| | | | |
|---|---|---|---|
| 0x9c | KNOX “C” | 1430 | 743 |
| 0x9d | KNOX “D” | 1430 | 820 |
| 0x9e | KNOX “*” | 1052 | 820 |
| 0x9f | KNOX “#” | 1279 | 820 |
| 0xa0 | Dial Tone | 440 | 350 |
| 0xa1 | Ring tone | 480 | 440 |
| 0xa2 | Busy Tone | 620 | 480 |
| 0xa3 | Call Progress Tone | 490 | 350 |
| 0xa4-0xff | Inactive | NA | NA |

**Table 1-13 VOCODE_TONE Field Options**

Note that the bytes for 16 bit Frequency is optional. It is present only if bit 2 of Tone Configuration Byte is 1.

## 1.2.6 DPMR_REPEAT_HEADER

This command is to set dpmr repeat header number in power saving communication (Only for SCT3926)

| Field Identifier | Call Details |
|---|---|
| 1 Byte | 1 Byte |
| 0x09 | Bit 0-3: repeat header number<br>Bit 4-7: reserved. |
| Default | 0 |

## 1.2.7 ANALOG_FEATURE

This field configures for special feature. Note that these special feature can only be used when the analog vocoder (ID = 16) is loaded.

| Field Identifier | ANALOG_FEATURE |
|---|---|
| 1 Byte | 2 Bytes |
| 0x010 | Bit 0: DTMF enable flag<br>0: DTMF detection is disabled<br>1: DTMF detection is enabled<br>Bit 1: DTMF detection position<br>0: DTMF detection is done before the de-emphasis filter<br>1: DTMF detection is done after the de-emphasis filter<br>Bit 2: MSK enable flag<br>0: MSK transmission and detection |

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|---|---|
| | disabled<br>1: MSK transmission and detection is enabled.<br>Bit 3: Compander enable flag<br>0: Compander is disabled<br>1: Compander is enabled.<br>Bit 4-15: reserved. |
| Default | 3 |

**Table 1-14 ANALOG_FEATURE Field Format**

## 1.2.8 VOCOER_SEL

This field forces vocoder type.

SCT3258 supports multiple vocoders, with default as AMBE+2C vocoder from DVSI.

| Field Identifier | Vocoder Type |
|---|---|
| 1 Byte | 1 Byte |
| 0x010 | 0: AMBE +2，external<br>1: Tsinghua ASELP vocoder<br>2: Reserved<br>3: AMBE+2C, internal<br>6: ST24 vocoder<br>16: Analog calls<br>255: Vocoder selected with Version field.<br>Others, reserved |
| Default | 3 |

**Table 1-15 VOCODER_SEL Field Format**

If vocoder type is 0xFF, the vocoder selection is done with the Version field as specified by ETSI TS 102 490 or TS 102 658. At the TX side, MCU sets Version field either with DPMR_M_V_F_E command or with CALL_START command. At the receiver side, SCT3258 extract Version field from far end and set the vocoder type accordingly.

## 1.2.9 VOCODER_INI

This command initializes the vocoder.

| Field Identifier | Initialization type |
|---|---|
| 1 Byte | 1 Byte |
| 0x012 | N/A |

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| Default | 0 |
|---|---|

**Table 1-16 VOCODER_INI Field Format**

## 1.2.10 CHIP_RESET

This command resets SCT3258. This command has the effect as NMI signal (non-maskable interrupt), which restarting the firmware in SCT3258 without reloading the firmware.

| Field Identifier |
|---|
| 1 Byte |
| 0x14 |

**Table 1-17 CHIP_RESET Field Format**

## 1.2.11 CHIP_LOWPWR

This command puts SCT3258 in low power mode.

| Field Identifier | Low Power Mode |
|---|---|
| 1 Byte | 1 byte |
| 0x015 | Bit 7:6, Reserved<br>Bit 5: Power down external vocoder<br>Bit 4: Power down external codec<br>Bit 3:0: Power mode of SCT3258<br>0: Normal<br>1: Stop TX and RX processing<br>2: Reserved<br>3: Power down certain peripheral of SCT3258 including serial port, DMA and timers.<br>4: SCT3258 enters IDLE mode<br>5: SCT3258 enters Sleep mode<br>6: SCT3258 enters Halt mode |
| Default | 0 |

**Table 1-18 CHIP_LOWPWR Field Format**

## 1.2.12 PARITY_ENABLE

This command enables parity checks on all packets:

| Field Identifier | Parity Enable/Disable |
|---|---|
| 1 Byte | 1 Byte |
| 0x016 | 0: Disable parity checks |

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| | |
|---|---|
| | 1: Enable parity checks |
| Default | 0 |

**Table 1-19 PARITY_ENABLE Field Format**

Note that unlike other fields, PARITY_ENABLE field can only be the only field in a packet.

## 1.2.13 ACK_MESSAGE

This is a response packet to a received packet, when packet is not received correctly, and the packet field is not recognizable or unreliable.

Note that when received field is recognizable reliably even though there is error in the packet, SCT3258 respond with the command field instead of ACK_MESSAGE field.

Note that 0x17 0x0A does not indicate an error condition. This is sent when SCT3258 has completely sent the message to be sent to the far end.

| Field Identifier | Failure Code |
|---|---|
| 1 Byte | 1 Byte |
| 0x017 | 0: No error<br>1: Response to system reset. SCT3258 is ready to receive command from the host.<br>2: Field length exceeds maximum allowed<br>3: Parity error<br>4: Unknown command error<br>5: Packet length does not match with field Length<br>6: Error in processing field<br>7: I2C write error<br>8: SLD request time out error<br>9: TCH request time out error<br>10. Message to far end completion indication<br>11. Error in processing called ID or own ID<br>12. Chip not activated<br>13. Unsupported feature<br>14. Function not loaded<br>15. Odd byte error<br><br>Others: reserved |
| Default | 0 |

**Table 1-20 PACKET_FAILURE Field Format**

## 1.2.14 WORK_MODE

This command set the work mode for SCT3258.

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| Field Identifier | Work Mode | Modem Loop Mode | Audio Loop Back | Debug Mode |
|---|---|---|---|---|
| 1 Byte | 1 Byte | 1/2 Byte [7:4] | 1/2 Byte [3:0] | 1 Byte |
| 0x018 | See Table 1-19 for details. | 0: No Loop<br>1: Codec Loop back<br>2: 4 FSK Loop back<br>3: Packet Loop back<br>4: FEC Loop back<br>5: Vocoder Loop back<br>6: Decimator Loop back<br>Others: reserved | 0: No Loop<br>1: Codec Loop back<br>2: Decimator Loop back<br>3: Vocoder Loop back<br>4: FEC Loop back<br>5: Packet Loop back<br>6: 4 FSK Loop back<br>Others: reserved | See Table 1-20 for details |
| Default | 0 | 0 | 0 | 0 |

**Table 1-21 WORK MODE Field Format**

| Bit Position | Descriptions |
|---|---|
| 7 | Two Point Modulation Disable Flag<br>0: Enable two point modulation in TX mode<br>1: Disable two point modulation in TX mode |
| 6 | Special SCT3700 calibration loop enable<br>0: Normal mode<br>1: Enable the special SCT3700 calibration loop |
| 5-4 | Reserved |
| 3-0 | Work Mode<br>0: Idle<br>1: RX<br>2: TX<br>3: Full Duplex |

**Table 1-22 Details of Work Mode Byte**

| Byte Value | Descriptions |
|---|---|
| 0 | No debug |
| 1 | Mod port send fixed sine wave of 1000 Hz |
| 2 | Mod port send 4 FSK signal with FS1 pattern |
| 3 | Reserved |
| 4 | Reserved |
| 5 | BER Test mode |
| 6 | Reserved |
| 7 | Reserved |
| 8 | Reserved |
| 9 | TX sends 1031 Hz test Tone instead Voice from microphone |

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| | |
|---|---|
| 10 | TX sends silence instead Voice from microphone |
| 11 | DMR BER test mode (STD ID 511) |
| 14 | DPMR/DMR BER test mode (STD IB CALL), see note 1 |

**Table 1-23 Details of Debug Mode Byte**

Note 1: The STD IB CAL pattern changes the STB IB 511 pattern with every 100th bit inverted. This gives this pattern a 1% bit error rate

## 1.2.15 CARRIER_READY

This command informs SCT3258 that carrier is detected on the RX side. Upon receiving this message, SCT3258 start to look for preamble and frame SYNCs. This command is also used to inform SCT3258 whether to mute the audio if the called ID (for DPMR calls) or CTCSS/DCS code does match.

If carrier detection (SQ) is to be done by SCT3258 (enabled by CALLL_OPTION command), this command starts the carrier detection on SCT3258.

| Field Identifier | Carrier Ready Indication |
|---|---|
| 1 Byte | 1 Byte |
| 0x019 | Bit 7:3, Reserved<br>Bit 2: Mute Flag for muting audio if called ID or CTCSS/DCS code does not match.<br>0: Un-mute the audio even if the called ID (for DPMR calls) or CTCSS/DCS code (for a analog calls) does not match<br>1: Mute the audio if the called ID (for DPMR call) or CTCSS/DCS code (for analog calls) does not match, un-mute other wise.<br>Bit 1:0, Carrier ready flag<br>0: Carrier Lost<br>1: Carrier Ready<br>2: Carrier Ready, but the phase of the demodulated signal is reversed.<br>Others: reserved<br>Note： The reverse of demodulated signal polarity does not work in analog mode. Please configure the polarity with ANALOG_SUBAUDIO. |
| Default | 0 |

**Table 1-24 CARRIER_READY Field Format**

## 1.2.16 PROCESS_MODE

This command set the process mode of SCT3258. The process mode includes DPMR / DMR physical layer only (1), physical layer plus data link layer (2) and physical layer plus data link

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layer plus call control layer (3). This command is also used to enable automatic detection of analog or digital calls.

| Field Identifier | Process Mode |
|---|---|
| 1 Byte | 1 Byte |
| 0x01A | 2: Layer 2, DPMR / DMR Layer 2 Mode<br>3: Layer 3, Easy Mode<br>128: Analog Mode<br>131: Automatic detection of analog call or DPMR calls (layer 3)<br>Others: Reserved |
| Default | 2 |

**Table 1-25 PROCESS_MOD Field Format**

With layer two processing mode, SCT3258 is responsible to perform layer one and layer two processing of the DPMR / DMR protocol, including the FEC, interleaving and message interpretation.

Layer 2 processing complies with ETSI TS 102 490 or TS 102 658 in DPMR mode, and ETSI TS 102 361 in DMR mode,

With layer three processing mode, in addition to layer two processing, the SCT3258 also performs call control of the DPMR / DMR processing. ANNEX A Standard User Interface for CSF radio in DPMR mode, and ANNEX C DMR Numbering and Dialing plan are implemented in SCT3258. At the transmitter, SCT3258 supports wild character dialing for group and all calls for DPMR and DMR, abbreviated dialing and masked dialing for DPMR. At the receiver, when a call is received, SCT3258 does matching of individual call number and group number and informs the MCU of matching results.

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## 1.2.17 NEW_EVENT

This message is sent when a new event occurs and requires MCU attention. Note that some of the fields different in DPMR and DMR mode.

| Field Identifier | Event |
|---|---|
| 1 Byte | 1 Byte |
| 0x01B | 0: No event<br>1: DPMR: New Header frame received, with FEC error<br>DMR: Data Sync received with FEC error<br>2: DPMR: New End frame received<br>3: DPMR: New CCH received<br>4: DPMR New Header frame received with no FEC error<br>DMR: Data SYNC received |

| | |
|---|---|
| | 5: DPMR: Super Frame received, with no FEC error<br>DMR: Voice SYNC received<br>6: DPMR: FS3 received<br>7: DPMR: FS4 received<br>8: Lost synchronization<br>9. DPMR: FS1 found pre-alert<br>10. DPMR: FS2 found pre-alert<br>11. Reserved<br>12. Carrier detected<br>13. Carrier lost<br>14. Unsupported call received<br>15. Wrong vocoder<br>16-255 reserved |
| Default | 0 |

**Table 1-26 NEW_EVENT Field Format**

If a digital call is received and correctly decoded, SCT3258 reports NEW_EVENT 4 OR 5. If a digital call is received but has FEC error, SCT3258 reports NEW_EVENT 1. These event reports can also be disabled with REPORT_FIELD command.

In DPMR mode, if mixed call detection is enabled, and a digital call is received, NEW_EVENT 9 or 10 will be reported first, before complete FEC checking process, to alert MCU that a digital call is received. These events can be used to terminate analog call detection.

If SCT3258 is to do carrier detection (SQ), NEW_EVENT 12 and 13 are used to report carrier status.

During the course of a digital call, if synchronization pattern is not detected for a period of time (configured through command SYNC_MISS_COUNT), SCT3258 reports NEW_EVENT 8.

## 1.2.18 DEMOD_GAIN

This message set the demodulation gain for the demodulator. If the value is zero, auto calibration is used.

| Field Identifier | Event |
|---|---|
| 1 Byte | 2 Byte |
| 0x01C | 0: Auto calibration<br>Others: linear gain values in Q16.11 (16 bits total, 11 bits fraction, MSB first), with 2048 as 0 dB |
| Default | 0 |

**Table 1-27 DEMOD_GAIN Field Format**

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## 1.2.19 REPORT_FIELD

This message informs SCT3258 what field to report when receiving a call.

| Field Identifier | Mask Byte |
|---|---|
| 1 Byte | 1 Byte |
| 0x01D | Mask to indicate the field to report when receiving a new call. Each individual bit select one field. Multiple fields can be selected. |
| Default | 0x70 |

**Table 1-28 REPORT_FIELD Field Format**

The details of the bits are different depending on DPMR or DMR mode.

| Bit Position | Field to report |
|---|---|
| 7 | NEW_EVENT (FS1 or FS2) |
| 6 | DPMR_M_V_F_E |
| 5 | DPMR_HT_CI_PM |
| 4 | DIGC_CALLED_ID_BIN |
| 3 | DPMR_CALLED_ID_BCD |
| 2 | DIGC_OWN_ID_BIN |
| 1 | DPMR_OWN_ID_BCD |
| 0 | DPMR_CC |

**Table 1-29 Bit definition for Mask Byte in DPMR Mode**

| Bit Position | Field to report |
|---|---|
| 7 | NEW_EVENT (FS1 or FS2) |
| 6 | DMR_FLC |
| 5 | DMR_CSBK |
| 4-1 | Reserved |
| 0 | DMR SLOTTYPE OR EMB |

**Table 1-30 Bit definition for Mask Byte in DMR Mode**

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## 1.2.20 BER_REPORT

This command enquires for the BER test report. The BER includes test success flag, test duration, total number of errors, and error rate in ppm (pieces per million). A duration parameter in the command control the duration used in ppm calculation

| Field Identifier | BER Duration |
|---|---|
| 1 Byte | 1 Byte |

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| | |
|---|---|
| 0x01E | BER statistics duration<br>0: All errors in the test are used in BER ppm calculation<br>1-30, number of seconds used for ppm calculation. Error beyond the duration are discarded in BER ppm calculation |
| Default | 0 |

**Table 1-31 BER_REPORT Field Format**

The response packet returns the BER results.

| Field Identifier | Valid Indication | Test length | Errors | Error in PPM |
|---|---|---|---|---|
| 1 Byte | 1 Byte | 2 Bytes | 4 Bytes | 4 Bytes |
| 0x01E | 1: BER test successful<br>0: BER test unsuccessful. Result is not meaningful | Word (MSB first) to indicate the number of seconds in BER test | Long word (MSB first) to indicate the number of errors found. | Long word (MSB first) to indicate the error percentage (in ppm) |
| Default | 0 | 0 | 0 | 0 |

**Table 1-32 BER_REPORT response Field Format**

## 1.2.21 ANALOG_TONE

This command enables the SCT3258 to generate or stop a tone signals to be played out from the line out port or the MOD port. The tone signal is a repetition of a two-tone pattern, in the form as Tone1-Tone2-Tone1-Tone2….-Tone1-Tone2. Each tone is made up of two frequencies. The frequencies and amplitudes are programmable. If silence is desired for the second tone, the user should set the two frequencies or the two amplitudes of the second tone to zero.

| Field Identifier | TONE_CTRL | Frequency 11 | Frequency 12 | Frequency 21 | Frequency 22 |
|---|---|---|---|---|---|
| 1 Byte | 1 Byte | 2 Bytes | 2 Bytes | 2 Bytes | 2 Bytes |
| 0x01F | See Table 1-32 for Details | First frequency of the first tone, in Hz<br>If Bit 7 of the MSB is 1, siren is generated | Second frequency of the first tone, in Hz | First frequency of the Second tone, in Hz | Second frequency of the second tone, in Hz |
| Default | 0 | NA | NA | NA | NA |

| Amplitude11 | Amplitude 12 | Amplitude 21 | Amplitude 22 | Length 1 | Length 2 |
|---|---|---|---|---|---|

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| 1 Byte | 1 Byte | 1 Bytes | 1 Bytes | 2 Bytes | 2 Bytes |
|---|---|---|---|---|---|
| Amplitude of the first frequency of the first tone, in 8 bit linear scale with 6 bit fractions | Amplitude of the second frequency of the first tone, in 8 bit linear scale with 6 bit fractions | Amplitude of the first frequency of the second tone, in 8 bit linear scale with 6 bit fractions | Amplitude of the second frequency of the second tone, in 8 bit linear scale with 6 bit fractions | Length of the first tone in milliseconds. | Length of the second tone in milliseconds. |
| NA | NA | NA | NA | NA | NA |

**Table 1-33 LOCAL_TONE Field Format**

| Bit Position | Field to report |
|---|---|
| 7 | Remote Tone Flag<br>0: Local Tone to the line out port<br>1: Remote Tone to the MOD port |
| 6-0 | Repeat Number<br>0: Stop Tone<br>127: Infinite<br>Others: Repetition Number |

**Table 1-34 Definition for TONE_CTRL**

## 1.2.22 CALL_START

Call Start command. Upon receiving this command, SCT3258 begins to send call to the far end.

| Field Identifier | Call Details |
|---|---|
| 1 Byte | 1 Byte |
| 0x20 | DATA[0]={M[2:0], V[1:0], F[1:0], EP}<br>Call Mode, Version, Call Format,<br>Emergency Call flag |
| Default | 0 |

**Table 1-35 CALL_START Field Format**

With CALL_START command, the header type, HT is set to 0x0.

The M, F, and P fields are described in the following tables

| M | Communication Modes |
|---|---|
| 000 | Voice communication (no user data in SLD field) |
| 001 | Voice + slow data (user data in SLD field) |
| 010 | Data communication type 1 (Payload is user data without FEC) |
| 011 | Data communication type 2 (Payload is user data with FEC) |

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| | |
|---|---|
| 100 | Data communication type 3 (Packet data, ARQ method) |
| 101 | Voice and appended data (type 2) |
| 110 | Appended data |
| Others | Reserved |

**Table 1-36 Communication Mode Field Details**

| | |
|---|---|
| V | Version number |
| 00 | DPMR standard traffic |
| 01 | Reserved |
| 10 | Reserved |
| 11 | Manufacturer specific |

**Table 1-37 Version Field Details**

| | |
|---|---|
| F | Communication Format |
| 00 | Call ALL (Broadcast) |
| 01 | Peer-to-peer communication |
| 10 | BS uplink |
| 11 | BS downlink |

**Table 1-38 Communication Format Field Details**

| | |
|---|---|
| EP | Emergency Priority Flag |
| 0 | Normal call |
| 1 | Emergency call |

**Table 1-39 Emergency Priority Flag Field Details**

## 1.2.23 CALL_STOP

Call Stop command. Upon receiving this command, SCT3258 sends an END frame to the far end and then stop the call to the far end.

| Field Identifier |
|---|
| 1 Byte |
| 0x21 |

**Table 1-40 CALL_STOP Field Format**

This command is also used to end an analog call.

## 1.2.24 CONNECT

Call Connect command. Upon receiving this command, SCT3258 send connection command to the far end. It is also used send a call alert to the far end in the case of voice calls.

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| Field Identifier | Call Details |
|---|---|
| 1 Byte | 1 Byte |
| 0x22 | DATA[0]={M[2:0], V[1:0], F[1:0], EP}<br>Call Mode, Version, Call Format,<br>Emergency Call flag |
| Default | 0 |

**Table 1-41 CONNECT Field Format**

With CONNECT command, the header type, HT is set to 0x01.

The meaning of M, V, F and EP fields are described in Table 1-32 to Table 1-35.

If there is no frame after sending this command, we must wait SCT3258 report 17 0A then make the subsequent operation (The response time is 150ms+Preamble +Silence+PowerSaveHeader).

## 1.2.25 DISCONNECT

Call Disconnection command. Upon receiving this command, SCT3258 sends disconnect command to the far end.

| Field Identifier | Call Details |
|---|---|
| 1 Byte | 1 Byte |
| 0x23 | DATA[0]={M[2:0], V[1:0], F[1:0], EP}<br>Call Mode, Version, Call Format,<br>Emergency Call flag |
| Default | 0 |

**Table 1-42 DISCONNECT Field Format**

With DISCONNECT command, the header type, HT is set to 0x02.

The meaning of M, V, F and EP fields are described in section CALL_START.

If there is no frame after sending this command, we must wait SCT3258 report 17 0A then make the subsequent operation (The response time is 300ms+Preamble +Silence +PowerSaveHeader).

## 1.2.26 DPMR_ACK

ACK command: Upon receiving this command, SCT3258 send acknowledgment to the far end.

| Field Identifier | Call Details |
|---|---|
| 1 Byte | 1 Byte |
| 0x24 | DATA[0]={M[2:0], V[1:0], F[1:0], EP} |

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| | |
|---|---|
| | Call Mode, Version, Call Format,<br>Emergency Call flag |
| Default | 0 |

**Table 1-43 DPMR_ACK field Format**

With DPMR_ACK command, the header type, HT is set to 0x03.

The meaning of M, V, F and EP fields are described in section CALL_START.

If there is no frame after sending this command, we must wait SCT3258 report 17 0A then make the subsequent operation (The response time is 100ms+Preamble +Silence+PowerSaveHeader).

## 1.2.27 SYSTEM_INFO

System Info command, upon receiving this command, SCT3258 sends system or status request or response to the far end.

| Field Identifier | Call Details |
|---|---|
| 1 Byte | 1 Byte |
| 0x25 | DATA[0]={M[2:0], F[3:0], EP}<br>Call Mode, Version, Call Format,<br>Emergency Call flag |
| Default | 0 |

**Table 1-44 SYSTEM_INFO field Format**

SYSTEM_INFO command does not set header type, HT, field. The header type can be set with DPMR_HT_CI_PM or DPMR_HT command before sending SYSTEM_INFO.

The meaning of M, V, F and EP fields are described in in section CALL_START.

If there is no frame after sending this command, we must wait SCT3258 report 17 0A then make the subsequent operation (The response time is 150ms+Preamble +Silence+PowerSaveHeader).

## 1.2.28 ANALOG_SUBAUDIO

This field contains analog sub_audio message. Upon receiving this message, SCT3258 configure the sub-audio generation and detection setting of SCT3258 in analog mode. It is also used to report detection of specific CTCSS/DCS tones and tail detection results.

The number of bytes for this field varies depending on the whether arbitrary CTCSS/DCS generation and detection is required. If no arbitrary CTCSS/DCS is required, the field length is 3. If arbitrary DCS code is required, the field length is 5. The added two bytes are for the arbitrary

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DCS code in hex form. If arbitrary CTCSS code is required, the field length is 9. Two of the added six bytes are for the CTCSS frequencies, another four bytes are for the two coefficients for the narrow band filter for that particular frequency.

The CTCSS coefficients are calculated from the arbitrary CTCSS frequency to be detected. The formula is:

a1 = round (2048 * sqrt (1.9990 * (1-cos(2*w0))))

a2 = round (-2047*2^9 * cos (w0)) + 2^20

where w0 is calculated from the desired arbitrary CTCSS frequency, f0, by

w0 = f0/1909.9

For example for CTCSS frequency 67 Hz, the coefficients are calculated as follows:

a1 = round (2048*sqrt (1.9990 * (1-cos (2*67/1909.9)))) = 144
a2 = round ((-2047*2^9 * cos (67/1909.9))) + 2^20 = 1157

The MCU is responsible to calculate the coefficients and pass them to SCT3258.

| Field Identifier | Sub-audio Configuration / Status | CTCSS/DCS number | Arbitrary CTCSS frequency or DCS code | Coefficient A1 | Coefficient a2 |
|---|---|---|---|---|---|
| 1 Byte | 1 Byte | 1 Byte | 2 Bytes | 2 Bytes | 2 Bytes |
| 0x26 | See Table 1-44 for details | 0: No CTCSS /DCS<br>1-108: CTCSS/DCS code<br>255: Arbitrary CTCSS/DCS code<br>Others: reserved | For CTCSS: CTCSS frequency in steps of 0.1 Hz (MSB first)<br><br>For DCS, 0-0x1ff, corresponding to the DCS code, in HEX format , MSB first | Narrow band filter coefficient a1 for arbitrary CTCSS frequency | Narrow band filter coefficient a1 for arbitrary CTCSS frequency |
| Default | 0x40 | 0 | NA | NA | NA |

**Table 1-45 SUBAUDIO Field Format**

| Bit Position | Bit Name | Descriptions |
|---|---|---|

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| | | |
|---|---|---|
| 7 | CD_SEL | DCS and CTCSS select<br>0: CTCSS<br>1: DCS |
| 6 | EXP_SEL | Expanded CTCSS / DCS code Select<br>0: Standard CTCSS/DCS set (38 for CTCSS, 83 for DCS)<br>1: Expanded CTCSS DCS Set (51 for CTCSS, 107 for DCS) |
| 5 | Polarity | DCS polarity<br>0: Positive DCS polarity<br>1: Negative DCS polarity |
| 4 | Blind | Blind sub-audio detection enable<br>0: No blind sub-audio detection<br>1: Blind sub-audio detection enabled. |
| 3 | Auto Polarity Detection | DCS Auto polarity detection enable<br>0: DCS Auto polarity detection disabled<br>1: DCS Audio polarity detection enable |
| 2-1 | Tail Detection Configuration or Report | In a configuration packet, it is used to enabling tail generation and detection:<br>00: No Tail detection<br>01: 120 degree tail<br>10: 180 degree tail for CTCSS. Enable tail detection for DCS.<br>11: 240 degree tail<br><br>In a report packet, it is used to report tail detection results:<br>00: No Tail found<br>01: 120 degree tail found<br>10: 180 degree tail found<br>11: 240 degree tail found |
| 0 | FOUND | Sub audio find flag<br>0: No CTCSS/DCS found<br>1: CTCSS/DCS found |

**Table 1-46 Sub-Audio Configuration Details**

| Serial Number： | DCS code in Octal format | DCS code in Hex format |
|---|---|---|
| 1 | 017 | 00F |
| 2 | 023 | 013 |
| 3 | 025 | 015 |
| 4 | 026 | 016 |
| 5 | 031 | 019 |
| 6 | 032 | 01A |

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| | | |
|---|---|---|
| 7 | 036 | 01E |
| 8 | 043 | 023 |
| 9 | 047 | 027 |
| 10 | 050 | 028 |
| 11 | 051 | 029 |
| 12 | 053 | 02B |
| 13 | 054 | 02C |
| 14 | 065 | 035 |
| 15 | 071 | 039 |
| 16 | 072 | 03A |
| 17 | 073 | 03B |
| 18 | 074 | 03C |
| 19 | 114 | 04C |
| 20 | 115 | 04D |
| 21 | 116 | 04E |
| 22 | 122 | 052 |
| 23 | 125 | 055 |
| 24 | 131 | 059 |
| 25 | 132 | 05A |
| 26 | 134 | 05C |
| 27 | 143 | 063 |
| 28 | 145 | 065 |
| 29 | 152 | 06A |
| 30 | 155 | 06D |
| 31 | 156 | 06E |
| 32 | 162 | 072 |
| 33 | 165 | 075 |
| 34 | 172 | 07A |
| 35 | 174 | 07C |
| 36 | 205 | 085 |
| 37 | 212 | 08A |
| 38 | 223 | 093 |
| 39 | 225 | 095 |
| 40 | 226 | 096 |
| 41 | 243 | 0A3 |
| 42 | 244 | 0A4 |
| 43 | 245 | 0A5 |
| 44 | 246 | 0A6 |
| 45 | 251 | 0A9 |
| 46 | 252 | 0AA |
| 47 | 255 | 0AD |
| 48 | 261 | 0B1 |
| 49 | 263 | 0B3 |

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| | | |
|---|---|---|
| 50 | 265 | 0B5 |
| 51 | 266 | 0B6 |
| 52 | 271 | 0B9 |
| 53 | 274 | 0BC |
| 54 | 306 | 0C6 |
| 55 | 311 | 0C9 |
| 56 | 315 | 0CD |
| 57 | 325 | 0D5 |
| 58 | 331 | 0D9 |
| 59 | 332 | 0DA |
| 60 | 343 | 0E3 |
| 61 | 346 | 0E6 |
| 62 | 351 | 0E9 |
| 63 | 356 | 0EE |
| 64 | 364 | 0F4 |
| 65 | 365 | 0F5 |
| 66 | 371 | 0F9 |
| 67 | 411 | 109 |
| 68 | 412 | 10A |
| 69 | 413 | 10B |
| 70 | 423 | 113 |
| 71 | 431 | 119 |
| 72 | 432 | 11A |
| 73 | 445 | 125 |
| 74 | 446 | 126 |
| 75 | 452 | 12A |
| 76 | 454 | 12C |
| 77 | 455 | 12D |
| 78 | 462 | 132 |
| 79 | 464 | 134 |
| 80 | 465 | 135 |
| 81 | 466 | 136 |
| 82 | 503 | 143 |
| 83 | 506 | 146 |
| 84 | 516 | 14E |
| 85 | 523 | 153 |
| 86 | 526 | 156 |
| 87 | 532 | 15A |
| 88 | 546 | 166 |
| 89 | 565 | 175 |
| 90 | 606 | 186 |
| 91 | 612 | 18A |
| 92 | 624 | 194 |

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| | | |
|---|---|---|
| 93 | 627 | 197 |
| 94 | 631 | 199 |
| 95 | 632 | 19A |
| 96 | 646 | 1A6 |
| 97 | 654 | 1AC |
| 98 | 662 | 1B2 |
| 99 | 664 | 1B4 |
| 100 | 703 | 1C3 |
| 101 | 712 | 1CA |
| 102 | 723 | 1D3 |
| 103 | 731 | 1D9 |
| 104 | 732 | 1DA |
| 105 | 734 | 1DC |
| 106 | 743 | 1E3 |
| 107 | 754 | 1EC |
| 108 | 0 | 000 |

**Table 1-47 DCS Extend Table**

| Serial Number: | DCS code in Octal format | DCS code in Hex Format |
|---|---|---|
| 1 | 023 | 013 |
| 2 | 025 | 015 |
| 3 | 026 | 016 |
| 4 | 031 | 019 |
| 5 | 032 | 01A |
| 6 | 043 | 023 |
| 7 | 047 | 027 |
| 8 | 051 | 029 |
| 9 | 054 | 02C |
| 10 | 065 | 035 |
| 11 | 071 | 039 |
| 12 | 072 | 03A |
| 13 | 073 | 03B |
| 14 | 074 | 03C |
| 15 | 114 | 04C |
| 16 | 115 | 04D |
| 17 | 116 | 04E |
| 18 | 125 | 055 |
| 19 | 131 | 059 |
| 20 | 132 | 05A |
| 21 | 134 | 05C |
| 22 | 143 | 063 |
| 23 | 152 | 06A |
| 24 | 155 | 06D |
| 25 | 156 | 06E |

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| | | |
|---|---|---|
| 26 | 162 | 072 |
| 27 | 165 | 075 |
| 28 | 172 | 07A |
| 29 | 174 | 07C |
| 30 | 205 | 085 |
| 31 | 223 | 093 |
| 32 | 226 | 096 |
| 33 | 243 | 0A3 |
| 34 | 244 | 0A4 |
| 35 | 245 | 0A5 |
| 36 | 251 | 0A9 |
| 37 | 261 | 0B1 |
| 38 | 263 | 0B3 |
| 39 | 265 | 0B5 |
| 40 | 271 | 0B9 |
| 41 | 306 | 0C6 |
| 42 | 311 | 0C9 |
| 43 | 315 | 0CD |
| 44 | 331 | 0D9 |
| 45 | 343 | 0E3 |
| 46 | 346 | 0E6 |
| 47 | 351 | 0E9 |
| 48 | 364 | 0F4 |
| 49 | 365 | 0F5 |
| 50 | 371 | 0F9 |
| 51 | 411 | 109 |
| 52 | 412 | 10A |
| 53 | 413 | 10B |
| 54 | 423 | 113 |
| 55 | 431 | 119 |
| 56 | 432 | 11A |
| 57 | 445 | 125 |
| 58 | 464 | 134 |
| 59 | 465 | 135 |
| 60 | 466 | 136 |
| 61 | 503 | 143 |
| 62 | 506 | 146 |
| 63 | 516 | 14E |
| 64 | 532 | 15A |
| 65 | 546 | 166 |
| 66 | 565 | 175 |
| 67 | 606 | 186 |
| 68 | 612 | 18A |

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| | | |
|---|---|---|
| 69 | 624 | 194 |
| 70 | 627 | 197 |
| 71 | 631 | 199 |
| 72 | 632 | 19A |
| 73 | 654 | 1AC |
| 74 | 662 | 1B2 |
| 75 | 664 | 1B4 |
| 76 | 703 | 1C3 |
| 77 | 712 | 1CA |
| 78 | 723 | 1D3 |
| 79 | 731 | 1D9 |
| 80 | 732 | 1DA |
| 81 | 734 | 1DC |
| 82 | 743 | 1E3 |
| 83 | 754 | 1EC |
| 84 | 0 | 000 |

**Table 1-48 DCS Standard Table**

| Serial Number： | CTCSS frequency |
|---|---|
| 1 | 67 |
| 2 | 71.9 |
| 3 | 74.4 |
| 4 | 77 |
| 5 | 79.7 |
| 6 | 82.5 |
| 7 | 85.4 |
| 8 | 88.5 |
| 9 | 91.5 |
| 10 | 94.8 |
| 11 | 97.4 |
| 12 | 100 |
| 13 | 103.5 |
| 14 | 107.2 |
| 15 | 110.9 |
| 16 | 114.8 |
| 17 | 118.8 |
| 18 | 123 |
| 19 | 127.3 |
| 20 | 131.8 |
| 21 | 136.5 |
| 22 | 141.3 |
| 23 | 146.2 |

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| | |
|---|---|
| 24 | 151.4 |
| 25 | 156.7 |
| 26 | 162.2 |
| 27 | 167.9 |
| 28 | 173.8 |
| 29 | 179.9 |
| 30 | 186.2 |
| 31 | 192.8 |
| 32 | 203.5 |
| 33 | 210.7 |
| 34 | 218.1 |
| 35 | 225.7 |
| 36 | 233.6 |
| 37 | 241.8 |
| 38 | 250.3 |
| 39 | 134.4 |

**Table 1-49 CTCSS Standard Table**

| Serial Number： | ctcss 52 digit decimal |
|---|---|
| 1 | 63 |
| 2 | 67 |
| 3 | 69.3 |
| 4 | 71.9 |
| 5 | 74.4 |
| 6 | 77 |
| 7 | 79.7 |
| 8 | 82.5 |
| 9 | 85.4 |
| 10 | 88.5 |
| 11 | 91.5 |
| 12 | 94.8 |
| 13 | 97.4 |
| 14 | 100 |
| 15 | 103.5 |
| 16 | 107.2 |
| 17 | 110.9 |
| 18 | 114.8 |
| 19 | 118.8 |
| 20 | 123 |
| 21 | 127.3 |
| 22 | 131.8 |
| 23 | 136.5 |

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| | |
|---|---|
| 24 | 141.3 |
| 25 | 146.2 |
| 26 | 151.4 |
| 27 | 156.7 |
| 28 | 159.8 |
| 29 | 162.2 |
| 30 | 165.5 |
| 31 | 167.9 |
| 32 | 171.3 |
| 33 | 173.8 |
| 34 | 177.3 |
| 35 | 179.9 |
| 36 | 183.5 |
| 37 | 186.2 |
| 38 | 189.9 |
| 39 | 192.8 |
| 40 | 196.6 |
| 41 | 199.5 |
| 42 | 203.5 |
| 43 | 206.5 |
| 44 | 210.7 |
| 45 | 218.1 |
| 46 | 225.7 |
| 47 | 229.1 |
| 48 | 233.6 |
| 49 | 241.8 |
| 50 | 250.3 |
| 51 | 254.1 |
| 52 | 134.4 |

**Table 1-50 CTCSS Extend Table**

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## 1.2.29 CALL_MATCH

This field is used in a SCT3258 generated report message when a call is received, as a result from a CARRIER_READY command from the MCU. It is generated if a DPMR or a DMR call is received and that the receiver can start to SYNC with the transmitter.

| Field Identifier | Call Details |
|---|---|
| 1 Byte | 1 Byte |
| 0x27 | See Table 1-48 for details |
| Default | 0 |

**Table 1-51 CALL_MATCH Field Format**

| Bit Position | Bit Name | Descriptions |
|---|---|---|
| 7-4 | Group ID | For group calls, the matched group ID |
| 3 | CC Match | Color code match status<br>0: Color code matched<br>1: Color code not matched |
| 2 | Un-match Reason | The reason for call not matching<br>0: Valid Called ID not received<br>1: Called ID not matched<br>Others: reserved |
| 1 | Group Call | Group or individual call flag when a call is matched<br>0: Individual call<br>1: Group call |
| 0 | Called ID Match Flag | Called ID Match Flag<br>0: Called ID matched with receiver Own ID, or one of the receiver group ID<br>1: Called ID is not matched |

**Table 1-52 Call Match Details**

## 1.2.30 ADDRESSING_MODE

This field is used to configure the addressing mode of SCT3258 for DPMR. As described in the DPMR spec (TS 102 490 or 658).), with initial addressing mode, the user uses 254 binary ID's. With configured addressing mode, the user can use a dial pad to dial 7 digital numbers.

The field format is described below.

| Field Identifier | Addressing Mode Details |
|---|---|
| 1 Byte | 1 Byte |
| 0x28 | See Table 1-50 for details |
| Default | 0 |

**Table 1-53 ADDRESSING_MODE Field Format**

| Bit Position | Bit Name | Descriptions |
|---|---|---|
| 7-4 | Reserved | Reserved |
| 3-1 | Dial mask Length | 0-6: The length of the Dial mask for Configured Addressing mode<br>Others: invalid. |
| 0 | Mode Select | Address mode select<br>0: Initial Addressing mode<br>1: Configured Address mode (Standard User Interface) |

**Table 1-54 Addressing Mode Details**

The number of digital the user is allowed to dial is (7- Dial_Mask_Length). For example, 3 mean that 3 numbers are masked. The user is able to dial a maximum of 4 numbers. By default,

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Dial_Mask_Length = 0, meaning no mask is used and the user is allowed to dial all 7 digits.

## 1.2.31 GROUP_ID

The command is for DPMR mode only. A separate command DMR_GROUP_ID is used for DMR mode.

Each DPMR terminal can be assigned an individual ID (own ID) and one or several group ID's. When an incoming called ID matched with the receiver's individual ID or group ID, the receiver responds. The GROUP_ID field is used to pass the group ID to SCT3258.

This field contains the group ID in 7 BCD codes and one group number.

| Field Identifier | Group ID and group number |
|---|---|
| 1 Byte | 4 Bytes |
| 0x29 | DATA[0]= {K1, K2}<br>DATA[1]= {K3, K4}<br>DATA[2]= {K5, K6}<br>DATA[3]= {K7, Group Number},<br>where group number can be 1-15 |
| Default | 0,0,0,0 |

**Table 1-55 GROUP_ID Field Format**

K1, … , K7 are 0-9 for BCD numbers, or 10 for '*'.

The command fields support up to 15 group numbers.

## 1.2.32 SCRAMBLER_SEED

This command enables/disables the voice scrambler for encryptions and selects the 32 bit scrambler seed.

| Field Identifier | Scrambler Type | Scrambler seeds |
|---|---|---|
| 1 Byte | 1 Bytes | 0/4 bytes |
| 0x02A | 0: Disable the Scrambler<br>1: Enable 32 bit scrambler (DPMR Mode only)<br>4: Enabled 16 bit scrambler | 0 byte if scrambler type = 0;<br>4 bytes if scrambler type = 1;<br>DATA[0] = SEED[31:24]<br>DATA[1] = SEED[24:16]<br>DATA[2] = SEED[15:8]<br>DATA[3] = SEED[7:0]<br>2 bytes if scrambler type = 4;<br>DATA[0] = SEED[15:8] |

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| | | |
|---|---|---|
| | | DATA[1] = SEED[7:0] |
| Default | 0 | 0,0,0,0 |

**Table 1-56 SCRAMBLER_SEED Field Format**

Note: The 32 bit scrambler is indeed a 16 bit scrambler. The first 16-bit is used as scrambler seed which the end user can program. The second 16-bit is an XOR mask, which can be used by radio manufacturers to purposely make their radio to be incompatible with radio made by other manufactures. The second 16-bit should have at least 5 "1" to be effective.

For the 16 bit scrambler, when this command is set in TX, please set the PM in HT_CI_PM command to enable the encryption function.

## 1.2.33 MOD_GAIN

This message set the modulation gain for the modulator.

| Field Identifier | Event |
|---|---|
| 1 Byte | 2 Byte |
| 0x02C | Linear gain applied to the modulator in Q16.11 (16 bits total, 11 bits fraction, MSB first), with 2048 as 0 dB |

**Table 1-57 MOD_GAIN Field Format**

## 1.2.34 CODEC_SELECT

This command selects the codec that works with SCT3258. Current supported codecs include TLC320AIC3204 from TI and WM8758B from Wolfson.

| Field Identifier | Codec Selection |
|---|---|
| 1 Byte | 1 Bytes |
| 0x02D | 1: TLC320AIC3204<br>2: WM8758B<br>3: ES8338<br><br>Others: reserved. |
| Default | 2 |

**Table 1-58 CODEC_SELECT Field Format**

## 1.2.35 SET_I2C_ADDRESS

SCT3258 also supports other types of codecs, as long as it has an I2C interface for control and a serial interface for data transfer which conforms with SCT3258 format and timing. The I2C address of the codec can be set the command SET_I2C_ADDRESS

| Field Identifier | I2C Address |
|---|---|

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| 1 Byte | 1 Bytes |
|---|---|
| 0x02E | I2C address to pass to codec |
| Default | 0x34 |

**Table 1-59 SET_I2C_ADDRESS Field Format**

## 1.2.36 SYNC_MISS_COUNT

Once a call is set up, SCT3258 monitor the link conditions and continuously check the presence of synchronization word, (FS2 in the case DPMR and Voice SYNC pattern in DMR). It reports SYNC miss if the number of sync miss is greater than a threshold. This command set the threshold.

| Field Identifier | Sync Miss Count |
|---|---|
| 1 Byte | 1 Bytes |
| 0x02F | Threshold for sync miss counter before reporting to MCU.<br>0-254: threshold of sync miss to exceed (greater than) before report sync miss<br>255: never report sync miss |
| Default | 255 |

**Table 1-60 SYNC_MISS_COUNT Field Format**

The message to report Sync miss is NEW_EVENT with field value of 0x08.

## 1.2.37 Q_PRO_STR

This command enquires for product code.

| Field Identifier |
|---|
| 1 Byte |
| 0x030 |

**Table 1-61 Q_PRO_STR Field Format**

The response packet returns a string: “SCT3258” or other product code.

| Field Identifier | Number of Bytes | Data |
|---|---|---|
| 1 Byte | 1 Byte | Variable Number of bytes |
| 0x30 | Number of bytes | “SCT3258F” |

**Table 1-62 Q_PRO_STR Field Options**

## 1.2.38 Q_HW_VER

This command enquires for hardware version.

| Field Identifier |
|---|

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| 1 Byte |
|---|
| 0x031 |

**Table 1-63 Q_HW_VER Field Format**

The response packet returns a string: “V0.01.01” or other hardware version.

| Field Identifier | Number of Bytes | Data |
|---|---|---|
| 1 Byte | 1 Byte | Variable Number of bytes |
| 0x31 | Number of Bytes | “V0.02.02” |

**Table 1-64 Q_HW_VER Field Options**

## 1.2.39 Q_SW_VER

This command enquires for software version:

| Field Identifier |
|---|
| 1 Byte |
| 0x032 |

**Table 1-65 Q_SW_VER Field Format**

The response packet returns a string: “V0.01.01” or other software version.

| Field Identifier | Number of Bytes | Data |
|---|---|---|
| 1 Byte | 1 Byte | Variable Number of bytes |
| 0x32 | Number of Bytes | “V1.00.04” |

**Table 1-66 Q_SW_VER Field Options**

## 1.2.40 Q_CID_SN

This command enquires for customer ID and serial number.

| Field Identifier |
|---|
| 1 Byte |
| 0x033 |

**Table 1-67 Q_CID_SN Field Format**

The response packet returns customer ID in 6 BCD numbers and serial number in 10 BCD numbers.

| Field Identifier | Customer ID | Serial Number |
|---|---|---|
| 1 Byte | 3 Bytes | 5 Bytes |
| 0x33 | Customer ID in 6 BCD numbers | Serial Number in 10 BCD numbers |

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**Table 1-68 Q_CIN_SN Field Options**

## 1.2.41 DMR_GROUP_ID

Each DMR terminal can be assigned an individual ID (own ID) and one or several group ID's. When an incoming called ID matched with the receiver's individual ID or group ID, the receiver responds. The DMR_GROUP_ID field is used to pass the group ID to SCT3258.

This field contains the group ID in 8 BCD codes and one group number.

| Field Identifier | Group ID and group number |
|---|---|
| 1 Byte | 5 Bytes |
| 0x29 | DATA[0]= {K1, K2}<br>DATA[1]= {K3, K4}<br>DATA[2]= {K5, K6}<br>DATA[3]= {K7, K8}<br>DATA[4] = Group Number<br>where group number can be 1-15 |
| Default | 0,0,0,0 |

**Table 1-69 GROUP_ID Field Format**

K1, … , K8 are 0-9 for BCD numbers.

The command fields support up to 15 group numbers.

## 1.2.42 DC_OFFSET

This command applies a DC offset to the input signals from the ADC and to the output signal to the DAC. It also applies a digital gain to the signals to the DAC before DC offset is added.

The value of the DC offset is in 2's complements, with 0x7fff corresponds to the maximum positive DC offset, and 0x8000 corresponds to the maximum negative DC offsets. The electrical voltage of the DC offset depends on the codec types and the analog gain that is applied to the codec.

With zero analog gain, the maximum offset is capped by the supply voltage of the codec, and the minimum is capped by zero.

In the case of DAC output, a gain is applied to the output signal before DC offset is added. The gain is in linear scale, with 0x0800 corresponding to zero gain.

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| Field Identifier | DC offset for ADC Left Channel | DC offset for ADC Right Channel | DC offset for DAC Left Channel | DC offset for DAC right Channel | Digital Gain for DAC left channel | Digital Gain for DAC right Chanel |
|---|---|---|---|---|---|---|
| 1 Byte | 2 Byte | 2 Bytes | 2 Bytes | 2 Bytes | 2 Bytes | 2 Bytes |
| 0x039 | DC offset Value, MSB followed by LSB, 0x8000-0x7fff | DC offset Value, MSB followed by LSB, 0x8000-0x7fff | DC offset Value, MSB followed by LSB, 0x8000-0x7fff | DC offset Value, MSB followed by LSB, 0x8000-0x7fff | Digital Gain Value, MSB followed by LSB, In linear Scale, Q16.11 | Digital Gain Value, MSB followed by LSB, In linear Scale, Q16.11 |
| Default | 0 | 0 | 0 | 0 | 2048 | 2048 |

**Table 1-70 DC_OFFSET Field Format**

When paired with SCT3700, DC_OFFSET command is also used to read the DC levels and amplitudes of the signal in the RX path. In addtion, the DC levels and amplitudes of TX signal can be read by looping the TX to the RX and then read the RX DC levels and amplitudes.

To read DC levels and amplitudes, DC_OFFSET command should be sent with “Read/Write” bit in the TYPE field set to 1. The DC_OFFSET command used in “read” mode has the following format:

| Field Identifier | DC Time Constant |
|---|---|
| 1 Byte | 1 Byte |
| 0x39 | Time constant for averaging DC and amplitude. |
| Default | 0 |

Default time constant of 0 corresponds to average time of 31.25 ms, which is the time it takes to reach within 10% of the steady state energy value. If the constant is n, the average time is given by the following formula:

T = 31.25 * 2^n.

The Response to the read DC_OFFSET read command has the following format:

| Field Identifier | DC offset for the Left Channel | DC offset for the Right Channel | Amplitude for the Left Channel | Amplitude for the Right Channel | Total energy of IQ channel | Division out of AMP_I/AMP_Q |
|---|---|---|---|---|---|---|
| 1 Byte | 2 Byte | 2 Bytes | 2 Bytes | 2 Bytes | 4 Bytes | 1 Byte |

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| | | | | | | |
|---|---|---|---|---|---|---|
| 0x039 | DC offset Value, MSB followed by LSB, in Q16.15, 0x8000-0x7fff | DC offset Value, MSB followed by LSB, in Q16.15, 0x8000-0x7fff | Amplitude Value, MSB followed by LSB, in Q16.15 0x0-0x7fff | Amplitude Value, MSB followed by LSB, in Q16.15 0x0-0x7fff | I^2 +Q^2, used to do TX DC calibration, only for SCT3700, | Used for IQ calibration only for SCT3700 |
| Default | - | - | - | | | - |

## 1.2.43 ANALOG_CONFIG

This command changes the SCT3258 configuration in analog mode. The parameter that can be configured in analog mode include LPF selection for 12.5 kHz or 25 kHz channel spacing, pre-emphasis and de-emphasis filter settings, and various gain values at different stages in the signal chain. All gain values are linear gain with 16 bit resolution.

| Field Identifier | Mode Configure | TX gain Before Limiter | TX Gain after Limiter | CTCSS Gain | DCS Gain | RX Volume Gain |
|---|---|---|---|---|---|---|
| 1 Byte | 1 Byte | 2 Bytes | 2 Bytes | 2 Bytes | 2 bytes | 2 Bytes |
| 0x03C | See Table 1-67 for details | 16 bit linear gain, MSB first, in Q16.11 | 16 bit linear gain, MSB first, in Q16.12 | 16 bit linear gain, MSB first, in Q16.12 | 16 bit linear gain, MSB first, in Q16.12 | 16 bit linear gain, MSB first, in Q16.11 |
| Default | 2 | 2048 | 4096 | 4096 | 4096 | 2048 |

**Table 1-71 ANALOG_CONFIG Field Format**

| Bit Position | Bit Name | Descriptions |
|---|---|---|
| 7 | ALT_LPF | Additional attenuation at 3 kHz when BB_CH_SEL = 1 |
| 6 | DCS_HPF | DCS high pass filter enable flag, for removing DC<br>0: DCS high pass filter is disabled<br>1: DCS high pass filter is enabled |
| 5 | COMPANDER | Compander enable<br>0: Compander is disabled<br>1: Compander is enabled. |
| 4 | ANA_TONE_CFG | Analog Tone configure<br>0: No CTCSS or DCS when sending analog tones<br>1: Send CTCSS or DCS when sending analog tones |
| 3 | IF_CH_SEL | IF Channel Select<br>0: 12.5 kHz channel<br>1: 25 kHz channel |
| 2 | SKIP_EMP | Skip pre-emphasis and de-emphasis filters |

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| | | |
|---|---|---|
| | | 0: No skip pre-emphasis and de-emphasis filters<br>1: Skip pre-emphasis and de-emphasis filters |
| 1 | BB_CH_SEL | Base band channel filter select<br>0: LPF corner at 2.55k Hz<br>1: LPF corner at 3 kHz |
| 0 | FILTER_BYPASS | Filter by pass. All analog signal processing in the TX path and the RX path are bypassed if this bit is set. |

**Table 1-72 Mode Configure Details**

Note that IF_CH_SEL and BB_CH_SEL can be different. IF_CH_SEL selects 12.5KHz narrow band channel selection filter or 25 kHz wide band channel selection filter. BB_CH_SEL selects the base band filter LPF corner. The following table gives a guide line for IF_CH_SEL or BB_CH_SEL settings.

| IF_CH_SEL | BB_CH_SEL | Standard |
|---|---|---|
| 0 | 0 | 12.5kHz narrow band channel, compliant with Europe CE standards (ETSI EN 300 296) |
| 0 | 1 | 12.5kHz narrow band channel, compliant with US FCC standards (TIA 603) |
| 1 | 0 | N/A |
| 1 | 1 | 25 kHz wide band channel, compliant with Europe CE standards (ETSI EN 300 296) and US FCC standards (TIA 603). |

When BB_CH_SEL = 1, the user can configure ALT_LPF, to add an 1.5 dB attenuation at 3kHz. This is useful for two point modulation when the VCO boosts the frequency response at 3kHz, which may result in the total frequency response out of the spec.

## 1.2.44 CTC_PARAM

This command is used to configure CTCSS detection parameters. The CTCSS detection is done with two conditions, frequency and energy. The frequency has to be close to the designated CTCSS frequency, and the CTCSS energy has to above a certain threshold. To reduce the probability of false detection, both conditions have to be satisfied for a number of times before a CTCSS detection is declared. If one of the conditions is not satisfied for a number of times, CTCSS lost is declared. This command sets those thresholds.

Instead of using the standard phase reversal as CTCSS tail, SCT3258 can also send 55 Hz tone at the end of call as an indication for call end. “55 Hz” is used here as a general term for the special tone for call end indication. Its frequency is a programmable. CTC_PARAM command is also used to configure the mode and frequency of the 55 Hz tone. Also, as in the CTCSS tone case, a narrow band filter (NBF) is used to capture the 55 Hz tone. The coefficients for the NBF need to be set in this command if the 55 Hz tone detection is enabled.

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| Field Identifier | Energy_Hi | Energy_Lo | Freq_Miss | Eng_Miss | Freq_norm | Eng_Norm |
|---|---|---|---|---|---|---|
| 1 Byte | 2 Byte | 2 Bytes | 1 Bytes | 1 Bytes | 1 bytes | 1 Bytes |
| 0x03D | Energy high threshold for CTCSS found detection | Energy low Threshold for CTCSS lost detection | Number of times frequency conditions missed before declaring CTCSS lost | Number of times energy conditions missed before declaring CTCSS lost | Number of times frequency conditions satisfied before declaring CTCSS found under good condition | Number of times energy conditions satisfied before declaring CTCSS found under good condition |
| Default | 0x0384 | 0x0258 | 6 | 4 | 2 | 1 |

| Freq_Tough | Eng_Tough | Tail_180 | Tail_120 | CTC_Tail_DI | 55 Hz Tail Configuration |
|---|---|---|---|---|---|
| 1 Byte | 1 Bytes | 1 Bytes | 1 Bytes | 1 Bytes | 1 Byte |
| Number of times frequency conditions satisfied before declaring CTCSS found under tough condition | Number of times energy conditions satisfied before declaring CTCSS found under tough condition | 180 degree CTCSS tail detection threshold. Higher threshold make it more difficult to detect tail. | 120 degree CTCSS tail detection threshold. Higher threshold make it more difficult to detect tail. | CTCSS tail detection index, larger index value makes detection easier. A value of 0 is the hardest. | 55 Hz Tail Configuration, See Table below for details |
| 7 | 6 | 0x80 | 0xd0 | 0 | 0 |

| Tail Frequency | Tail NBF Coefficients | Signal to Total Energy Ratio | Reserved |
|---|---|---|---|
| 2 Bytes | 4 Bytes | 1 Byte | 2 bytes |
| Special Tail frequency, in steps of 0.1 Hz (MSB first). 55 Hz as default. | Narrow band filter coefficients for the special tail frequency. See Section 1.2.28 for details of the | Signal to Total Energy Ratio Threshold for CTCSS detection. The larger the threshold, the more difficult to detect | |

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| | coefficients | | |
|---|---|---|---|
| 2, 38 | 0, 118, 3, 179 | 0xb0 | 0 |

**Table 1-73 CTC_PARAM Field Format**

| Bit Position | Bit Name | Descriptions |
|---|---|---|
| 7 | ANA_END_SEND | Flag for sending end of transmission<br>0: No 17 0A sent after CTCSS/DCS tail is sent<br>1: 17 0A sent after CTCSS/DCS tail is sent |
| 6 | CTCSS_TOL | CTCSS detection tolerance<br>0: No tolerance if received CTCSS frequency is off spec<br>1: More tolerance if received CTCSS frequency is off spec. |
| 5 | TAIL_LENGTH | Flag for analog tail length for CTCSS and DCS<br>0: tail length 180 ms<br>1: tail length 250 ms |
| 4 | HPF_CORNER | Flag for analog High pass filter corner<br>0: HPF corner at 300 Hz<br>1: HPF corner at 450 Hz |
| 3 | CTC_TAIL_RX | 55 Hz tail detection enable in CTCSS call:<br>0: 55 Hz Tone detection enabled<br>1: 55 Hz Tone detection enabled |
| 2 | VOICE_TAIL_RX | 55 Hz tail detection enable in call with no sub-audio:<br>0: 55 Hz Tone detection enabled<br>1: 55 Hz Tone detection enabled |
| 1 | CTC_TAIL_TX | 55 Hz tail enable in CTCSS call:<br>0: No 55 Hz Tone are sent at the end of call<br>1: 55 Hz Tone are sent at the end of call |
| 0 | VOICE_TAIL_TX | 55 Hz tail enable in voice call with no sub-audio:<br>0: No 55 Hz Tone are sent at the end of call<br>1: 55 Hz Tone are sent at the end of call |

**Table 1-74 55 Hz Tail Configuration Details**

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## 1.2.45 VOCOER_IO_SET

This field selects the input and output settings for vocoders. It is most used for voice recording and play back. Currently, this command is only used in DPMR mode only.

| Field Identifier | Recording/Play Back Setting |
|---|---|

| 1 Byte | 1 Byte |
|---|---|
| 0x03e | Vocoder IO configuration, See table below |
| Default | 0 |

**Table 1-75 VOCODER_IO_SET Field Format**

| Bit Position | Bit Name | Descriptions |
|---|---|---|
| 7 | Reserved | Reserved |
| 6 | MOD_IN | Modulator input select:<br>0: from vocoder<br>1: from MCU (through CHAN_D commanad) |
| 5:4 | ENC_OUT | Voice encoder output select:<br>00: Voice encoder output to modem<br>01: Voice encoder output to MCU<br>10: Voice encoder output to modem and MCU |
| 3 | DEC_IN | Voice decoder input select:<br>0: from demodulator<br>1: from MCU (through CHAN_D commanad) |
| 2 | DEMOD_OUT | Demodulator output select<br>0: to voice decoder<br>1: to MCU (through CHAN_D commanad) and to voice decoder |
| 1 | Reserved | Reserved |
| 0 | Reserved | Reserved |

**Table 1-76 Vocoder IO Configuration Details**

With VOCODER_IO_SET command, the user can implement functions like local/remote recording and local/remote play back. The following table gives example of the settings.

| Function | Vocoder IO Settings |
|---|---|
| Local Recording | 0x20 |
| Remote Recording | 0x04 |
| Local Play back | 0x08 |
| Remote Play back | 0x40 |

**Table 1-77 Vocoder IO Settings for Different Functions**

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## 1.2.46 MODOUT_CONFIG

This command is used to configure the phase delay between the two channels in a two point modulation scheme. Adding a phase delay between the channels is useful to achieve a flat frequency response in two point modulation. The minimum step size is 10.4 us. And maximum delay is plus or minus 416 us. For example, if the phase delay time is set to 0x08,the signal at LOR is 83.2 us after the signal at LOL .If phase delay time is set to 0xf8, then the signal at LOR is 83.2 us a head of with the signal at LOL.

| Field Identifier | Phase delay time configure | Reserved |
|---|---|---|
| 1 Byte | 1 Byte | 4 Bytes |
| 0x041 | Minimum to -40 max to 40, with a step size of 10.4 us | Reserved |
| Default | 0 | 0,0,0,0 |

**Table 1-78 MODOUT_CONFIG Field Format**

## 1.2.47 IQCOMP_COEF

This command is used when paired with SCT700 only.

This command is used to configure the IQ compensation filter of for the RX path and the TX path. If SCT3258 works in IQ input or IQ output mode, the I path and the Q path of the RF front en should be perfectly matched, or the performance suffers. However, the I path and the Q path are never perfectly matched in the analog world. IQ compensation filter are used to re-aligned the IQ signals so that they are matched. The figure below shows the control data and control signals of IQ compensation filter.

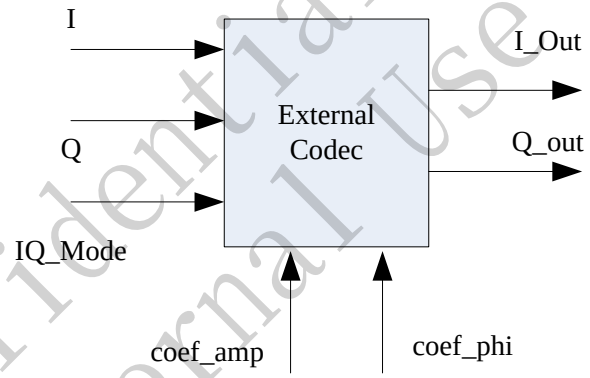



**Figure 1-1 IQ Compensation Filter**

The IQ compensation filter has two operation modes: the normal mode and the calibration mode. In normal mode (IQ_Mode = 0), the IQ compensation filter is enabled. In calibration mode (IQ_Mode = 1), the IQ compensation filter is disabled, allowing the algorithm to evaluate the amount of IQ imbalance, and calculate the filter coefficients coef_amp and coef_phase accordingly.

The default values of coef_amp and coef_phi are 0, which should be the value if I and Q path are perfect aligned and need to compensation.

In calibration mode, the value coef_phase has no effect, while coef_amp has an effect to the output. To complete disable IQ compensation filter, coef_amp should be set to zero, while IQ_Mode is set to 1.

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SCT3258 has internal algorithm to aid the calculation of the IQ compensation filter coefficients. The user can use the DC_OFFSET read command to see the effect of the calibration.

The structure of the TX IQ compensation filter and the RX IQ compensation filter are identical. Both compensation filters are configured with IQCOMP_COEF command.

The contents of the IQCOMP_COEF command are described in the table below.

| Field Identifier | Mode Configure | rx_coef_amp | rx_coef_phi | tx_coef_amp | tx_coef_phi |
|---|---|---|---|---|---|
| 1 Byte | 1 Byte | 2 Bytes | 2 Bytes | 2 Bytes | 2 bytes |
| 0x042 | See Table 1-80 for details | RX amplitude coefficient, 16 bit linear value, MSB first, in Q16.14 | RX phase coefficient, 16 bit linear value, MSB first, in Q16.14 | TX amplitude coefficient, 16 bit linear value, MSB first, in Q16.14 | TX phase coefficient, 16 bit linear value, MSB first, in Q16.14 |
| Default | 2 | 0 | 0 | 0 | 0 |

**Table 1-79 IQCOMP_COEF Field Format**

| Bit Position | Bit Name | Descriptions |
|---|---|---|
| 7 | Reserved | Reserved |
| 6 | RX_IQ_MODE | RX IQ compensation Filter work mode<br>0: Enable RX IQ compensation Filter, with coefficients described in the subsequent field parameters<br>1: RX IQ compensation Filter in calibration mode (only rx_coef_amp is effective) |
| 5-3 | Reserved | Reserved |
| 2 | TX_IQ_MODE | TX IQ compensation Filter work mode<br>0: Enable TX IQ compensation Filter, with coefficients described in the subsequent field parameters<br>1: TX IQ compensation Filter in calibration mode (only tx_coef_amp is effective) |
| 1-0 | Reserved | Reserved |

**Table 1-80Mode Configure Details**

If Bit 6 (RX_IQ_MODE) is set to 1, the RX phase coefficients in the command do not take effect. Likewise, the TX phase coefficients in the command does not take effect unless Bit 2 (TX_IQ_MODE) of the mode byte is 0.

IQCOMP_COEF command is also used to read the IQ compensation filter coefficients currently used by SCT3258, either calculated by SCT3258 or set previously by the MCU. The response

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packet to the read command is identical to configuration packet.

## 1.2.48 DIGC_DATA_FRAME

This field contains type 1, 2, 3 data, voice and appended data, transparent data and short appended data.

| Field Identifier | Frame Property | Data Length | Data Bytes |
|---|---|---|---|
| 1 Byte | 1 Byte | 1 Byte | Up to 180 bytes |
| 0x43 | See the following tables for detail | 0-180: Indicate the length of the Data<br>Others: reserved | DATA[0]<br>DATA[1]<br>…<br>DATA[DataLength-1]<br><br>Contain data, start from DATA[0], MSB first |
| Default | NA | NA | NA |

**Table 1-81 DIGC_DATA_FRAME Field Format**

The frame property field for DPMR and for DMR are different.

| Bit Position | Bit Name | Descriptions |
|---|---|---|
| 7 | DATA_VALID | Flag to indicate whether current frame has valid data<br>0: Current frame has no valid data. Far end can discard current frame<br>1: Current frame has valid data. |
| 6 | CONT_FLAG | Continuation flag<br>0: Data Continues after this frame<br>1: Data finish at this frame<br>SCT3258 does not process this field. It just passes this bit to the far end MCU. The MCU should not send more DIGC_DATA_FRAME commands after CONT_FLAG is set to one. |
| 5 | FEC_ERROR | FEC error flag<br>0: No FEC error in the received frame<br>1: FEC error in the received frame |
| 4 | Reserved | Reserved |
| 3-0 | Frame Type | Frame types:<br>0: Type 1 Data, Valid Data Length is 0 to 36<br>1: Type 2 Data, Valid Data Length is 0 to 20<br>2: Type 3 Data, Not implemented yet<br>3: Voice and Appended Data, Valid Data Length is 0 to 20 |

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| | | |
|---|---|---|
| | | 4: Transparent Data, Valid Data Length is 0 to 48<br>5. Short Appended Data, Valid Data Length is 9, 18, 27, 36 depending on the number of appended data frames.<br>Others, reserved |

**Table 1-82 Frame Property Details in DPMR Mode**

| Bit Position | Bit Name | Descriptions |
|---|---|---|
| 7 | Slot Num | 0: slot 1<br>1: slot 2 |
| 6 | FEC_ERROR | FEC error flag<br>0: No FEC error in the received frame<br>1: FEC error in the received frame |
| 5 | Reserved | Reserved |
| 4 | Data/Voice Burst | Data or voice burst selection<br>0: Data Burst<br>1: Voice Burst |
| 3-0 | Data Type | DMR data type (for data burst only)<br>0: PI header<br>1: Voice LC header<br>2: Terminator with LC<br>3: CSBK<br>4: MBC Header<br>5: MBC Continuation<br>6: Data Header<br>7: Rate 1/2 Data<br>8: Rate 3/4 Data<br>9: Idle<br>10: Rate 1 Data<br>11-15: Reserved |

**Table 1-83 Frame Property Details in DMR Mode**

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At the transmitter, the MCU passes the contents of data burst to SCT3258 through this field. SCT3258 adds FEC according to the data type and puts into a TDMA burst. At the receiver, SCT3258 applies FEC decoding to the received data burst and reports the contents to the MCU using the same field.

The content and the length of the frame is different according to the data type in DMR_DATA_FRAME. The details is describe in the below text.

When Data Type is PI header , Voice LC Header ,Terminator with LC, or the bit 4 is Voice Burst, the content of the frame is FULL LC PDU(without CRC). The length is 9 bytes.

| Information element | Length | Remark |
|---|---|---|
| Protect Flag (PF) | 1 | |
| Reserved | 1 | |
| Full Link Control Opcode (FLCO) | 6 | |
| Feature set ID (FID) | 8 | The FID shall be either SFID or MFID |
| Error Full LC Data | 56 | (see note 1) |
| NOTE 1: The data information element is defined by the feature protocol document TS 102 361-2 [5]. | | |

**Table 1-84 FULL LC PDU**

When Data Type is CSBK or MBC Header, the content of the frame is Control Signalling Block (CSBK) PDU (without CRC ). The length is 10 bytes.

| Information element | Length | Remark |
|---|---|---|
| Last Block | 1 | This bit shall be set to 1 |
| Protect Flag | 1 | |
| CSBK Opcode(CSBKO) | 6 | |
| FID | 8 | The FID shall be either SFID or MFID |
| CSBK Data | 64 | |
| NOTE : The data information element is defined by TS 102 361-2[5] | | |

**Table 1-85 CSBK PDU**

When Data Type is MBC Continuation, the content of the frame can be MBC Continuation Block or MBC Last Block.The length is 12 bytes.

| Information element | Length | Remark |
|---|---|---|
| Last Block | 1 | |
| MBC data | 95 | |

**Table 1-86 MBC Continuation Block**

| Information element | Length | Remark |
|---|---|---|
| Last Block | 1 | |
| MBC Data | 79 | |
| MBC CRC | 16 | Filled with 0, calculated by the chip automatically |

**Table 1-87 MBC Last Block**

When Data Type is DATA_HEADER, the content of the frame is DATA_HEAD_PDU. We introduce the C_HEAD_PDU only (without CRC, the other HEAD_PDU is in the TS 102 361-1 Protocol ). The length is 10 bytes.

| Information element | Length | Remark |
|---|---|---|
| Group or Individual | 1 | This bit is set to indicate that the destination LLID is for a group |

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| | | |
|---|---|---|
| Response Requested (A) | 1 | |
| Header Compression (HC) | 1 | |
| Pad Octet Count (POC) | 1 | |
| Format | 4 | Data packet identification |
| SAP Identifier | 4 | |
| Pad Octet Count (POC) | 4 | |
| Logical Link ID (LLID) | 24 | Destination |
| Logical Link ID (LLID) | 24 | Source |
| Full Message Flag (FMF) | 1 | |
| Blocks to Follow (BF) | 7 | |
| Re-Synchronize flag (S) | 1 | |
| Send sequence Number (N(S)) | 3 | |
| Fragment Sequence Number (FSN) | 4 | |

**Table 1-88 C_HEADER_PDU**

When Data Type is Rate 1/2 Data, the contents of the frame can be Confirm_R_1_2_DATA_CONTINU_PDU, UNConfirm_R_1_2_DATA_CONTINU_PDU, Confirm_R_1_2_DATA_LAST_PDU or UNConfirm_R_1_2_DATA_LAST_PDU.The length is 12 bytes. See details from table 1-65 to 1-68.

| Information element | Length | Remark |
|---|---|---|
| Data Block Serial Number (DBSN) | 7 | |
| C-DATA CRC | 9 | Filled with 0, calculated by the chip automatically |
| User Data | 80 | The user data field may contain pad octets |

**Table 1-89 Confirm_R_1_2_DATA_CONTINU_PDU**

| Information element | Length | Remark |
|---|---|---|
| User Data | 96 | The user data field may contain pad octets |

**Table 1-90 UNConfirm_R_1_2_DATA_CONTINU_PDU**

| Information element | Length | Remark |
|---|---|---|
| Data Block Serial Number (DBSN) | 7 | |
| C-DATA CRC | 9 | Filled with 0, calculated by the chip automatically |
| User Data | 48 | The user data field may contain up to 6 pad octets |

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| | | |
|---|---|---|
| Message CRC | 32 | Filled with 0, calculated by the chip automatically |

**Table 1-91 Confirm_R_1_2_DATA_LAST_PDU**

| Information element | Length | Remark |
|---|---|---|
| User Data | 64 | (see note) |
| Message CRC | 32 | Filled with 0, calculated by the chip automatically |
| NOTE:The user data field may contain up to 8 pad octets. | | |

**Table 1-92 UNConfirm_R_1_2_DATA_LAST_PDU**

When Data Type is Rate 3/4 Data, the contents of the frame can be Confirm_R_3_4_DATA_CONTINU_PDU, UNConfirm_R_3_4_DATA_CONTINU_PDU, Confirm_R_3_4_DATA_LAST_PDU or UNConfirm_R_3_4_DATA_LAST_PDU.The length is 12 bytes. See details from table 1-69 to 1-72.

| Information element | Length | Remark |
|---|---|---|
| Data Block Serial Number (DBSN) | 7 | |
| C-DATA CRC | 9 | Filled with 0, calculated by the chip automatically |
| User Data | 128 | The user data field may contain pad octets |

**Table 1-93 Confirm_R_3_4_DATA_CONTINU_PDU**

| Information element | Length | Remark |
|---|---|---|
| User Data | 144 | The user data field may contain pad octets |

**Table 1-94 UNConfirm_R_3_4_DATA_CONTINU_PDU**

| Information element | Length | Remark |
|---|---|---|
| Data Block Serial Number (DBSN) | 7 | |
| C-DATA CRC | 9 | Filled with 0, calculated by the chip automatically |
| User Data | 96 | The user data field may contain up to 12 pad octets |
| Message CRC | 32 | Filled with 0, calculated by the chip automatically |

**Table 1-95 Confirm_R_3_4_DATA_LAST_PDU**

| Information element | Length | Remark |
|---|---|---|
| User Data | 112 | The user data field may contain up to 14 pad octets |
| Message CRC | 32 | Filled with 0, calculated by the chip automatically |

**Table 1-96 UNConfirm_R_3_4_DATA_LAST_PDU**

When Data Type is IDLE, the content of the frame is IDLE_PDU. The length is 12 bytes.See details in table 1-73.

| Information element | Length | Remark |
|---|---|---|

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| | | |
|---|---|---|
| User Data | 96 | |

**Table 1-97 IDLE_PDU**

When Data Type is Rate 1 Data, the contents of the frame can be Confirm_R_1_DATA_CONTINU_PDU, UNConfirm_R_1_DATA_CONTINU_PDU, Confirm_R_1_DATA_LAST_PDU or UNConfirm_R_1_DATA_LAST_PDU.The length is 24 bytes. See details from table 1-74 to 1-77.

| Information element | Length | Remark |
|---|---|---|
| Data Block Serial Number (DBSN) | 7 | |
| C-DATA CRC | 9 | Filled with 0, calculated by the chip automatically |
| User Data | 176 | The user data field may contain pad octets |

**Table 1-98 Confirm_R_1_DATA__CONTINU_PDU**

| Information element | Length | Remark |
|---|---|---|
| User Data | 192 | The user data field may contain pad octets |

**Table 1-99 UNConfirm_R_1_DATA__CONTINU_PDU**

| Information element | Length | Remark |
|---|---|---|
| Data Block Serial Number (DBSN) | 7 | |
| C-DATA CRC | 9 | Filled with 0, calculated by the chip automatically |
| User Data | 144 | The user data field may contain up to 18 pad octets |
| Message CRC | 32 | Filled with 0, calculated by the chip automatically |

**Table 1-100 Confirm_R_1_DATA__LAST_PDU**

| Information element | Length | Remark |
|---|---|---|
| User Data | 160 | (see note) |
| Message CRC | 32 | Filled with 0, calculated by the chip automatically |
| NOTE:The user data field may contain up to 20 pad octets. | | |

**Table 1-101 UNConfirm_R_1_DATA__LAST_PDU**

## 1.2.49 DPMR_PREAMBLE_LENGTH

DPMR standard allows the length of the preamble to be more than the default 72 bits (15 milliseconds). This command sets the length of additional preambles and length of silence (before preamble) in increment of 20 milliseconds

| Field Identifier | Preamble Length | Silence Length |
|---|---|---|

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| 1 Byte | 1 Bytes | 1 Byte |
|---|---|---|
| 0x046 | Length additional Preambles in 20 millisecond unit | Silence length before preamble in 20 millisecond unit |
| Default | 0 | 0 |

**Table 1-102 DPMR_PREAMBLE_LENGTH Field Format**

## 1.2.50 DPMR_CALL_OPTION

This field is used to set the DPMR specific call options

| Field Identifier | DPMR Call Options |
|---|---|
| 1 Byte | 5 Bytes |
| 0x47 | Data[0]:<br>Bit 7: Slot timing flag<br>0: The message sent to far end can be sent freely<br>1: The message sent to far end has to be sent at slot boundary. The first bit of the preamble will be sent 30 milliseconds from the last bit of the last received message<br>Bit 6: SQ detection enable<br>0: SQ detection is not done by SCT3258<br>1: SQ detection is done by SCT3258<br>Bit 5: IQ receive enable<br>0: SCT3258 in IF IQ mode<br>1: SCT3258 in base band mode<br>Bit 4: RF control enable<br>0: SCT3258 does not control RF chip (SCT3700)<br>1: SCT3258 controls RF chip.<br>Bit 3-0: Number of the power saving headers to precede the next message. Zero to disable power saving headers<br><br>Data[1]: voice delay constant and transmit control<br>Bit 7: IQ transmit enable<br>0: SCT3258 transmitting in IQ mode<br>1: SCT3258 transmitting in base band mode<br>Bit 6: Enable Wrong Vocoder report<br>Bit 5:0: voice delay constant<br>0-40: voice delay constant in steps of 40 milliseconds<br>Data[2]: Report options<br>Bit 7: 1B 09, 1B 0A report<br>0: No report of 1B 09 and 1B 0A<br>1: Report 1B 09 and 1B 0A |

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| | |
|---|---|
| | Others: reserved |
| Default | 0 |

**Table 1-103 DPMR_CALL_OPTION Field Format**

Note: In order to send extended power saving headers, DPMR_CALL_OPTION must be called each time when a communication is to start

## 1.2.51 DPMR_END_UE

This field contains the 17 bits of unencoded END data.

| Field Identifier | 17 bit of unencoded END |
|---|---|
| 1 Byte | 3 Bytes |
| 0x4B | DATA[0]={ET[1:0], ARQ[1:0], TX_WAIT[3:0]}<br>DATA[1]={STATUS[4:0], 3b000}<br>DATA[2]=0 |
| Default | 0,0,0 |

**Table 1-104 DPMR_END_UE Field Format**

The details of ET, ARQ, TX_WAIT and status are given in the following tables.

| | |
|---|---|
| 00 | Normal end frame |
| 01 | End frame with status message |
| 10 | Reserved |
| 11 | Reserved |

**Table 1-105 Details ET Field**

| | |
|---|---|
| 00 | No ACK request to called station |
| 01 | ACK request to called station |
| 10 | Reserved |
| 11 | Reserved |

**Table 1-106 ARQ Field Details**

| | |
|---|---|
| 0000 | No specified time |
| 0001 | 40 ms (half a frame) |
| 0010 | 80 ms (one frame) |
| 0011 | 160 ms (two frames) |
| 0100 | 320 ms (one superframe) |
| Other | Other Reserved |

**Table 1-107 Details of Tx WAIT Field**

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## 1.2.52 DIGC_FS1

This field contains the 48 bits of FS1 data for DPMR. It is also used to generate a fixed 48 bit test pattern for both DPMR and DMR mode.

| Field Identifier | 48 bit of encoded FS1 |
|---|---|
| 1 Byte | 6 Bytes |
| 0x4C | DATA[0]=FS1[47:40]<br>DATA[1]=FS1[39:32]<br>…<br>DATA[5]=FS1[7:0] |
| Default | 0x57,0xff,0x5f,0x75,0xd5,0x77 |

**Table 1-108 DIGC_FS1 Field Format**

## 1.2.53 DIGC_CALLED_ID_BIN

This field contains the called ID in 24 bit binary form.

| Field Identifier | 24 bit binary called ID |
|---|---|
| 1 Byte | 3 Bytes |
| 0x50 | DATA[0]=CALL_ID_BIN[23:16]<br>DATA[1]= CALL_ID_BIN [15:8]<br>DATA[2]= CALL_ID_BIN [7:0] |
| Default | 0,0,0 |

**Table 1-109 DIGC_CALLED_ID_BIN Field Format**

## 1.2.54 DIGC_OWN_ID_BIN

This field contains the own ID in 24 bit binary form.

| Field Identifier | 24 bit binary own ID |
|---|---|
| 1 Byte | 3 Bytes |
| 0x51 | DATA[0]=OWN_ID_BIN[23:16]<br>DATA[1]= OWN_ID_BIN [15:8]<br>DATA[2]= OWN_ID_BIN [7:0] |
| Default | 0,0,0 |

**Table 1-110 DIGC_OWN_ID_BIN Field Format**

## 1.2.55 DPMR_CALLED_ID_BCD

This field contains the called ID in 7 BCD codes.

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| Field Identifier | Called ID in 7 BCD codes |
|---|---|
| 1 Byte | 4 Bytes |
| 0x52 | DATA[0]= {K1, K2}<br>DATA[1]= {K3, K4}<br>DATA[2]= {K5, K6}<br>DATA[3]= {K7, 0x4b0000} |
| Default | 0xaa,0xaa,0xaa,0xa0 |

**Table 1-111 DPMR_CALLED_ID_BCD Field Format**

If the user dials less than 7 digits, the user should use 0x0F to fill the place for un-dialed digits. Upon receiving the message with less than 7 digits, SCT3258 will fill the rest of the digits to complete 7 digits if layer 3 process mode is enabled.

## 1.2.56 DPMR_OWN_ID_BCD

This field contains the own ID in 7 BCD codes.

| Field Identifier | Own ID in 7 BCD codes |
|---|---|
| 1 Byte | 4 Bytes |
| 0x53 | DATA[0]= {K1, K2}<br>DATA[1]= {K3, K4}<br>DATA[2]= {K5, K6}<br>DATA[3]= {K7, 0x4b0000} |
| Default | 0x00,0x00,0x00,0x00 |

**Table 1-112 DPMR_OWN_ID_BCD Field Format**

## 1.2.57 DPMR_M_V_F_E

This field contains communication mode (M), version (V), and communication format (F) fields.

| Field Identifier | M and F fields |
|---|---|
| 1 Byte | 1 Byte |
| 0x54 | DATA[0]={M[2:0], V[1:0], F[1:0], EP} |
| Default | 0 |

**Table 1-113 DPMR_M_V_F Field Format**

The meaning of M, V, F and EP are described in section CALL_START.

## 1.2.58 DPMR_PROTOCOL_OPTION

This field is used to set the DPMR specific protocol options

| Field Identifier | DPMR Protocol Options |
|---|---|

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| 1 Byte | 1 Byte |
|---|---|
| 0x55 | Bit 7: Protocol Support<br>0: Support TS 102 490<br>1: Support TS 102 490 and TS 102 658<br>Bit 6: NDR Support<br>0: No Support for NDR<br>1: Support for NDR<br>Bit 5 : TS 102 658 Mode 3 Support<br>0: No Support for Mode 3<br>1: Support for Mode 3<br>Bit 4: Maintenance Message Report<br>0: Repeated Maintenance Message only reported only once.<br>1: Every reported messages are reported<br>Bit 3: Reserve (Should set to zero)<br>Bit 2: Reserve<br>Bit 1 : All call mapping<br>0: Old way of all call mapping: TS 102 658 (V2.1.1 or earlier)<br>1: New way of all call mapping<br>Bit 0 : 12.5 kHz DPMR support.<br>0: 6.25 kHz DPMR<br>1: 12.5 kHz DPMR |
| Default | 0 |

**Table 1-114 DPMR_PROTOCOL_OPTION Field Format**

## 1.2.59 DPMR_SLD

This field contains 18 bits of slow data.

| Field Identifier | M and F fields |
|---|---|
| 1 Byte | 4 Bytes |
| 0x56 | DATA[0]=SLD[17:10]<br>DATA[1]=SLD[9:2]<br>DATA[2]={SLD[1:0], 0x000000} |
| Default | 0x80,0x40,0x00 |

**Table 1-115 DPMR_SLD Field Format**

The meaning of SLD field is defined in the TS 102 658 or TS 102 490:

| Bit position | Meaning |
|---|---|
| 17 | Continuation Flag 1<br>0: User data continuous after the following byte<br>1: User data is terminated by the following byte |

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| | |
|---|---|
| 16-9 | User data byte 1 |
| 8 | Continuation Flag 2<br>0: User data continuous after the following byte<br>1: User data is terminated by the following byte |
| 7-0 | User data byte 2 |

**Table 1-116 Details SLD Field**

Note that SLD filed can also be modified by WORK_MODE command. Upon receiving command WORK_MODE with the mode set to TX or duplex, the SLD field is reset to default 0x80, 0x40, 0x00. In a typical voice call with slow data, the MCU always send WORK_MODE command before sending the first DPMR_SLD command.

## 1.2.60 DPMR_HT_CI_PM

This field contains header type (HT) and call information (CI) fields.

| Field Identifier | HT and CI fields |
|---|---|
| 1 Byte | 2 Bytes |
| 0x57 | DATA[0]={HT[3:0], CI[10:7]}<br>DATA[1]={CI[6:0], PM} |
| Default | 0,0 |

**Table 1-117 DPMR_HT_CI_PM Field Format**

Details of HT, CI and PM can be found in the following tables.

HT indicates header type.

| | |
|---|---|
| 0000 | Communication start header (a superframe follows) |
| 0001 | Connection request header (an END frame follows) |
| 0010 | Unconnect request header (an END frame follows) |
| 0011 | ACK (this a single frame, ACK or NACK is differentiated by the CI bits setting) |
| 0100 | System request header (an END frame follows) or Maintenance Message header |
| 0101 | ACK header reply to a system request (a superframe follows) |
| 0110 | System delivery header (a superframe follows) |
| 0111 | Status response header (an END frame follows) |
| 1000 | Status request header |
| Other | Reserved |

**Table 1-118 Header Type Details**

PM is the channel preservation flag used for base station to send out channel preservation messages. When sending a voice call, this bit is used to enable the encryption.

| | |
|---|---|
| PM | Channel Preservation Flag |

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| | |
|---|---|
| 0 | Channel is not preserved |
| 1 | Channel is preserved |

**Table 1-119 Channel Preservation Flag Field Details**

CI contains 3 bits of CI type and 8 bits of CI information.

| CI Type | CI Information |
|---|---|
| 3 bits | 8 bits |

**Table 1-120 CI Format**

The meaning of CI depends on the header types.

| Use | Purpose |
|---|---|
| Power save | Indicate normal or extended header type |
| T1 or T2 Data | Indicate the type of data (supplementary service) |
| T3 Data (Packet) | Indicate data frame size and number of frames |
| Acknowledgements | Indicate ACK or NACK and reason |
| System request<br>System response<br>Delivery Header | CI Type defines the purpose<br>CI Information is not used and set to 0000 0000 |

**Table 1-121 CI Usuage**

The details of CI for different header types are described in the following sub-sections,

### 1.2.60.1 Call Information for Power Save

CI Type (3 bits):

| CI Type | Definition |
|---|---|
| 111 | Extended wake-up Header |
| Other value | Normal Header type |

**Table 1-122 CI Type for Power Save**

If the extended wake-up Header is used then the last 4 information bits will show how many Headers frames follow the current one (i.e. counting down to zero).

| | |
|---|---|
| 0000 0000 | Normal Header frame |
| 0000 0001<br>----------<br>0000 1111 | Ext Header frame 1<br>----------<br>Ext Header frame 15 |
| Other | Reserved |

**Table 1-123 CI Information for Power Save**

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### 1.2.60.2 Call Information for Types 1 and 2 data

Data communications (types 1 and 2):

| CI Type | CI Information | |
|---|---|---|
| 001 | **Format** | **Reserved** |
| | 4 bits | 4 bits |

**Table 1-124 CI Type for Type 1 and Type 2 Data**

Reserved bits are set to 0000.

Format:

| | |
|---|---|
| 0000 | Status message |
| 0001 | Precoded message |
| 0010 | Free text message (radio generated data) |
| 0011 | Short file transfer |
| 0100 | User defined data 1 |
| 0101 | User defined data 2 |
| 0110 | User defined data 3 |
| 0111 | User defined data 4 |
| Other | Reserved |

**Table 1-125 CI Information for Type 1 and Type 2 Data**

### 1.2.60.3 Call Information for Type 3 (packet) data

Information bits for Packet data format (Type 3):

| CI Type | CI Information | |
|---|---|---|
| 011 | **pdS** Frame Size | **pdM** Data Size |
| | 4 bits | 4 bits |

**Table 1-126 CI for Type 3 Data**

Details of pdS (packet data size) and pdM (number of transmitted frames) are details by the tables below.

| pdS | Frame time (ms) | Data size bits |
|---|---|---|
| 0 | 80 | 288 |
| 1 | 160 | 672 |
| 2 | 240 | 1056 |
| 3 | 320 | 1440 |
| Other | Reserved | Reserved |

**Table 1-127 Packet Data Frame Sizes (pdS)**

| pdM | Number of Data frames |
|---|---|

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| | |
|---|---|
| 0 | 1 frame |
| 1 | 2 frames |
| 2 | 3 frames |
| 3 | 4 frames |
| 4 | 5 frames |
| 5 | 6 frames |
| 6 | 7 frames |
| 7 | 8 frames |
| Other | Reserved |

**Table 1-128 Number of Transmitted Frames (pdM)**

### 1.2.60.4 Call Information for System Transactions

System request/answer/delivery header:

| CI | Type Definition |
|---|---|
| 000 | Reserved |
| 001 | Dynamic group request/answer/delivery |
| 010 | Reserved |
| 011 | Reserved |
| 100 | ESN request/reply |
| 101 | MFID request/reply |
| 110 | Contact station address(via Interconnect, IP) |
| 111 | Reserved |

**Table 1-129 CI Type for System Transactions**

| CI Info | Definition |
|---|---|
| 0000 0000 | All bits set to zero (the data size is indicated in the CCH SLD field) |

**Table 1-130 CI Information for System Transactions**

### 1.2.60.5 Call Information for Acknowledgments

Acknowledgment:

| CI Type | Definition |
|---|---|
| 000 | Reserved |
| 001 | ACK (Rx OK) |
| 010 | NACK (data error, resend request) |
| 011 | NACK (request denied) |
| Other | Reserved |

**Table 1-131 CI Type for Acknowledgements**

| CI Info | Definition |
|---|---|

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| | |
|---|---|
| 0 | |
| 1 to 255 | ACK/NACK status (rejection reason defined by user) |

**Table 1-132 CI Information for Acknowledgements**

### 1.2.60.6 Call Information for appended data

CI type:

| CI | Type Definition | Meaning |
|---|---|---|
| 000 | Short Data | Service Requested is Short Data |
| 011 | Call Diversion | Call Diversion Service |

**Table 1-133 CI type for Appended data**

Short data CI Info:

| CI Info | length | Meaning |
|---|---|---|
| UAD(0 - 3) | [7:6],2 bits | Appended Short Data Number of appended UDTs required to transport short data |
| SYMB | [5:0],6 bits | Number of symbols in the short data , |

**Table 1-134 CI Information for Short data**

Call Diversion CI Info:

| CI Info | Length | Meaning |
|---|---|---|
| UAD(00) | [7:6],2 bits | Number of Appended Data = 1 |
| SYMB | [5:0],6 bits | N/A for call diversion |

**Table 1-135 CI Information for Call Diversion**

### 1.2.60.7 Call Information for Maintenance Message

| CI type | Type Definition | CI Info |
|---|---|---|
| 000 | IDLE Message | N/A |
| 001 | Guard Message | See table below |
| | Preservation message | See note |

**Table 1-136 Call Information for Maintenance Message**

| CI Info | length | Meaning | | |
|---|---|---|---|---|
| Reserved | [7:4],4bits | Reserved | | |
| Guard Kind | [3:0],4 bits | 0000 | Reserved | |
| | | 0001 | DIS_PTT | Disable Target MS or Talkgroup PTT |
| | | 0010 | EN_PTT | Enable Target MS or Talkgroup PTT |

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| | | | | |
|---|---|---|---|---|
| | | 0011 | ILLEGALLY PARKED | Clear down from the payload channel MS whose address does not match,Source or Target Address |
| | | 0100 to 1111 | Reserved | |

**Table 1-137 Call Information for Guard Message**

Note: Preservation message is not identified by CI. It is identified with PM as 1. In preservation message, CI keeps value of previous traffic channel down link messages, along with CALLED ID, OWN ID, M, V, F, and EP fields.

## 1.2.61 DPMR_CI

This field contains 11 bits call information. It is used for reporting CI information by SCT3258 to the MCU upon receiving an extended header.

| Field Identifier | CI field |
|---|---|
| 1 Byte | 2 Byte |
| 0x5A | DATA[0]={CI[10:3]}<br>DATA[1]={CI[2:0], 0x5b00000} |
| Default | 0,0 |

**Table 1-138 DPMR_CI Field Format**

## 1.2.62 DPMR_CC

This field contains the 24 bits of CC data for the transmitter and the receiver.

| Field Identifier | 24 bit of encoded CC |
|---|---|
| 1 Byte | 3 Bytes |
| 0x5C | DATA[0]=CC[23:16]<br>DATA[1]=CC[15:8]<br>DATA[2]=CC[7:0] |
| Default | 0x57, 0x75,0x77 |

**Table 1-139 DPMR_CC Field Format**

## 1.2.63 DPMR_SEND_SF

Send super frame command. Upon receiving this command, SCT3258 assemble a super frame and send to the far end.

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| Field Identifier |
|---|
| 1 Byte |
| 0x5F |

**Table 1-140 DPMR_SEND_SF Field Format**

## 1.2.64 DPMR_SEND_HEADER

Send header frame command. Upon receiving this command, SCT3258 assemble a header frame and send to the far end.

| Field Identifier |
|---|
| 1 Byte |
| 0x60 |

**Table 1-141 DPMR_SEND_HEADER Field Format**

## 1.2.65 DPMR_SEND_END

Send end frame command. Upon receiving this command, SCT3258 assemble an end frame and send to the far end.

| Field Identifier |
|---|
| 1 Byte |
| 0x61 |

**Table 1-142 DPMR_SEND_END Field Format**

## 1.2.66 DPMR_SEND_AD

Send DPMR_SEND_AD command. Upon receiving this command,SCT3258 assemble a series of appended data frmae，include HEADER FRAME ,APPENDED DATA FRAME (according as appended data number),END FRAME.

| Field Identifier | Call Details |
|---|---|
| 1 Byte | 1 Byte |
| 0x62 | DATA[0]={M[2:0], V[1:0], F[1:0], EP}<br>Call Mode, Version, Call Format,<br>Emergency Priority |
| Default | 0 |

**Table 1-143 DPMR_SEND_AD Field Format**

DPMR_SEND_AD can only be used in TS 102 658 PROTOCOL.

The meaning of M, V, F and EP fields are described in the section CALL_START.

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Note: Please set the content of appended data frame with DIGC_DATA_FRAME command before sending appended data frame with this command.

## 1.2.67 DIGI_MIC_GAIN

This message set the modulation gain for the microphone used in the digital mode only.

| Field Identifier | Event |
|---|---|
| 1 Byte | 2 Byte |
| 0x068 | Linear gain applied to the microphone in Q16.11 (16 bits total, 11 bits fraction, MSB first), with 2048 as 0 dB. This gain is used in digital mode only |

**Table 1-144 DIGI_MIC_GAIN Field Format**

## 1.2.68 DIGI_SPEAKER_GAIN

This message set the modulation gain for the speaker used in the digital mode only.

| Field Identifier | Event |
|---|---|
| 1 Byte | 2 Byte |
| 0x069 | Linear gain applied to the speaker in Q16.11 (16 bits total, 11 bits fraction, MSB first), with 2048 as 0 dB. This gain is used in digital mode only |

**Table 1-145 DIGI_SPEAKER_GAIN Field Format**

## 1.2.69 I2C_OPERATION

This command is used to read and write raw I2C command to the I2C devices that are connected to SCT3258. Examples of the connected devices include codec, security keys and RF transceiver.

This command differs from the codec bypass packet described in 1.1.3 in that codec bypass packet can only be used for certain codecs, while I2C_OPERATION can be used for any connected I2C devices.

| Field Identifier | I2C device Address | Device register address | I2C Data Length | I2C data |
|---|---|---|---|---|
| 1 Byte | 1 Byte | 1 Byte | 1 Byte | I2C Data Length |
| 0x06A | I2C device address | Device register address | Number of Bytes to Write. | Data[0]<br>Data[1]<br>… |

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| | | | | |
|---|---|---|---|---|
| | | | | Data[I2C Data Length] |
| Default | 0 | 0 | 0 | N/A |

**Table 1-146 I2C_OPERATION Write Field Format**

To read from I2C device, the user should set TYPE field to 0x20, and the I2C_OPERATION syntax is as follows:

| Field Identifier | I2C device Address | Device register address | I2C Data Length |
|---|---|---|---|
| 1 Byte | 1 Byte | 1 Byte | 1 Byte |
| 0x06A | I2C device address | Device register address | Number of Bytes to Read |
| Default | 0 | 0 | 0 |

**Table 1-147 I2C_OPERATION Read Field Format**

The response packet returns the I2C data

| Field Identifier | Data read from I2C |
|---|---|
| 1 Byte | I2C Data Length |
| 0x6A | Data[0]<br>Data[1]<br>…<br>Data[I2C Data Length-1] |
| Default | 0 |

**Table 1-148 I2C_OPERATION Read Response Field Format**

## 1.2.70 MISC_GAIN

This command set the various gain levels, including the input and output gain for IQ signal, and CTCSS/DCS receive gains, and frequency of the signal used for calibration, and DC filter settings.

| Field Identifier | IQ TX Gain | IQ RX Gain | CTCSS/DCS Receive Gain | Calibration Signal Frequency | DC Filter Configuration | Reserved |
|---|---|---|---|---|---|---|
| 1 Byte | 2 Bytes | 2 Bytes | 2 Bytes | 2 Bytes | 1 Byte | 3 Bytes |
| 0x06B | 16 bit linear gain, MSB first, in Q16.11 | 16 bit linear gain, MSB first, in Q16.11 | 16 bit linear gain, MSB first, in Q16.11 | 16 bit linear gain, MSB first, in Q16.11 | See table below for detail. | |
| Default | 2048 | 2048 | 2048 | 2048 | 0 | |

**Table 1-149 MISC_GAINS Field Format**

| Bit Position | Bit Name | Descriptions |
|---|---|---|

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| | | |
|---|---|---|
| 7-2 | Reserved | |
| 1 | DC Filter enable | 0: DC filter is disabled<br>1: DC filter is enabled. |
| 0 | Manual DC Mode | 0: Auto DC Mode, DC filter is enabled at low input level and disabled at high input level for digital calls, and always enabled for analog call.<br>1: Manual DC Mode. See bit 1 for details. |

**Table 1-150 DC Configuration Detail**

The IQ RX gain is applied to the input IQ signals (when IQ receive is enabled) after it is received by SCT3258. The IQ TX gain is applied to the output IQ signals (when IQ transmit is enabled) before it is sent to the codec.

The CTCSS/DCS receive gain is used in analog mode, and is applied to the CTCSS/DCS signals after the 300 HF LPF. It is used to adjust the level of DCS subaudio signals without affecting the level of audio signals.

The calibration signals are I / Q sinusoidal signals generated by SCT3258 TX path, which can be looped back by SCT3700 to SCT3258 RX for I Q calibration. The frequency of the signals can be programmable with this command. A default value of 2048 corresponds to a frequency of 350 Hz.

## 1.2.71 SQ_LEVEL

This command is used for calibration of SQ detection. It is also used to read the RSSI and OOBE levels.

| Field Identifier | GAIN_ADJUST | EXT LNA_GAINL | Forced RF Gain | OOBE_INTV | OOBE_HI_D | OOBE_LO_D | OOBE_HI_A | OOBE_LO_A |
|---|---|---|---|---|---|---|---|---|
| 1 Byte | 1 Byte | 1Byte | 1 Byte | 1 Byte | 2 Bytes | 2 Bytes | 2 Bytes | 2 Bytes |

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| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 0x06C | Gain Adjustment, from -128 dB to +127 dB | External LNA value. Refer to table below. | Forced RF Gain Setting. See table below for Details | OOBE detection interval in units of 1.333 ms. | Digital OOBE high threshold, 16 bit linear value | Digital OOBE low threshold, 16 bit linear value | Analog OOBE high threshold, 16 bit linear value | Analog OOBE low threshold, 16 bit linear value |
| Default | 0 | 0 | 0 | 90 | 0x4000 | 0x0400 | 0x4000 | 0x0400 |

**Table 1-151 SQ_LEVEL Field Format**

| Bit Position | Bit Name | Descriptions |
|---|---|---|
| 7 | Forced RF Gain Enabled | 0: Forced RF Gain Disable. RF gain controlled by AGC algorithm<br>1: Forced RF Gain Enabled. RF gain are setting by bit 6-0. |
| 6-0 | Gain Details | RF Gain Details: It is different for RDA1847 or SCT3700<br>For RDA1847,<br>0-15, Gain Index<br>Others, reserved<br><br>For SCT3700<br>Bit 6-4: LNA gain setting<br>001: highest gain<br>010: highest gain -6 dB<br>011: highest gain -12 dB<br>100: highest gain -24 dB<br>101: highest gain -36 dB<br>110: highest gain -48 dB<br>Others: reserved<br><br>Bit 3-0: PGA settings:<br>0000: lowest gain<br>0001-1100, 2-24 dB, 2 dB a step<br>Others, reserved |

**Table 1-152 Force RF Gain Details**

| Bit Position | Bit Name | Descriptions |
|---|---|---|
| 7:0 | External LNA gain Setting | Only For SCT3700<br>Bit 7:6: Reduction of AGC target level.<br>00: Maximum AGC target level used<br>01: Maximum AGC target level – 6 dB |

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| | | |
|---|---|---|
| | | 10: Maximum AGC target level – 12 dB<br>11: Maximum AGC target level – 18 dB<br><br>Bit 5:0: External LNA gain in db if it is used.<br>If the external LNA is not used, the value is always zero. If the external LNA is used, it is the difference of external gain value when external LNA is enabled and when it is disabled. Please refer to Chapter 2.6 for its usage. |

**Table 1-153 External LNA setting Details**

GAIN ADJUST is used to account for all unspecified gains in the analog path. It may be different with different analog implementations. It may be even different from radio to radio. The RSSI is correct only if the value is set correctly. If external LNA is used, it need to be disabled when calibrating GAIN_ADJUST.

If external LNA is not used, "EXT LNA GAIN" should be assigned to its default value of zero.

If external LNA is used, the user should set EXT LNA GAIN appropriately to reflect the characteristic of the external LNA. It involves two part. Bit 5-0 is used to set the value of LNA gain in dB. It reflect the receive gain increase when the external LNA is enabled from when it is disabled. Also, if the external LNA is used, the user should lower the maximum SCT3700 AGC gain to avoid saturating the RX signals. The amount of maximum AGC target level decrease are set by bit 7-6 of "EXT LNA GAIN" fields. The following is a guide line:

00: if the external LNA gain is less than 6 dB.
01: if the external LNA gain range in 6-12db
10: if the external LNA gain range in 12-18db
11: if the external LNA gain range in 18-24db

OOBE indicates the energy of FM demodulated signals at frequency band above 3 kHz. OOBE value is small when RF carrier signal exists, and large when no carrier signals present. It is a good indication for SQ (or carrier) detection. Typically, when carrier signals present, OOBE level is below 8.

OOBE level has a high threshold and a low threshold for SQ detection, with the high threshold larger than the low threshold. To detect the presence of carrier, the OOBE level is compared with the low threshold. Carrier detection is reported if the OOBE level is above the low threshold. Once carrier is found, the OOBE level is compared with the high threshold. If the OOBE level is lower than the high threshold, carrier lost is reported.

The OOBE levels for analog and digital calls are different. As a result, SCT3258 has two sets of high and low OOBE threshold for digital and analog separately.

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SQ_LEVEL command is also used to read narrow, wide band RSSI and OOBE levels. The response packet has the following format.

| Field Identifier | NB_RSSI | WB_RSSI | OOBE Level | Inband Level |
|---|---|---|---|---|
| 1 Byte | 1 Byte | 1 Byte | 2 Byte | 2 Byte |
| 0x06C | Narrow band RSSI + 137 | Wide band RSSI + 137 | OOBE level | Inband signal energy level |
| Default | | | | |

There are two RSSI estimators in the SCT3258. One of them is narrow band RSSI, used to estimate the energy of signals in the channel of interests. The other is wide band RSSI, used to estimate the energy of the signal in the channel of interests and surrounding channels. All RSSI values in this command are of dB scale.

All RSSI values are expressed as unsigned 8 bit integer. The real RSSI is this value minus 137. For example, for WB_RSSI value of 17, the real wide band RSSI is 17 – 137 = -120 dBm.

## 1.2.72 SPI_OPERATION

This command is used to write raw SPI command to the SPI devices that are connected to SCT3258. Examples of the connected devices include SCT3700.

| Field Identifier | SPI Register Address | SPI register Contents |
|---|---|---|
| 1 Byte | 1 Byte | 1 Byte |
| 0x06D | SPI Register Address | SPI register Contents |
| Default | 0 | 0 |

**Table 1-154 SPI_OPERATION Write Field Format**

The SPI register table for SCT3700 can be found in SCT3700 data sheet.

## 1.2.73 DMR_SLC

This command is used to report the short LC data from the base station.

| Field Identifier | Offset |
|---|---|
| 1 Byte | 4 Bytes |
| 0x06E | Contains 28 bit of SLC data |
| Default | 0 |

**Table 1-155 DMR_SLC Field Format**

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## 1.2.74 DMR_CALL_SLOT

This field is used to set the slot number for the transmitting or receiving calls.

| DMR_CALL_SLOT | Call | Timing Update | Direct TDMA | Rev | Slot mode |
|---|---|---|---|---|---|
| 1 BYTE | BIT[7] | BIT[6] | BIT[5] | BIT[4] | BIT[3:0] |
| 0x6F | 0:Receive Call<br>1:Transmit Call | 0: keep current TX timng<br>1: Update TX timing according RX timing | 0: Direct TDMA Mode disabled<br>1: Direct TDMA mode enabled | Reserved | TDMA mode:<br>0:random receive<br>1:receive TDMA slot1<br>2: receive TDMA slot2<br>BS mode<br>0: No voice<br>1:play slot1 voice<br>2: play slot2 voice |

**Table 1-156 DMR_CALL_SLOT Field Format**

Note that the slot number is in terms of mobile station slot number, or downlink slot number.

## 1.2.75 EQULIZER_FILTER

This command configures the coefficients for the audio equalizer filters. A total of up to three equalizer filters can be enabled. Each filter is a second order IIR filter implemented in straight form. The transfer function of the filters are shown in the following equation:

H(z) = (b0 + b1 * z^(-1) + b2*z^(-2)) / (1 + a1 * z^(-1) + a2 * z^(-2))
The sampling rate of the filter is 24 kHz.

This command gives the filter coefficients. All coefficients are in Q16.14 format (16 bit total, with 14 bit fraction), with MSB first

By default, no equalizer filter is enabled.

| Field Identifier | Filter Selection Byte | a1 | a2 | b0 | b1 | b2 |
|---|---|---|---|---|---|---|
| 1 Byte | 1 Byte | 2 Bytes | 2 Bytes | 2 Bytes | 2 Bytes | 2 Bytes |

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| | | | | | | |
|---|---|---|---|---|---|---|
| 0x70 | Bit 7-6: reserved<br>Bit 5: TX or RX selection<br>0: Filter at RX side<br>1: Filter at TX side<br>Bit 4: Enable flag<br>1: Selected filter enabled<br>0: Selected filter not enabled<br>Bit 3-0: Filter selection<br>1: Filter 1 selected<br>2: Filter 2 selected<br>3: Filter 3 selected<br>Others: reserved | a1 in Q16.14 MSB first | a2 in Q16.14 MSB First | b0 in Q16.14 MSB First | b1 in Q16.14 MSB first | b2 in Q16.14 MSB First |
| Default | 0 | - | - | - | - | - |

**Table 1-157 EQUALIZER FILTER Field Format**

## 1.2.76 DMR_FLC

This message set the information for the FLC in a DMR control packet.

| Field Identifier | FLC |
|---|---|
| 1 Byte | 3 Byte |
| 0x076 | Data[0] = {PF, REV, FLCO[5:0]}<br>Data[1] = {FID[7:0]}<br>Data[2] = {SeversOption[7:0]} |

**Table 1-158 DMR_FLC Field Format**

Details of the elements are defined in the DMR standard (TS 102 361)

**PF field is defined by:**

| PF | Protect Flag |
|---|---|
| 0 | Reserved |

**Table 1-159 PF Settings**

**FLCO field is defined by:**

| FLCO | Full Link Control Opcode |
|---|---|
| 000000 | Group Voice Call（Grp_V_Ch_Usr） |
| 000011 | Unit to Unit Voice Call（UU_V_Ch_Usr） |

**Table 1-160 FLCO Settings**

**FID field is defined by:**

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| FID | Feature set ID |
|---|---|
| 0000 0000 | Standardized feature set ID for the services |
| 0000 0001 | Reserved for future standardization |
| 0000 0010 | Reserved for future standardization |
| 0000 0011 | Reserved for future standardization |
| 0000 0100 | Manufacture's specific feature set ID(MFID) |
| Etc. | Etc |
| 0111 1111 | Manufacture's specific feature set ID(MFID) |
| 1xxx xxxx | Reserved for future MFID's allocation(MFID) |

**Table 1-161 FID Settings**

**Service Option field is defined by:**

| Service option element | Bit position | Value remark |
|---|---|---|
| Emergency | Bit[7] | 0:Non-Emergency service<br>1:Emergency service |
| Privacy | Bit[6] | 0:privacy is not defined |
| Reserved | Bit[5:4] | 00:Reserved for future |
| Broadcast | Bit[3] | 0:Non-broadcast service |
| Open voice call mode(OVCM) | Bit[2] | 0:Non_OVCM call<br>1:OVCM call |
| Priority level | Bit[1:0] | 00: NO Priority<br>01: Priority 1<br>10: Priority 2<br>11: Priority 3<br>Note: Priority 3 is the highest Priority |

**Table 1-162 Service Option Settings**

## 1.2.77 DMR_CC

This message set the color code index for DMR calls

| Field Identifier | CC |
|---|---|
| 1 Byte | 1 Byte |
| 0x077 | 0: CC0 is used<br>1: CC1 is used<br>…<br>15: CC15 is used |

**Table 1-163 DMR_CC Field Format**

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## 1.2.78 DMR_CALL_START

This command is used to start a call. Upon receiving this command, SCT3258 begins to send call to the far end.

| Field Identifier |
|---|
| 1 Byte |
| 0x078 |

**Table 1-164 DMR_CALL_START Field Format**

## 1.2.79 DMR_CALL_OPTION

This field is used to set the DMR specific call options

| Field Identifier | DPMR Call Options |
|---|---|
| 1 Byte | 5 Bytes |
| 0x79 | Data[0]:<br>Bit 7: unused slot fill pattern<br>0: unused slot filled with 0<br>1: unused slot filled with random bits.<br>Bit 6: SQ detection enable<br>0: SQ detection is not done by SCT3258<br>1: SQ detection is done by SCT3258<br>Bit 5: IQ receive enable<br>0: SCT3258 in IF IQ mode<br>1: SCT3258 in base band mode<br>Bit 4: RF control enable<br>0: SCT3258 does not control RF chip (SCT3700 or RDA 1847)<br>1: SCT3258 controls RF chip.<br>Bit 3: RF slot timing control (SCT3700 Only)<br>0: No control of SCT3700 TX signal during transmission<br>1: Turn on and off SCT3700 TX signal aligned with the base band slot timing.<br>Bit 2: RF_TIMING RX mode enable<br>0: RF_TIMING port outputs slot timing information in TX mode only<br>1: RF_TIMING port outputs slot timing information in TX mode and RX mode<br>Bit 1-0: Traffic mode<br>0: DMR continuous transmission mode |

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| | |
|---|---|
| | 1: DMR direct mode with no reverse channel<br>2: DMR direct mode with reverse channel<br><br>Data[1]: voice delay constant and IQ transmit mode<br>Bit 7: IQ transmit enable<br>0: SCT3258 transmitting in IQ mode<br>1: SCT3258 transmitting in base band mode<br>Bit 6: Reserved<br>Bit 5:0: voice delay constant<br>0-40: voice delay constant in steps of 40 milliseconds<br>Others: reserved<br><br>Data[2]:<br>Bit 7-4: Number of good CACH received before reporting DMR_SLOT_FOUND with the slot verified flag set.<br>Bit 3-0: Number of repeated LC header<br><br>Data[3]:<br>Bit 7-6: Reserved<br>Bit 5-0: Bit mask for reporting repeated identical messages.<br>Bit 4: 0, report identical SLC once<br>1, report every SLC<br>Bit 3: 0: Report identical IDLE data frames once<br>1: Report all IDLE data frames<br>Bit 2: 0: Report identical CSBK data frames once<br>1: Report all CSBK data frames<br>Bit 1: 0: Report identical LC headers once<br>1: Report all LC headers<br>Bit 0: 0: Report identical LC terminators once<br>1: Report all LC terminators<br>Data[4]: Reserved |
| Default | 0 |

**Table 1-165 DMR_CALL_OPTION Field Format**

## 1.2.80 DMR_OFFSET

This command set the timing offset of the RF_TIMING signal to the internal slot timing signal, which is aligned with base station timing. With default parameters, the RF_TIMING signal is identical to the internal slot timing signal, which has a 60 millisecond period and 50% duty cycle, with the rising edge aligned with the end of a BS slot. With DMR_OFFSET command, the

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RF_TIMING signal can be ahead or lag of the internal slot timing signals.

| Field Identifier | Offset | Length |
|---|---|---|
| 1 Byte | 2 Bytes | 2 Bytes |
| 0x07A | 16 bit signed integer, MSB first. The unit is 41.7 microseconds. The range is -720 to 720, or -30 milliseconds to +30 milliseconds | 16 bit unsigned integer, MSB first. The unit is 41.7 microseconds. The range is 240 to 1200, or 10 milliseconds to 50 milliseconds |
| Default | 0 | 720 |

**Table 1-166 DMR_OFFSET Field Format**

## 1.2.81 DMR_SLOT_TYPE

This field is used report the slot type when a call is received.

| Slot Type | Colour Code | Data Type |
|---|---|---|
| 1 BYTE | BIT[7:4] | BIT[3:0] |
| 0x7B | Value:0-15 | 0000:PI header<br>0001:Voice LC header<br>0010:Terminatior with LC<br>0011:CSBK<br>0100:MBC header<br>0101:MBC continuation<br>0110:data header<br>0111:rate 1/2 data<br>1000:rate 3/4 data<br>1001:idle<br>1010:Rate1 Data<br>Others:Reserved for future use |

**Table 1-167 DMR_SLOT_TYPE Field Format**

## 1.2.82 DMR_EMB

This field is used report the EMB field when a call is received.

| DMR_EMB | Colour Code | PI | LCSS | Rev |
|---|---|---|---|---|
| 1 BYTE | BIT[7:4] | BIT[3] | BIT[2:1] | BIT[0] |

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| | | | | |
|---|---|---|---|---|
| 0x7C | Value:0-15 | Reserved | 00:Single fragment LC or first fragment CSBK signalling<br>01:First fragment of LC signalling<br>10:Last fragment of LC or CSBK signaling<br>11:Continuation fragment of LC or CSBK | Reserved |

**Table 1-168 DMR_EMB Field Format**

## 1.2.83 DMR_CALLED_ID_BCD

This field contains the called ID in 8 BCD codes.

| Field Identifier | Called ID in 7 BCD codes |
|---|---|
| 1 Byte | 4 Bytes |
| 0x7D | DATA[0]= {K1, K2}<br>DATA[1]= {K3, K4}<br>DATA[2]= {K5, K6}<br>DATA[3]= {K7, K8} |
| Default | 0xaa,0xaa,0xaa,0xaa |

**Table 1-169 DMR_CALLED_ID_BCD Field Format**

K1 should be either 0 or 1. An 8 digit BCD should not exceed “16777016”. If the user wishes to dial a 7 digit BCD number “1234567”, he/she should fill the K1-K8 as “01234567”.

## 1.2.84 DMR_OWN_ID_BCD

This field contains the own ID in 8 BCD codes.

| Field Identifier | Own ID in 7 BCD codes |
|---|---|
| 1 Byte | 4 Bytes |
| 0x7E | DATA[0]= {K1, K2}<br>DATA[1]= {K3, K4}<br>DATA[2]= {K5, K6}<br>DATA[3]= {K7, K8} |
| Default | 0x00,0x00,0x00,0x00 |

**Table 1-170 DMR_OWN_ID_BCD Field Format**

## 1.2.85 DMR_SLOT_FOUND

This field is used by SCT3258 to inform the MCU that synchronization with the far end BS or MS

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is achieved and CACH is decoded.

| DMR_SLOT FOUND | Slot Found Details |
|---|---|
| 1 BYTE | 1 Byte |
| 0x7F | Details in the table below |

**Table 1-171 DMR_SLOT_FOUND Field Format**

| Bit Position | Bit Name | Bit Descriptions |
|---|---|---|
| 7 | Sync Type | MS Sync or BS Sync<br>0: MS Sync<br>1: BS Sync |
| 6 | Slot Verify | Slot verify flag<br>0: Slot not verified<br>1: Slot Verified |
| 5 | Slot invert | Slot invert flag<br>0: slot not inverted from last report<br>1: slot inverted from last report |
| 4 | Reserved | Reversed |
| 3 | AT | Access Type<br>0:inbound channel is idle<br>1:inbound channel is busy<br>note:in continuous transmission mode, for both voice and data,the AT bit is set to 1; |
| 2 | TC | TDMA Channel selection<br>0:Following outbound burst is channel 1<br>1:Following outbound burst is channel 2 |
| 1:0 | LCSS/TDMA SLOT | When BS SYNC is found (bit 7 = 1),<br>These two bits indicate LCSS<br>00:Single fragment LC or first fragment CSBK signalling see note<br>01:First fragment of LC signalling<br>10:Last fragment of LC or CSBK signaling<br>11:Continuation fragment of LC or CSBK Signaling<br>When MS SYNC is found (bit 7 = 0), this two bit indicates the types of MS SYNC found<br>0: no TDMA direct SYNC<br>1: TDMA SYNC 1 found<br>2: TDMA SYNC 2 found |

**Table 1-172 Slot Found Details**

This field is used only by SCT3258 to report the CACH information when BS SYNC is found, or types of MS SYNC if MS SYNC is found. Identical CACH are not reported to reduce the traffic between SCT3258 and the MCU (after CACH is verified). A CACH verify scheme is used in

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reporting DMR_SLOT_FOUND. Bit 7-4 of Byte 2 of DMR_CALL_OPTION command specifies the number of good CACH received before reporting DMR_SLOT_FOUND with the slot verified flag set. Before the number of good CACH has reached the specified limit, SCT3258 reports every CACH that is found, without setting the slot verified flag. When the specified limit is reached, SCT3258 reports the last CACH with the slot verified flag set. If the slot number of the last reported CACH matches the previously reported CACH, it indicates that no slot inversion is needed. If the slot number of the last reported CACH does not match the previously reported CACH, it indicates that slot inversion is needed. In this case, the user should treat all previous report CACH as if the slot number is inverted (1 as 2, and 2 as 1).

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# 2 Application Guides

## 2.1 Code Downloading

After SCT3258 reset, it first down loads a boot loader from the MCU through the HPI and then execute the boot loader. The boot loader then loads the rest of the application from the embedded flash and start execution.

### 2.1.1 Boot loader Downloading

The boot loader is a program with fixed 1024 words. It is downloaded to SCT3258 after SCT3258 reset.

After the initial reset, SCT3258 enters into boot loading mode. In boot loading mode, SCT3258 is automatically configured after reset to load a 1024-word block of instructions from the HPI port. SCT3258 then begins executing that instruction block immediately.

The HPI port is configured as 16-bit mode, low-byte first, and either Intel mode or Motorola mode. PIO1 is sampled and used to select either Intel (LOW input) or Motorola (HIGH input).

The host must download exact 1024 words for this boot process to work.

Note that the boot loader for DPMR mode and DMR mode is different. Depending on the boot loader types, SCT3258 enters DPMR mode or DMR mode. In DPMR mode, commands for DMR only (type 5) are not recognized. In DMR mode, command for DPMR only (type 3) are not recognized.

### 2.1.2 Application Downloading

After the boot loader is downloaded to SCT3258 and executed, SCT3258 reads the application firmware in the embedded flash and start execution.

## 2.2 DPMR Call Processing

SCT3258 support two ways of processing digital calls: Easy Mode (layer 3), DPMR Layer 2 Mode (layer 2).

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## 2.2.1 Easy Mode

The Easy Mode is for users who don't have the in-depth understanding the DPMR protocol. MCU uses simple CALL_START, CALL_STOP, CONNECT, DISCONNECT messages to start, stop a voice calls, or initiate or disconnect a data calls. Internally, SCT3258 handles all call processing according the DPMR Protocol .

With Easy Mode, in addition to layer two processing, the SCT3258 also performs call control of the DPMR processing. "ANNEX A" Standard User Interface for CSF radio are implemented in SCT3258. At the transmitter, SCT3258 supports wild character dialing, abbreviated dialing and masked dialing. At the receiver, when a call is received, SCT3258 matches incoming called number with its own individual call number and group number and informs the MCU of matching results.

With Easy Mode, SCT3258 strictly complies with the DPMR protocol (ETSI TS 102 490 or 658). The ability to change any aspect of the protocol is disabled.

The process mode can be set or changed with PROCESS_MODE command. For Easy Mode, Mode 3 is selected with PROCESS_MODE command.

### 2.2.1.1 Transmitting

In Easy Mode, the MCU passes called ID and own ID to SCT3258. SCT3258 saves this information in the internal registers. Upon receiving a CALL_START packet, it formatted the information according to the DPMR protocol and send to the far end.

For example, a voice can be started with the following packet.

| | |
|---|---|
| Packet Header | START_BYTES |
| | LENGTH |
| | TYPE |
| Fields | DPMR_CALLED_ID |
| | DPMR_OWN_ID |
| | CALL_START |
| Parity | FBYTE |
| | PARITY_BYTE |

**Table 2-1 Example Packet For Starting a Voice Call in Easy Mode**

If a field has been sent from the MCU to SCT3258 before, and does not change in the next packet, this field can be omitted from the packet. For example, to redial a previous dial number, the MCU only needs to send the following packet to SCT3258

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| | |
|---|---|
| Packet Header | START_BYTES |
| | LENGTH |
| | TYPE |
| Fields | CALL_START |
| Parity | FBYTE |
| | PARITY_BYTE |

**Table 2-2 Example of Redial Packet**

The call can be ended with a packet with CALL_STOP field.

| | |
|---|---|
| Packet Header | START_BYTES |
| | LENGTH |
| | TYPE |
| Fields | CALL_STOP |
| Parity | FBYTE |
| | PARITY_BYTE |

**Table 2-3 Example of Call Stop Packet**

#### 2.2.1.2 Receiving

Upon receiving a call from the far end, SCT3258 extracts useful information according to the DPMR protocol and reports to the MCU.

The example below shows the packet sent to the MCU after receiving a new header frame.

| | |
|---|---|
| Packet Header | START_BYTES |
| | LENGTH |
| | TYPE |
| Fields | CALL_HT_CI |
| | DPMR_CALLED_ID |
| | DPMR_OWN_ID |
| | DPMR_M_V_F_E |
| | DPMR_CC |
| | CALL_MATCH |
| Parity | FBYTE |
| | PARITY_BYTE |

**Table 2-4 Example of Packet from DSP after Receiving a Call**

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The type of FIELD to report can be changed with REPORT_FIELD command.

The fields can be sent in one packet or in multiple packets.

## 2.2.2 DPMR Layer 2 Mode

With DPMR Layer 2 Mode, SCT3258 support DPMR up to layer 2 the does not handle the call processing layer of the DPMR protocol. All call processing supports are provided by external MCU. In this case, SCT3258 just performs the tasks of transporting DPMR related messages between the MCU's of the two terminals. As a result, the following process will be performed by the MCU:

**Air Interface Call Control Layer (Layer 3)**

1. Establishing, maintaining and termination of calls
2. Individual or group call transmission and receptions
3. Destination addressing
4. Automatic matching of Called ID of incoming call to own ID and group ID

As comparison, these processes are completed by SCT3258 with Easy Mode.

With DPMR Layer 2 Mode, the MCU can access primitive elements of the DPMR architecture. For example, to start a call, the MCU send individual elements in the protocol to SCT3258, with DPMR_SEND_HEADER and DPMR_SEND_SF, whereas in easy mode, the user sends a simple CALL_START to SCT3258.

With easy mode, the user also has the ability to change synchronization pattern with DPMR_FS1, DPMR_FS2, DPMR_FS3 and DPMR_FS4.

### 2.2.2.1 Transmitting

In DPMR Layer 2 Mode, the MCU pass un-encoded HT, END and CCH frames, or individual information elements such as Called_ID or Own_ID to SCT3258. SCT3258 saves this information in the internal registers. Upon receiving a DPMR Packet with DPMR_SEND_HEADER, DPMR_SEND_END or DPMR_SEND_SF, the MCU performs channel coding and interleaving, forms the complete HI, END and Super Frame structure, and send to the far end.

For example, upon receiving the DPMR Packet below, SCT3258 builds a CCH0 to CCH3 structure.

| | |
|---|---|
| Packet Header | START_BYTES |
| | LENGTH |

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| | |
|---|---|
| | |
| | TYPE |
| Fields | DPMR_CCH_UE (0) |
| | DPMR_CCH_UE (1) |
| | DPMR_CCH_UE (2) |
| | DPMR_CCH_UE (3) |
| Parity | FBYTE |
| | PARITY_BYTE |

**Table 2-5 Example 1 of a Packet from MCU in DPMR Layer 2 Mode**

Then if the MCU send a packet with DPMR_SEND_SF, it forms a super frame with the saved information, and sends it to the far end.

Similarly the host message to build a HI frame and END frame can be like the one below:

| | |
|---|---|
| Packet Header | START_BYTES |
| | LENGTH |
| | TYPE |
| Fields | DPMR_HI_UE |
| Parity | FBYTE |
| | PARITY_BYTE |

**Table 2-6 Example of a HI Packet from MCU**

| | |
|---|---|
| Packet Header | START_BYTES |
| | LENGTH |
| | TYPE |
| Fields | DPMR_END_UE |
| Parity | FBYTE |
| | PARITY_BYTE |

**Table 2-7 Example of END Packet from MCU**

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### 2.2.2.2 Receiving

Upon receiving a Super Frame, Header Frame or an End Frame, SCT3258 extracts the CCH0 – CCH3, HI or END fields, performs de-interleaving and channel decoding on the frames. SCT3258 extracts individual elements to the MCU.

The example below shows the packet sent to the MCU after receiving a new header frame.

| | |
|---|---|
| Packet Header | START_BYTES |

| | |
|---|---|
| | LENGTH |
| | TYPE |
| Fields | DPMR_CALLED_ID |
| | DPMR_OWN_ID |
| | DPMR_M_F |
| | DPMR_HT_CI_PM |
| Parity | FBYTE |
| | PARITY_BYTE |

**Table 2-8 Example of Packet from DSP after Receiving a Header Frame**

## 2.3 DMR Call Processing

SCT3258 support two ways of processing digital calls: Easy Mode (Layer 3 Mode), DMR Layer 2 Mode.

In Easy Mode (Layer 3 Mode), SCT3258 digital mode support layer 1-3 processing of the DMR protocol complied with ETSI TS 102 361. The MCU uses simple DMR_CALL_START, CALL_STOP messages to start, stop a voice calls.

In Layer 2 Mode, SCT3258 implements Layer 1 and Layer 2 of the DMR protocol, complied with ETSI TS 102 361. Layer 3 and above are implemented on the MCU. The interface between SCT3258 and the MCU are based on Layer 2 and Layer 3 interface.

Note that if SCT3700 supported is enabled, only Layer 2 Mode is supported.

### 2.3.1 DMR Easy Mode

In Easy Mode (Layer 3 Mode), the MCU uses simple DMR_CALL_START, CALL_STOP messages to start, stop a voice calls. Internally, SCT3258 handles all call processing according the DMR Protocol. “ANNEX C of TS 102 361” numbering plan is implemented in SCT3258. At the transmitter, SCT3258 supports individual call and group call dialing. At the receiver, when a call is received, SCT3258 matches incoming called number with its own individual call number and group number and informs the MCU of matching results.

### 2.3.2 DMR Layer 2 Mode

In DMR Layer 2 Mode, SCT3258 interacts with the control processor (the MCU) to complete the DMR voice and data calls. The layer 1 and layer 2 of the DMR are implemented in SCT3258, while layer 3 and above are implemented in the MCU. The information exchange between the MCU and SCT3258 is through DIGC_DATA_FRAME commands or indications. The MCU passes the data types and the payload information the SCT3258 through DIGC_DATA_FRAME to

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SCT3258. SCT3258 performs FEC according to the data types, SCT3258 adds FEC, formulates them into 30 ms data burst and adds 4FSK modulation before sending to the far end. At the receiver, SCT3258 decode the received data burst, and report to the MCU through DIGC_DATA_FRAME indications.

In DMR Layer 2 mode, care must be taken to make sure that the DIGC_DATA_FRAME messages are sent no more and no less than 60 millisecond at a time. Examples of layer 2-3 message exchange are described in the example session below.

Two types of commands are used by the MCU to control SCT3258. The first type is hardware related for configuring SCT3258 in certain mode ready for communication. The second type is protocol related, where commands are used to pass layer 2 and layer 3 information. In particular DIGC_DATA_FRAME command is used to start or stop a voice or data call.

## 2.3.3 DMR MS Call Flow

The diagram below shows at typical DMR voice call from one mobile station to another mobile station without a repeater.

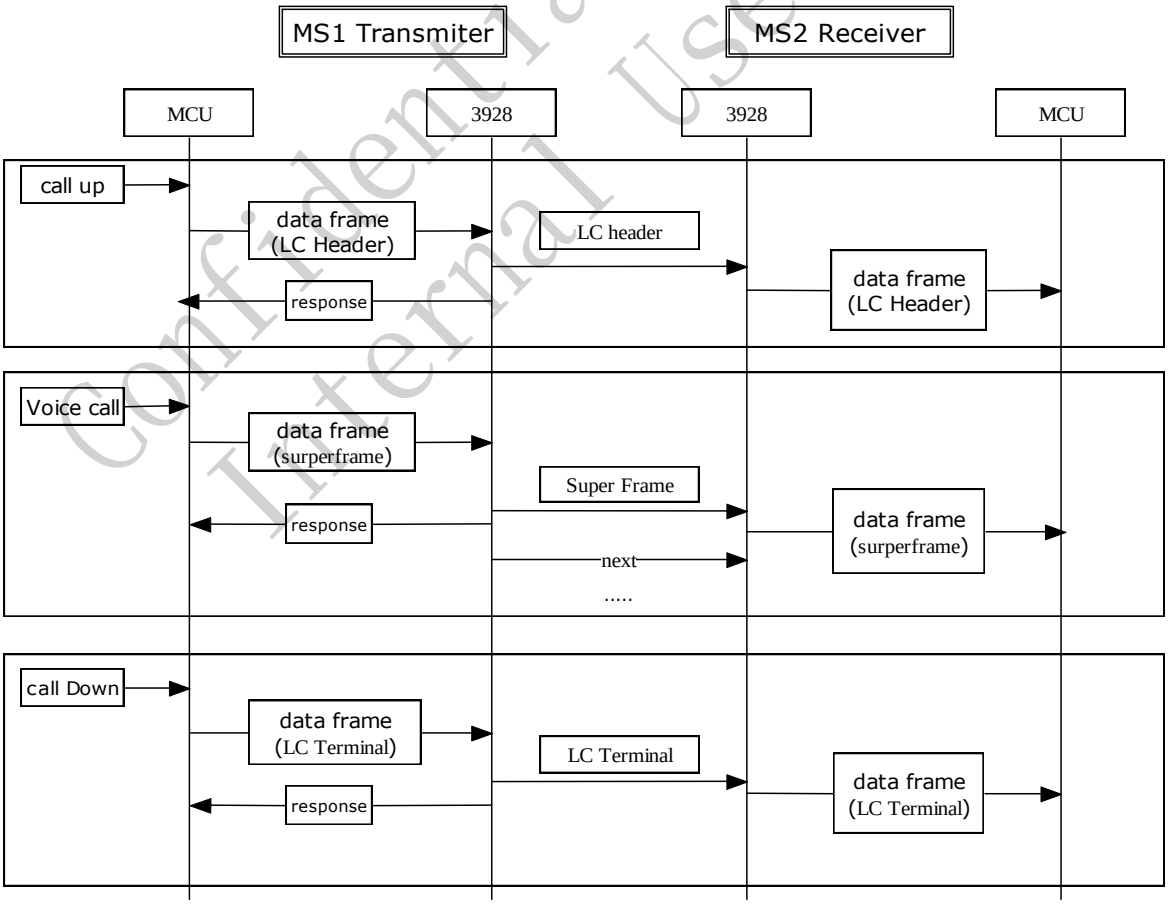



**Figure 2-1 DMR MS Call Flow**

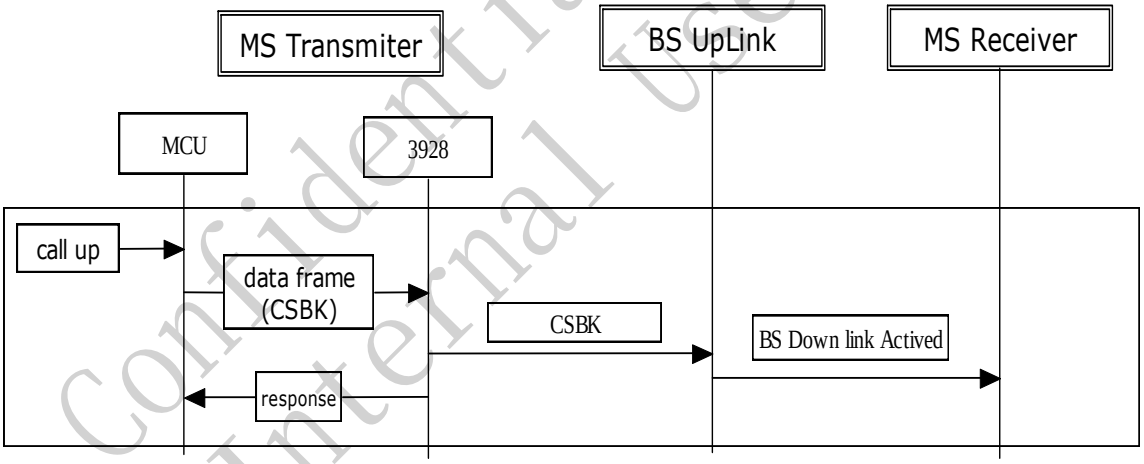
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The MCU informs SCT3258 the types of data frame to send through command DMR_DATA_FRAME and then waits for response from SCT3258. SCT3258 put the data frame for transmission and sends a response in the form of (0x17 0x0a). The next DMR_DATA_FRAME won't be sent unless a response (0x17 0x0A) is received. The interval between two DMR_DATA_FRMAE is around 60 milliseconds.

Only one DMR_DATA_FRAME command with LC contents is needed to start the voice communication at the TX side. The subsequent super frames use the same LC contents. Similarly, DMR_DATA_FRAME is only reported once upon receiving super frames at the RX side. The MCU knows the call end condition from either by receiving a terminal frame, or by receiving SCT3258 response of 1B 08.

## 2.3.4 Base Station Activation

In order to place voice or data calls through the base station or repeater, the mobile station should first acquire synchronization by monitor signals from the base station. If the base station is in idle state, the mobile station should first send a base station activation message to activate the base station. This is done by MCU sending a DMR_DATE_FRAME with CSBK contents. The diagram below shows the base station activation procedure.




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**Figure 2-2 Base Station Activation**

## 2.3.5 Listen Before Transmitting

If the base station is activated, it constantly broadcasts base station signals. The mobile station should first acquire timing synchronization from the base station before it can start or receive a call. Timing synchronization is done by monitoring the SYNC pattern and the CACH signals from the base station.

The MCU on the mobile station first sends command DMR_CALL_SLOT to inform SCT3258

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which slot to monitor, and waits SCT3258 for DMR_SLOT_FOUND response. As soon as SCT3258 acquire synchronization, it sends DMR_SLOT_FOUND response, first with slot verified flag unset, then after the preset number of CACH is received, with the slot verified flag set. At this point, the mobile station is aligned with the base station. SCT3258 can report any data frames from the far end through DMR_DATA_FRAME message. The MCU can also issue DMR_DATA_FRAME command to SCT3258 to send to the far end.

The diagram below shows a timing acquisition process.

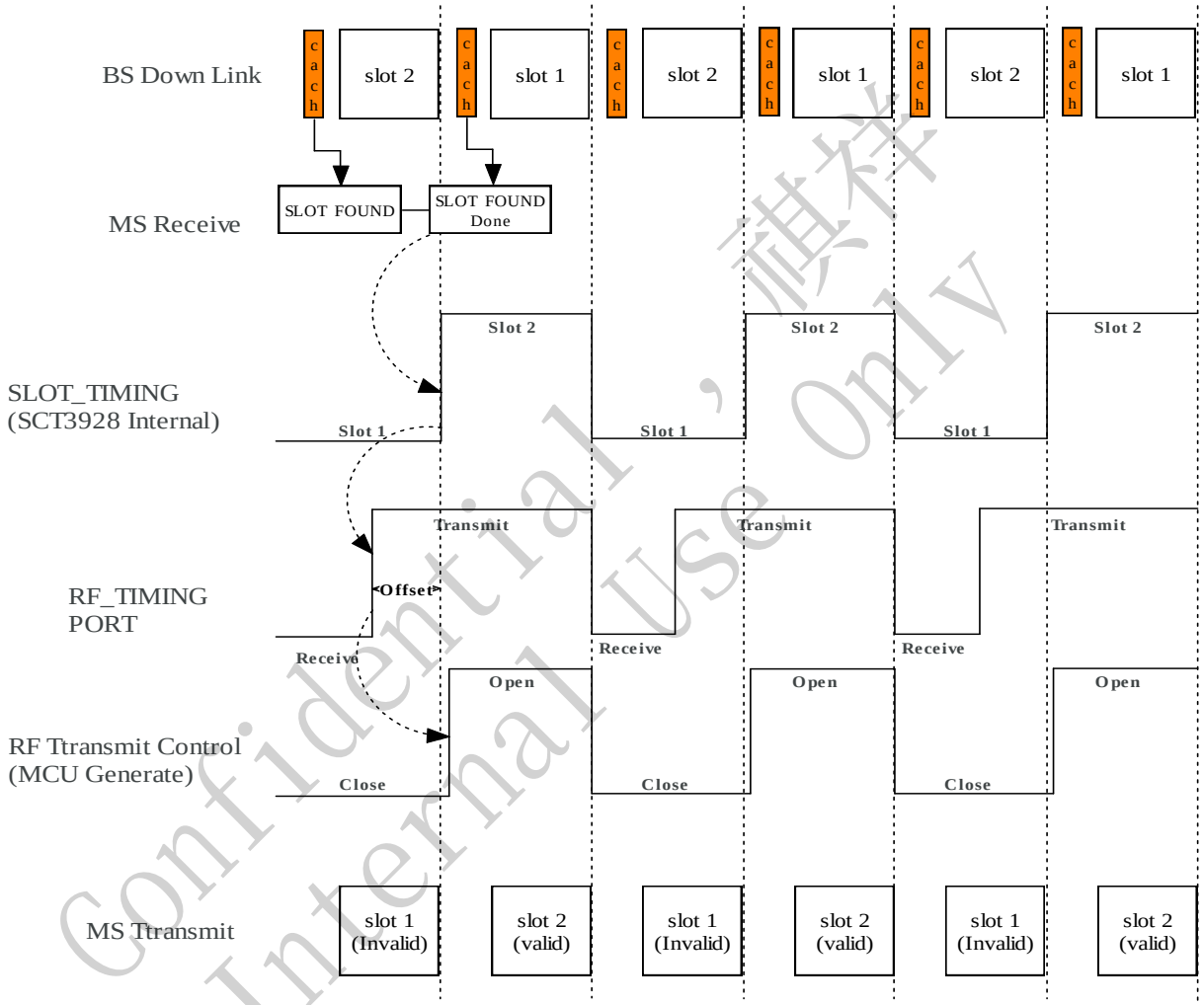



**Figure 2-3 Signal Timing Relationship**

After SCT3258 acquires synchronization from the base station, it generates an internal 60 millisecond cyclic timing signal, SLOT_TIMING. In the TX mode, SCT3258 controls the timing on the modulation output port so that it is in synchronization with the base station timing. In the meantime, it also outputs a SLOT_TIMING signal through the RF_TIMING port to control the RF circuit. The RF_TIMING signal can be identical to the SLOT_TIMNG. It can also be ahead or lag the SLOT_TIMING signal. The offset are programmable through command DMR_OFFSET. The MCU can then generate necessary timing signals to control the RF transceiver and the power amplifier. In the case of SCT3700, the switching of RF transceiver and controlled by the SCT3258.

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# 2.4Typical Call Sessions with SCT3258 in DPMR Mode

## 2.4.1 Digital Voice Call

A voice call involves the transmitter and the receiver. In both side, the MCU is the main controller, controlling the RF circuit, the MMI and SCT3258. Before the call, the MCU should set the RF circuit in TX mode for the transmitter and in RX mode for the receiver. The sub-section below only describes the interaction between the MCU and SCT3258.

### 2.4.1.1 Initial Setting for both the Transmitter and the Receiver

1. Set Vocoder type with VOCODER_SEL
2. Set SCT3258 to Easy mode with PROCESS_MODE command
3. Set Color Code corresponding to the RF Channel with DPMR_CC command
4. Set addressing mode to either “Initial Address Mode” or “Configured Addressing Mode” with ADDRESS_MODE command
5. Set Own ID with DIGC_OWN_ID_BIN or DPMR_OWN_ID_BCD (“Configured Addressing Mode” only)

### 2.4.1.2 Initial Setting for the Receiver

1. Set the demodulator gain with DEMOD_GAIN command
2. Set the types of field to report when receiving a call with REPORT_FIELD command

### 2.4.1.3 Initial Setting for the Transmitter

1. Set the modulator gain with MOD_GAIN command

### 2.4.1.4 Transmitter Side, Start a Call

1. Set SCT3258 in TX or DUPLEX with WORK_MODE command
2. Set called ID and own ID with DPMR_CALLED_ID and DPMR_OWN_ID
3. Start the call with CALL_START, for peer-to-peer voice call

### 2.4.1.5 Receiver Side, Start a Call

1. Set SCT3258 in RX or DUPLEX with WORK_MODE command
2. When detecting carrier, inform SCT3258 that carrier is ready with CARRIER_READY command

After the call from transmitter arrives, SCT3258 reports useful information. The MCU decides

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whether to accept or discard the call.

#### 2.4.1.6 Transmitter Side, End a Call

1. Drop the call with CALL_STOP command
2. When the END message is sent out, SCT3258 responds with ACK_MESSAGE (0x17 0x0A)

#### 2.4.1.7 Receiver Side, End a Call

1. Drop the carrier with CARRIER_READY command

### 2.4.2 Voice Call with Slow Data

1. Start the call the same way as voice only call described in the previous section except that the communication mode(the M in M_V_F_EP) is set to Call with Slow Data
2. After the calls start, the MCU first send a packet with DPMR_SLD field.
3. When SCT3258 begin transmitting, it fills slow data in the super frame and transmits slow data along with the voice frames. As soon as the slow data is being transmitted, it sends a DPMR_SLD response packet to the MCU, at 80 millisecond interval.

| | |
|---|---|
| Packet Header | START_BYTES |
| | LENGTH |
| | TYPE (Read Near End) |
| Fields | DPMR_SLD |
| Parity | FBYTE |
| | PARITY_BYTE |

**Table 2-9 Example Packet for Request Near End Slow Data**

4. Upon receiving DPMR_SLD response, the MCU write DPMR_SLD to SCT3258, also at 80 millisecond interval.
5. At the receiver, upon receiving super frame with SLOW data, SCT3258 extract slow data and report to the MCU with DPMR_SLD packet at 80 millisecond interval.
6. When all data has been transmitted, the MCU has to wait the response packet of the last frame DPMR_SLD.
7. Drop the call with CALL_STOP command
8. When the END message is sent out, SCT3258 responds with ACK_MESSAGE (0x17 0x0A)

### 2.4.3 Voice Call Recording

1. Enable voice recording function with command VOCODER_IO_SET at the transmitter for near end voice call recording (IO_SET = 0x20), or at the receiver for far end voice call recording (IO_SET = 0x04)

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2. Start the call the same way as voice only call described in the previous section.
3. When the call starts, SCT3258 start to send encoded voice data with CHAN_D field, at 80 millisecond interval.

| | |
|---|---|
| Packet Header | START_BYTES |
| | LENGTH |
| | TYPE (Read Near End) |
| Fields | CHAN_D |
| Parity | FBYTE |
| | PARITY_BYTE |

**Table 2-10 Example Packet for Voice Recoding**

4. Upon receiving CHAN_D, the MCU should process the message and save its contents. The MCU does not send a response to CHAN_D.
5. The process repeats at 80 millisecond interval.
6. Drop the call with CALL_STOP command
7. When the END message is sent out, SCT3258 responds with ACK_MESSAGE (0x17 0x0A)
8. Disable the voice recording by setting IO_SET = 0x00

## 2.4.4 Voice Call Play Back

1. Enable voice play back function with command VOCODER_IO_SET at the transmitter for far end voice play back (IO_SET = 0x40), or at the receiver for near end voice play back (IO_SET = 0x08)
2. Start the call the same way as voice only call described in the previous section.
3. When the call starts, MCU starts to send encoded voice data with CHAN_D field, at 80 millisecond interval.

| | |
|---|---|
| Packet Header | START_BYTES |
| | LENGTH |
| | TYPE (Read Near End) |
| Fields | CHAN_D |
| Parity | FBYTE |
| | PARITY_BYTE |

**Table 2-11 Example Packet for Play Back**

4. Upon receiving CHAN_D, SCT3258 send back the response to CHAN_D.
5. The process repeats at 80 millisecond interval.
6. Drop the call with CALL_STOP command
7. When the END message is sent out, SCT3258 responds with ACK_MESSAGE (0x17 0x0A)
8. Disable the voice recording by setting IO_SET = 0x00

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## 2.4.5 Data Call with Type 1 or Type 2 Data

1. Start the call the same way as voice only call described in the previous section except that the communication mode (the M in M_V_F_EP) is set to Type 1 or Type 2 data call. The DPMR_HT_CI has to be configured as section 1.2.53.
2. After the calls start, the MCU first send a packet with DIGC_DATA_FRAME field, which includes up to 36 bytes (in 80 ms) of data. In the DIGC_DATA_FRAME fields, user should also indicate whether valid data are present and the length of the valid data. The user should also indicate whether this is the last data frame.
3. When SCT3258 begin transmitting, it fills data in the TCH frame and starts transmitting data. As soon as data is being transmitted, it sends a DIGC_DATA_FRAME response packet to the MCU, at 80 millisecond interval.

| | |
|---|---|
| Packet Header | START_BYTES |
| | LENGTH |
| | TYPE (Read Near End) |
| Fields | DIGC_DATA_FRAME |
| Parity | FBYTE |
| | PARITY_BYTE |

**Table 2-12 Example Packet for Request Near End Type 1 and Type 2 Data**

4. Upon receiving DPMR_DTAT_FRAME response, the MCU write DIGC_DATA_FRAME with more data to SCT3258, also at 80 millisecond interval.
5. At the receiver, upon receiving DPMR pay load, SCT3258 extracts data from TCH frames and report to the MCU with DIGC_DATA_FRAME packet at 80 millisecond interval.
6. When all data has been transmitted, the MCU responds to data query with DIGC_DATA_FRAME with indication that no more data is to transmit.
7. Drop the call with CALL_STOP command
8. When the END message is sent out, SCT3258 responds with ACK_MESSAGE (0x17 0x0A)

## 2.4.6 Voice Call with Appended Data

The Voice Call with Appended Data starts as voice call and ends as a Type 2 Data Call.

1. Start the call the same way as voice only call described in the previous sections except that the communication mode(the M in M_V_F_EP) is set to Voice Call with Appended Data. The DPMR_HT_CI has to be configured as section 1.2.53.
2. Proceed to voice communication as a normal voice call
3. When the user releases the PTT, the MCU first send a packet with DIGC_DATA_FRAME field, which includes up to 36 bytes (in 80 ms) of data. In the DIGC_DATA_FRAME fields, user should also indicate whether valid data are present and the length of the valid data. The user should also indicate whether this is the last data frame.

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4. When SCT3258 begin transmitting, it fills data in the TCH frame and starts transmitting data. As soon as data is being transmitted, it sends a DIGC_DATA_FRAME response packet to the MCU, at 80 millisecond interval.
5. Upon receiving DPMR_DTAT_FRAME response the MCU write DIGC_DATA_FRAME with more data to SCT3258, also at 80 millisecond interval.
6. At the receiver, upon receiving DPMR pay load, SCT3258 extracts data from TCH frames and report to the MCU with DIGC_DATA_FRAME packet at 80 millisecond interval.
7. When all data has been transmitted, the MCU responds to data query with DIGC_DATA_FRAME with indication that no more data is to transmit.
8. Drop the call with CALL_STOP command
9. When the END message is sent out, SCT3258 responds with ACK_MESSAGE (0x17 0x0A)

## 2.4.7 Data Call Type 3 (for future release)

Type 3 Data calls involve a call set up and disconnect procedure as described below:

1. Station A: Send connection request
2. Station B: Send ACK
3. Station A: Send data frames
4. Station B: Send ACK
5. Station A: Send more data frames
6. Station B: Send ACK

…

7. Station A: Send Disconnect request

The interactions of MCU and SCT3258 to complete a data call are described below.

### 2.4.7.1 Initialization of Data Call Type 3

The initialization of Type3 data call is the same as voice calls.

### 2.4.7.2 Connection Request

1. Set Station A in TX mode and Station B in RX mode with WORK_MODE command
2. Set call information (CI) for appropriate data format with DPMR_CI_PM command
3. Send CONNECT command in station A
4. Set Carrier Ready in station B
5. SCT3258 in station B report HT_CI to station B MCU
6. Station B MCU becomes aware of a data call request, and get ready to send an ACK
7. When the CONNECT message is sent out in station A, SCT3258 responds with ACK_MESSAGE (10)

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### 2.4.7.3 Connection Acknowledge

1. Set Station B in TX mode and Station A in RX mode with WORK_MODE command
2. Set CI to indicate ACK or NACK (with NACK reason) with DPMR_CI_PM.
3. Send DPMR_ACK command in station B
4. Set Carrier Ready in station A
5. SCT3258 in station A report HT_CI to the MCU
6. Station A MCU becomes aware of the acknowledgement from station B, and get ready to send data.
7. When the Connect Acknowledgement message is sent out in station B, SCT3258 responds with ACK_MESSAGE (10)

### 2.4.7.4 Sending Type 3 Data Frames

1. Set Station A in TX mode and Station B in RX mode with WORK_MODE command
2. Set CI for appropriate data format with DPMR_CI_PM command
3. Start the call with CALL_START
4. After the calls start, the MCU first send a packet with DIGC_DATA_FRAME field with Type 3 data indication, which includes up to 48 bytes (in 80 ms) of data. In the DIGC_DATA_FRAME fields, user should also indicate whether valid data are present and the length of the valid data. The user should also indicate whether this is the last data frame.
5. SCT3258 fills data in the data frame and start transmission. As soon as data is being transmitted, it sends a DIGC_DATA_FRAME query packet to the MCU to request for more data. This process continues at 80 milli-second interval.
6. Upon receiving DIGC_DATA_FRAME query, the MCU write DIGC_DATA_FRAME to SCT3258, also at 80 millisecond interval.
7. At station B, upon receiving data frames, SCT3258 extract data and report to the MCU with DIGC_DATA_FRAME packet at 80 millisecond interval.
8. Station B MCU assembles the data frames from the DIGC_DATA_FRAME fields.
9. When all data in current data frame has been transmitted, the MCU responds to data query with DIGC_DATA_FRAME with indication that no more data is to transmit.
10. Drop the call with CALL_STOP command
11. When the END message is sent out, SCT3258 responds with ACK_MESSAGE (0x17 0x0A)

### 2.4.7.5 Data Acknowledgement

1. Set Station B in TX mode and Station A in RX mode with WORK_MODE command
2. Set CI to indicate ACK or NACK (with NACK reason) with DPMR_CI_PM.
3. Send DPMR_ACK command in station B
4. Set Carrier Ready in station A
5. SCT3258 in station A report HT_CI to the MCU
6. Station A MCU becomes aware of the acknowledgement from station B, and get ready to send more data packet.
7. When the Data Acknowledgement message is sent out in station B, SCT3258 responds with

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ACK_MESSAGE (0x17 0x0A) to MCU B.

#### 2.4.7.6 Disconnect

1. Set Station A in TX mode and Station B in RX mode with WORK_MODE command
2. Send DISCONNECT command in station A.
3. Set Carrier Ready in station B
4. SCT3258 in station B report HT_CI to the MCU
5. Station B MCU becomes aware of the Disconnect request from station A, and start disconnect procedure
6. Station B disconnect the cal
7. When the DISCONNECT message is sent out in station A, SCT3258 responds with ACK_MESSAGE (0x17 0x0A)

### 2.4.8 Short Appended Data (TS 102 658 Only)

ETSI TS 102 658 defines a special message for transmitting short data. It is constructed by a connection request HEADER frame, one to four appended data frames, and an END frame. To send Short Appended Data, the MCU can follow the procedure below.

#### 2.4.8.1 Send Short Appended Data

Set Station A in TX mode with WORK_MODE command

1. Set call information (CI) for appropriate data format with DPMR_CI_PM command
2. Send the contents of the short appended data with DIGC_DATA_FRAME command with type set to Short Appended Data.
3. Send SEND_AD command in station A
4. When the CONNECT message is sent out in station A, SCT3258 responds with ACK_MESSAGE (10)
5. The MCU set Station A in RX mode to get ready to receive an ACK.

#### 2.4.8.2 Receive Short Appended Data

1. Set Station B in RX mode with WORK_MODE command
2. Set Carrier Ready in station B
3. SCT3258 in station B receives short appended data and report to station B MCU with DIGC_DATA_FRAME
4. Set Station B in TX mode to get ready to send an ACK.

### 2.4.9 Analog Voice Call in DPMR Mode

SCT3258 support dual mode operation with digital and analog calls. For analog call, SCT3258 supports CTCSS/DCS sub-audio signals.

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#### 2.4.9.1 Receiver Side, Start a Call

1. Set SCT3258 to analog call mode with PROCESS_MODE command
2. Set SCT3258 in RX or DUPLEX with WORK_MODE command
3. Set CTCSS/DCS settings with SUB_AUDIO command
4. When detecting carrier, inform SCT3258 that carrier is ready with CARRIER_READY

After the call from transmitter arrives, SCT3258 reports whether CTCSS/DCS is matched or not with SUB_AUDIO command.

#### 2.4.9.2 Transmitter Side, Start a Call

1. Set SCT3258 to analog call mode with PROCESS_MODE command
2. Set SCT3258 in TX or DUPLEX with WORK_MODE command
3. Set CTCSS/DCS settings with SUB_AUDIO command
4. Start the call with CALL_START command

#### 2.4.9.3 Receiver Side, End a Call

1. Drop the carrier with CARRIER_READY command

#### 2.4.9.4 Transmitter Side, End a Call

1. Drop the call with CALL_STOP command

### 2.4.10 Handling of Maintenance Message (TS 102 658 Only)

With ETSI TS 102 658, the BS may send multiple maintenance messages such as preservation messages, Idle messages or Guard messages. SCT3258 may report each occurrence of these messages, or elect to only report the same message once to save traffic between SCT3258 and the MCU. The election is controlled by command DPMR_PROTOCOL_OPTION.

### 2.4.11 Automatic Analog and DPMR Call Detection

To set up for automatic analog and DPMR call detection, the receiver is set up the same ways as a digital calls, with the following exceptions:

1. Set SCT3258 to Easy mode with PROCESS_MODE command, with mixed call bit enabled: i. e., the process mode as 131.
2. Set CTCSS/DCS settings with SUB_AUDIO command

The transmitter side is set up exactly as normal analog or DPMR call.

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With automatic analog and DPMR call detection, SCT3258 monitors the RF channel, and simultaneously detects FS1/FS2 pattern for DPMR calls, and CTCSS/DCS signaling for analog calls. If DPMR call is found, it sends CALL_MATCH message (even if the called ID or CC code does not match). If CTCSSS/DCS signaling is found, it send SUB_AUDIO message.

## 2.4.12 Audio Muting for Un-matched Calls

By defaults, SCT3258 enables audio as soon as a DPMR or an analog call is received, even if called ID or CTCSS/DCS code does not match. It is up to the MCU to decide whether to take the call or not. If the MCU decides to take to call, it enables audio PA. If not, it turns off the audio PA, and send carrier lost to SCT3258.

Optionally, the SCT3258 can mute the audio until a correct call match is achieved. This can be done by setting Mute Flag when sending the Carrier Ready command at the receiver. When Mute Flag is set, SCT3258 sends zeros to its audio out line until a called ID of the incoming call matches the own ID of itself, or the CTCSS/DCS code of the transmitter and receiver match.

# 2.5 Typical Call Sessions with SCT3258 in DMR Mode

## 2.5.1 DMR Voice Call in DMR Easy Mode

A voice call involves the transmitter and the receiver. In both side, the MCU is the main controller, controlling the RF circuit, the MMI and SCT3258. Before the call, the MCU should set the RF circuit in TX mode for the transmitter, and in RX mode for the receiver. The sections below only describe the interaction between the MCU and SCT3258.

Note that if SCT3700 support is enabled, DMR Easy mode is not supported.

### 2.5.1.1 Initial Setting for both the Transmitter and the Receiver

1.Set Vocoder type with VOCODER_SEL, 0 for AMBE3000 (external), 1 for Tsinghua, 3 for AMBE+2C

Command ID: VOCODER_SEL
Send Command: 84 A9 61 00 02 00 10 03
Receive Data:84 A9 61 00 02 00 10 00

2.Set SCT3258 to digital mode with PROCESS_MODE command

Command ID: PROCESS_MODE
Send Command: 84 A9 61 00 02 00 1A 03
Receive Data:84 A9 61 00 02 00 1A 00

3.Set Color Code corresponding to the RF Channel with DMR_CC command

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Send Command: 84 A9 61 00 02 08 77 01
Receive Data:84 A9 61 00 02 08 77 00

4.Configure the call option with DMR_CALL_OPTION
Command ID: DMR_CALL_OPTION
Send Command: 84 A9 61 00 06 08 79 41 00 00 00 00
Receive Data:84 A9 61 00 02 08 79 00
(In this particular command, SCT3258 is set to do SQ detection; slotted mode is used, with unused slot filled with zeros; SLOT verified number is 1).

5.Set Own ID with DIGC_OWN_ID_BIN or DMR_OWN_ID_BCD
Command ID: DIGC_OWN_ID_BIN
Send Command: 84 A9 61 00 04 08 50 XX XX XX
Receive Data:84 A9 61 00 02 08 50 00

#### 2.5.1.2 Initial Setting for the Receiver

1. Set the demodulator gain with DEMOD_GAIN command
2. Set the types of field to report when receiving a call with REPORT_FIELD command
Command ID: REPORT_FIELD
Send command: 84 A9 61 00 02 00 1D XX
Receive Data: 84 A9 61 00 02 00 1D 00

#### 2.5.1.3 Initial Setting for the Transmitter

1. Set the modulator gain with MOD_GAIN command

#### 2.5.1.4 Transmitter Side, Start a Call

1.Set SCT3258 in TX with WORK_MODE command
Command ID: WORK_MODE
Send Command: 84 A9 61 00 04 00 18 02 00 00
Receive Data: 84 A9 61 00 02 00 18 00

2.Set called ID and own ID with DIGC_CALLED_ID and DIGC_OWN_ID
Command ID: DIGC_OWN_ID_BIN
Send Command: 84 A9 61 00 04 08 50 XX XX XX
Receive Data: 84 A9 61 00 02 08 50 00

Command ID: DMR_CALLED_ID_BIN
Send Command: 84 A9 61 00 04 08 51 XX XX XX
Receive Data: 84 A9 61 00 02 08 51 00

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3.Set FLC information

Command ID: DMR_FLC

Send Command: 84 A9 61 00 04 08 76 00 00 00

Receive Data:84 A9 61 00 02 08 76 00

(In this particular command, FLCO as group call, FID is standard)

4.Start the call with DMR_CALL_START, for peer-to-peer voice call

Command ID: DMR_CALL_START

Send Command: 84 A9 61 00 01 08 78

receive command: 84 a9 61 00 02 08 78 00

### 2.5.1.5 Receiver Side, Start a Call

1.Set SCT3258 in RX or DUPLEX with WORK_MODE command

Command ID: WORK_MODE

Send Command: 84 A9 61 00 04 00 18 01 00 00

Receive Data:84 A9 61 00 02 00 18 00

2.When detecting carrier, inform SCT3258 that carrier is ready with CARRIER_READY command

Send Command: 84 A9 61 00 02 00 19 {0x01 or 0x02}

Receive Data:84 A9 61 00 02 00 19 00

After the call from transmitter arrives, SCT3258 reports useful information. The MCU decides whether to accept or discard the call.

### 2.5.1.6 Receiver Side, after a Call is Received

Receive Data:84 A9 61 00 02 30 7F 87

- DMR_SLOT_FOUND message. The content is TACT in the CACH, BS call, SLOT verify not completed

Receive Data:84 A9 61 00 02 30 1B 04

- Indicate voice SYNC is received

Receive Data:84 A9 61 00 02 30 7B 11

- DMR_SLOT_TYPE message. CC is 1. Data type is Voice LC Header

Receive Data:84 A9 61 00 04 30 76 00 00 00

- DMR_FLC message

Receive Data:84 A9 61 00 04 30 50 xx xx xx

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- DIGC_CALLED_ID_BIN

Receive Data:84 A9 61 00 04 30 51 xx xx xx

- DIGC_OWN_ID_BIN

Receive Data:84 A9 61 00 05 30 7D xx xx xx xx

- DIGC_CALLED_ID_BCD

Receive Data:84 A9 61 00 05 30 7E xx xx xx xx

- DIGC_OWN_ID_BCD

Receive Data:84 A9 61 00 02 30 27 00

- CALL_MATCH message, indicating a valid DMR call is received, and called ID and CC are matched.

#### 2.5.1.7 Transmitter Side, End a Call

1. Drop the call with CALL_STOP command
   Command ID: DMR_CALL_STOP
   Send Command: 84 A9 61 00 01 03 21
   receive command: 84 A9 61 00 02 03 21 00

2. When the END message is sent out, SCT3258 responds with ACK_MESSAGE (0x17 0x0A)

#### 2.5.1.8 Receiver Side, End a Call

When the transmitter stops the call, SCT3258 receives LC terminator from the far end, and report it with DMR_SLOT_TYPE to the MCU.

Receive Data:84 A9 61 00 02 30 7B 11

- DMR_SLOT_TYPE message. CC is 1. Data type is Voice LC Terminator.

2. After receiving DMR_SLOT_TYPE, the receiver can then drop the carrier with CARRIER_READY command
   Command ID: DMR_CARRIER_READY
   Send Command: 84 A9 61 00 02 00 19 00
   Receive Data:84 A9 61 00 02 00 19 00

### 2.5.2 DMR Voice Call in DMR Layer 2 Mode

A voice call involves the transmitter and the receiver. In both side, the MCU is the main controller, controlling the RF circuit, the MMI and SCT3258. Before the call, the MCU should set the RF circuit in TX mode for the transmitter and in RX mode for the receiver. The sub-section below

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only describes the interaction between the MCU and SCT3258.

#### 2.5.2.1 Initial Setting for both the Transmitter and the Receiver

1. Set Vocoder type with VOCODER_SEL. An up to 2-3 seconds delay is required to down load the vocoder from the flash.
   Command ID: VOCODER_SEL
   Send Command: 84 A9 61 00 02 00 10 01
   Receive Data:84 A9 61 00 02 00 10 00

2. Set SCT3258 to layer 2 mode with PROCESS_MODE command
   Command ID: PROCESS_MODE
   Send Command: 84 A9 61 00 02 00 1A 02
   Receive Data:84 A9 61 00 02 00 1A 00

3. Set Color Code corresponding to the RF Channel with DMR_CC command
   Command ID: DMR_CC
   Send Command: 84 A9 61 00 02 08 77 01
   Receive Data:84 A9 61 00 02 08 77 00

4. Set DMR call options with DMR_CALL_OPTION command
   Command ID: DMR_CALL_OPTION
   Send Command: 84 A9 61 00 06 08 79 41 00 10 00 00
   Receive Data:84 A9 61 00 02 08 79 00

(In this particular command, SCT3258 is set to do SQ detection; slotted mode, with unused slot filled with zeros; SLOT verified number is 1).

#### 2.5.2.2 Initial Setting for the Receiver

- Set the demodulator gain with DEMOD_GAIN command
- Set the types of field to report when receiving a call with REPORT_FIELD command

#### 2.5.2.3 Initial Setting for the Transmitter

1. Set the modulator gain with MOD_GAIN command

#### 2.5.2.4 Receiver Side, Start a Call

1. Set SCT3258 in RX or DUPLEX with WORK_MODE command
   Command ID: WORK_MODE_RX
   Send Command: 84 A9 61 00 04 00 18 01 00 00
   Receive Data:84 A9 61 00 02 00 18 00
2. Set which slot to receive with DMR_CALL_SLOT command

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Send Command: 84 A9 61 00 02 08 6F 01
Receive Data:84 A9 61 00 02 08 6F 00
(For mobile to mobile call, the slot number is 0)

3. When detecting carrier, inform SCT3258 that carrier is ready with CARRIER_READY command
Command ID: DMR_CARRIER_READY
Send Command: 84 A9 61 00 02 00 19 {0x01 or 0x02}
Receive Data:84 A9 61 00 02 00 19 00

After the call from transmitter arrives, SCT3258 reports useful information. The MCU decides whether to accept or discard the call.

### 2.5.2.5 Transmitter Side, Start a Call

1. Set which slot to transmit with DMR_CALL_SLOT command. If MS call is to be placed, the slot number is 0. Otherwise, slot 1 or slot 2 is selected
Command ID: DMR_CALL_SLOT
Send Command: 84 A9 61 00 02 08 6F 81
Receive Data: 84 A9 61 00 02 08 6F 00

2. Set SCT3258 in TX mode with WORK_MODE command
Command ID: WORK_MODE_TX
Send Command: 84 A9 61 00 04 00 18 02 00 00
Receive Data: 84 A9 61 00 02 00 18 00

3. Send LC with DIGC_DATA_FRAME (CC can be sent in the same packet)
Command ID: DIGC_DATA_FRAME
Send Command: 84 A9 61 00 0E 08 77 01 43 01 09 00 00 00 00 00 01 00 00 01
Receive Command: 84 A9 61 00 01 20 43

4. Start the call by sending super frame with DIGC_DATA_FRAME
Command ID: PROTOCOL_DATA_FRAME
Send Command: 84 A9 61 00 0E 08 77 01 43 11 09 00 00 00 00 00 01 00 00 01
Receive Command: 84 A9 61 00 01 20 43

### 2.5.2.6 Receiver Side, after a Call is Received

Receive Data: 84 A9 61 00 02 30 7F 87

- DMR_SLOT_FOUND message. The content is TACT in the CACH, BS call, SLOT verify not completed

Receive Data:84 A9 61 00 02 30 7F A9

- DMR_SLOT_FOUND message. The content is TACT in the CACH, BS call, SLOT

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verify completed, with no need to invert slot numbers in previous DMR_SLOT_FOUND message. At this point, the mobile station is synchronized with the base station. If the mobile station wishes to start a call, it needs to check the AT bit in the TACT of CACH, and proceed if it is not busy.

Receive Data: 84 A9 61 00 0E 30 77 01 43 01 09 00 00 00 00 00 01 00 00 01

- This message contains two fields, DMR_CC and DIGC_DATA_FRAME. CC is 0x01. The content in the data frame is data burst at SLOT1, length of 9, and Voice LC header.

Receive Data: 84 A9 61 00 0E 30 77 01 43 11 09 00 00 00 00 00 01 00 00 01

- This message contains two fields, DMR_CC and DIGC_DATA_FRAME. CC is 0x01. The content in the data frame is voice super frame at SLOT1, length of 9, and Voice LC header. Once SCT3258 receives and reports one voice super frame, it won't report subsequent voice super frame. But it continues to track the synchronization of the base station, and report 0x1B 0x08 if the synchronization is lost, which the MCU can use as an indication of call lost.

#### 2.5.2.7 Transmitter Side, End a Call

1. Stop the call by sending Terminator LC. Once SCT3258 has sent all message to the far end, it reports 17 0A to the MCU

   Command ID: PROTOCOL_DATA_FRAME
   Send Command: 84 A9 61 00 0c 08 43 02 09 00 00 00 00 00 01 00 00 01
   Receive Command: 84 A9 61 00 01 20 43
   Receive Command: 84 A9 61 00 02 20 17 0A

#### 2.5.2.8 Receiver Side, End a Call

When the transmitter stops the call, SCT3258 receives LC terminator from the far end, and report it with DIGC_DATA_FRAME to the MCU

Receive Data:84 A9 61 00 0E 30 77 01 43 02 09 00 00 00 00 00 01 00 00 01

- This message contains two fields, DMR_CC and DIGC_DATA_FRAME. CC is 0x01. The content in the data frame is data burst at SLOT1, length of 9, and Voice LC Terminator.

1. Drop the carrier with CARRIER_READY command

   Command ID: DMR_CARRIER_READY
   Send Command: 84 A9 61 00 02 00 19 00
   Receive Data:84 A9 61 00 02 00 19 00

### 2.5.3 Analog Voice Call in DMR Mode

SCT3258 support dual mode operation with digital and analog calls. For analog call, SCT3258

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supports CTCSS/DCS sub-audio signals.

#### 2.5.3.1 Receiver Side, Start a Call

1. Set SCT3258 to analog call mode with PROCESS_MODE command
2. Set SCT3258 in RX or DUPLEX with WORK_MODE command
3. Set CTCSS/DCS settings with SUB_AUDIO command
4. When detecting carrier, inform SCT3258 that carrier is ready with CARRIER_READY

After the call from transmitter arrives, SCT3258 reports whether CTCSS/DCS is matched or not with SUB_AUDIO command.

#### 2.5.3.2 Transmitter Side, Start a Call

1. Set SCT3258 to analog call mode with PROCESS_MODE command
2. Set SCT3258 in TX or DUPLEX with WORK_MODE command
3. Set CTCSS/DCS settings with SUB_AUDIO command
4. Start the call with DMR_CALL_START command

#### 2.5.3.3 Receiver Side, End a Call

1. Drop the carrier with CARRIER_READY command

#### 2.5.3.4 Transmitter Side, End a Call

1. Drop the call with CALL_STOP command

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## 2.6 Gain Calibration

This is the first step of calibration. It calibrates the receiver path to calculate SQ_LEVEL correctly without being affected by gains variation in the analog path..

For SCT3700, follow these steps:

1. Feed -60dbm FM signal (eg. by HP8920) to receiver and make chip run in receiver mode
2. Adjust “GAIN_ADJUST” till WB_RSSI equals to -60dbm

Note that if external LNA is used, it needs to be disable when doing GAIN calibration.

**Command ID: READ_RSSI**

**Send Command: 84 A9 61 00 01 20 6C 00**

**Receive Data: 84 A9 61 00 07 23 6C 00 3B 00 37 00 00**

For example, 0x3B-137 = -78dbm, which means there is -18db mismatch. So we set

GAIN_ADJUST to 0xee, and send

**Command ID: WRITE_RSSI**

**Send Command: 84 A9 61 00 0D 00 6C EE 00 00 30 10 00 04 00 10 00 04 00**

**Receive Data:84 A9 61 00 02 00 6C 00**

After that when we read again, it results,

**Command ID: READ_RSSI**

**Send Command: 84 A9 61 00 01 20 6C 00**

**Receive Data: 84 A9 61 00 07 23 6C 00 4C 00 36 00 00**

This time 0x4c-137 = -61dbm is almost correct, +/-1db is ignorable.

3. Enable external LNA if it is used

**Command ID: 3700CONFIG**

**Send Command: 84 A9 61 00 03 00 6D 0F 00 00**

**Receive Data:84 A9 61 00 02 00 6D 00**

4. Adjust "EXT LNA_GAIN" till WB_RSSI equals to -60dbm

**Command ID: readrssi**

**Send Command: 84 A9 61 00 01 20 6C 00**

**Receive Data: 84 A9 61 00 07 23 6C 00 6D 00 18 00 00**

0x6D-137 = -28dbm, which means 32db mismatch. So we set EXTLNA_VAL[5:0]=0x20, EXTLNA_VAL[7:6]=3 which is determined by 0x20-(0x4C-0x3B)=15dB

| Bit7-Bit6 | Value |
|---|---|
| 11 | 12<value |
| 10 | 6< value≤12 |
| 01 | 0<value≤6 |

**Command ID: WRITE_RSSI**

**Send Command: 84 A9 61 00 0D 00 6C EE E0 00 30 10 00 04 00 10 00 04 00**

**Receive Data:84 A9 61 00 02 00 6C 00**

Then read WB_RSSI again, where 0x4D-137 = -60dbm.

**Command ID: READ_RSSI**

**Send Command: 84 A9 61 00 01 20 6C 00**

**Receive Data: 84 A9 61 00 07 23 6C 00 4D 00 18 00 00**

Up till now, RX gain calibration is done.

## 2.7 DC and IQ Calibration

DC and IQ calibration of SCT3700 can be performed after gain calibration is done. Make sure that

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“GAIN ADJUST” and “EXT LNA GAIN” are set correctly before proceeding to this section.

Before starting, the user should be familiar with these 3 commands.

Command ID: DC_OFFSET_READ
Send Command: 84 A9 61 00 02 20 39 05
Receive Command:84 A9 61 00 0f 20 39 “RXDCI” “RXDCQ” “AMPI” “AMPQ” “TXDC_high” ”TXDC_low” “AMPCOMP”

Command ID: DC_OFFSET_WRITE
Send Command: 84 A9 61 00 0d 00 39 “ADC_DCI” “ADC_DCQ” “DAC_DCI” “DAC_DCQ” “DAC_GAINI” “DAC_GAINQ”
Receive Command:84 A9 61 00 00 02 39 00

Command ID: IQCOMP_COEF_WRITE
Send Command: 84 A9 61 00 0A 00 42 “cali_setting” “rx_coef_amp” “rx_coef_phi” “tx_coef_amp” “tx_coef_pha”
Receive Command:84 A9 61 00 00 02 42 00

1. Boot SCT3258. Don’t select any vocoder.
   Put SCT3700 into calibration loop back mode with command
   Command ID: WORK_MODE_CALIBRATOIN
   Send Command: 84 A9 61 00 04 00 18 43 00 00
   Receive Command:84 A9 61 00 02 00 18 00

2. Set the SCT3700 RF to transmit a carrier at frequency Fc+Fif (Fif = 1050 kHz for example).

3. Set the SCT3700 RF to receiver at frequency Fc. After the down conversion by the RF front end, IQ signals of frequency Fif are at the input of SCT3258.

4. Disable IQ calibration filter.
   Command ID: IQCOMP_COEF_WRITE
   Send Command: 84 A9 61 00 0A 00 42 **44** 00 00 00 00 00 00 00 00
   Receive Command:84 A9 61 00 02 00 42 00

5. Disable IF dc filter
   Command ID: MISC_GAIN
   Send Command: 84 A9 61 00 0D 00 6B 08 00 08 00 08 00 00 00 01 00 00 00
   Receive Command:84 A9 61 00 02 00 6B 00
   Clear all the dc offset
   Command ID: DC_OFFSET_WRITE
   Send Command: 84 A9 61 00 0d 00 39 00 00 00 00 00 00 00 00 00 00 00

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Receive Command:84 A9 61 00 00 02 39 00

6. Force PGA gain to “0”

Command ID: RSSI_WRITE

Send Command: 84 A9 61 00 0d 00 6C “GAIN ADJUST” “EXT LNA GAIN” 80 00 10 00 04 00 10 00 04 00

Receive Data:84 A9 61 00 02 00 6C 00

Note that “GAIN ADJUST” and “EXT LNA GAIN” should be set to the values obtained from last section.

7. Read the DC offset “RXDCI” and “RXDCQ” with command DC_OFFSET read command (100 milliseconds average in this example)

Command ID: DC_OFFSET_READ

Send Command: 84 A9 61 00 02 20 39 05

Receive Command:84 A9 61 00 0f 20 39 “**RXDCI**” “**RXDCQ**” “AMPI” “AMPQ” “TXDC_high” ”TXDC_low” “AMPCOMP”

8. Write the above DC compensation value “**RXDCI**” and “**RXDCQ**” with DC_OFFSET write command

Command ID: DC_OFFSET_WRITE

Send Command: 84 A9 61 00 0d 00 39 “**ADC_DCI**” “**ADC_DCQ**” “DAC_DCI” “DAC_DCQ” “DAC_GAINI” “DAC_GAINQ”

Receive Command:84 A9 61 00 00 02 39 00

The user can start with ADC_DCI = RXDCI, ADC_DCQ = RXDCQ. DAC_DCI and DAC_DCQ should be zero.

9. Repeat 6 and 7 until minimum RXDCI and RXDCQ is obtained. This completes the RX DC calibration for this PGA level.

10. Put RX IQ compensation filter in calibration mode by issue IQCOMP_COEF command with RX_IQ_MODE bit set to 1 and TX_IQ_MODE bit to 1

Command ID: IQCOMP_COEF_WRITE

Send Command: 84 A9 61 00 0A 00 42 **44** “rx_coef_amp” 00 00 00 00 00 00

Receive Command:84 A9 61 00 00 02 42 00

Start rx_coef_amp = 0;

Then configure rx_coef_amp = “AMPCOMP” from step 15.

11. Next, start doing RX IQ calibration for this PGA level. Please note whenever you send the “39” command, you should make sure that you use the correct “**ADC_DCI**” “**ADC_DCQ**” accordingly.

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12. Read the amplitude of the signal at I and Q path with command DC_OFFSET read command (100 milliseconds average in this example). Here we should configure the "DAC_GAINI" "DAC_GAINQ" to make sure amplitude big enough (**"AMPI"** / "**AMPQ**"is larger than 0x1000 and smaller than 0x7000).

Command ID: DC_OFFSET_READ

Send Command: 84 A9 61 00 02 20 39 05

Receive Command:84 A9 61 00 0f 20 39 "DCI" "DCQ" "AMPI" "AMPQ" "TXDC_high" "TXDC_low" "**AMPCOMP**"

Send Command: 84 A9 61 00 0d 00 39 "**ADC_DCI**" "**ADC_DCQ**" "DAC_DCI" "DAC_DCQ" **"DAC_GAINI" "DAC_GAINQ"**

Receive Command:84 A9 61 00 00 02 39 00

13. Write AMPCOMP until AMPI and AMPQ are almost equal.

Command ID: IQCOMP_COEF_WRITE

Send Command: 84 A9 61 00 0A 00 42 **44** "AMPCOMP" 00 00 00 00 00 00

Receive Command:84 A9 61 00 00 02 42 00

14. Put RX IQ compensation filter in normal mode by issue IQCOMP_COEF command with RX_IQ_MODE bit set to 0

Command ID: IQCOMP_COEF_WRITE

Send Command: 84 A9 61 00 0A 00 42 **00** "rx_coef_amp" "rx_coef_phi" 00 00 00 00

Receive Command:84 A9 61 00 00 02 42 00

Start rx_coef_phi = 0;

Then configure rx_coef_phi = "AMPCOMP" from step 18.

15. Read the amplitude of the signal at I and Q path with command DC_OFFSET read command (100 milliseconds average in this example)

Command ID: DC_OFFSET_READ

Send Command: 84 A9 61 00 02 20 39 05

Receive Command:84 A9 61 00 09 20 39 "DCI" "DCQ" "AMPI" "AMPQ""TXDC_high" "TXDC_low" "**AMPCOMP**"

16. Repeat step 17, 18 until AMPI and AMPQ are equal.

17. To complete whole RX DC and RX IQ calibration for all PGA levels, please repeat step 7 to step 16 each time with different PGA setting. PGA gain totally has 6 steps from 0-5.

Command ID: forceagcgain

Send Command: 84 A9 61 00 0d 00 6C "GAIN ADJUST" "EXT LNA GAIN" 80 00 10 00 04 00 10 00 04 00

Repeat step 7-16

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Command ID: RSSI_WRITE

Send Command: 84 A9 61 00 0d 00 6C “GAIN ADJUST” “EXT LNA GAIN” 81 00 10 00 04 00 10 00 04 00

Repeat step 7-16

Command ID: RSSI_WRITE

Send Command: 84 A9 61 00 0d 00 6C “GAIN ADJUST” “EXT LNA GAIN” 82 00 10 00 04 00 10 00 04 00

Repeat step 7-16

Command ID: RSSI_WRITE

Send Command: 84 A9 61 00 0d 00 6C “GAIN ADJUST” “EXT LNA GAIN” 83 00 10 00 04 00 10 00 04 00

Repeat step 7-16

Command ID: forceagcgain

Send Command: 84 A9 61 00 0d 00 6C “GAIN ADJUST” “EXT LNA GAIN” 84 00 10 00 04 00 10 00 04 00

Repeat step 7-16

Command ID: forceagcgain

Send Command: 84 A9 61 00 0d 00 6C “GAIN ADJUST” “EXT LNA GAIN” 85 00 10 00 04 00 10 00 04 00

Repeat step 7-16

18. Next step is to calibrate TX DC offset. Set “DAC_GAINI” “DAC_GAINQ” to 0. This results in zero tx baseband I/Q out.

Command ID: DC_OFFSET_WRITE

Send Command: 84 A9 61 00 0d 00 39 “ADC_DCI” “ADC_DCQ” “DAC_DCI” “DAC_DCQ” **00 00 00 00**

Receive Command:84 A9 61 00 00 02 39 00

Command ID: DC_OFFSET_READ

Send Command: 84 A9 61 00 02 20 39 05

Receive Command:84 A9 61 00 0f 20 39 “RXDCI” “RXDCQ” “AMPI” “AMPQ” “TXDC_high” ”TXDC_low” “AMPCOMP”

The initial value for “DAC_DCI” and “DAC_DCQ” should be zero (00 00 00 00)

19. Adjust “DAC_DCI” and “DAC_DCQ” and until minimum DC leakage is obtained (indicated by ”TXDC_high” “TXDC_low”). “TXDC_high” and “TXDC_low” constitute a 32 bit value that indicates the DC leakage energy. The larger the value, the more the DC leakage.

Suggested procedure:

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You can adjust DAC_DCQ first to get minimum leakage. For example use 0x0100 as a coarse step and 0x0010 as a fine step.

Then adjust DAC_DCI to get minimum leakage.

Again fine adjust DAC_DCQ.

And Fine adjust DAC_DCI again till leakage is minimum.

This complete the SCT3700 RX IQ calibration as well and RX DC and TX DC calibration process. The next step is TX IQ calibration. However, in most cases, TX IQ calibration is not necessary as SCT3700 is used in zero IF mode. Skip the steps below unless there is large TX IQ mismatch and SCT3700 is used in low IF mode.

20. Set the SCT3700 RF to transmit a carrier at frequency Fc. Now both SCT3700 TX and RX are using the same carrier frequency.

21. Generate a 1050 Hz sinusoidal I Q signal at the SCT3258 TX side, with zero amplitude
    Command ID: MISC_GAIN
    Send Command: 84 A9 61 00 0D 00 6B 08 00 08 00 08 00 18 00 08 00 08 00
    Receive Command:84 A9 61 00 02 00 6B 00
22. Put TX IQ compensation filter by issue IQCOMP_COEF command with TX_IQ_MODE bit set to 1
    Command ID: IQCOMP_COEF_WRITE
    Send Command: 84 A9 61 00 0A 00 42 04 “rx_coef_amp” “rx_coef_phi” “tx_coef_amp” 00 00
    Receive Command:84 A9 61 00 00 02 42 00

23. Read the amplitude of the signal at I and Q path with command DC_OFFSET read command (100 milliseconds average in this example)
    Command ID: DC_OFFSET_READ
    Send Command: 84 A9 61 00 02 20 39 05
    Receive Command:84 A9 61 00 09 20 39 “DCI” “DCQ” “AMPI” “AMPQ”

24. Repeat 23, 24 until AMPI and AMPQ are equal

25. Put TX IQ compensation filter in normal mode by issue IQCOMP_COEF command with TX_IQ_MODE bit to 0
    Command ID: IQCOMP_COEF_WRITE
    Send Command: 84 A9 61 00 0A 00 42 00 “rx_coef_amp” “rx_coef_phi” “tx_coef_amp” “tx_coef_phi”
    Receive Command:84 A9 61 00 00 02 42 00

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26. Read the amplitude of the signal at I and Q path with command DC_OFFSET read command (100 milliseconds average in this example)

    Command ID: DC_OFFSET_READ

    Send Command: 84 A9 61 00 02 20 39 05

    Receive Command:84 A9 61 00 09 20 39 “DCI” “DCQ” “AMPI” “AMPQ”

27. Repeat 26, 27 until AMPI and AMPQ are equal. This completes the TX IQ calibration.

## 2.8 Error Handling

SCT3258 responds to every packet received from the MCU, within T_rsp time, even it has parity error. Currently T_rsp = 40 milliseconds.

When the received packet has parity error, SCT3258 responds with ACK_MESSAGE with parity error indication.

If the MCU does not receive a message from the SCT3258 within T_rsp time, or receives an ACK_MESSAGE with parity error indication, or the received response message has parity error, it can consider the message lost, and resend the message if necessary.

For messages that initiated by SCT3258 (not as response to MCU command), the MCU checks parity error if parity check is enabled. If parity error fails, it simply discards the message, without sending a parity error response. This generally falls into two categories:

1. The message is DIGC_DATA_FRAME or DPMR_SLD, with which SCT3258 query for new data to be transmitted to the far end. In the case, if parity error fails, but the MCU can figure from the context that it is a DIGC_DATA_FRAME or DPMR_SLD message, it send query result as if there is no parity error. If the MCU can not figure out from the context, it simply drops the packet without sending a response.
2. The message is a SCT3258 report message generated by SCT3258. These types of messages do not require responses from the MCU. When there is a parity error, the MCU simply drops the message. If the MCU needs any specific information, it can send a query message to the DSP.

## 2.9 DPMR Standard User Interface

SCT3258 supports the Standard User Interface described in ANNEX A of ETSI TS 102 490.

With the Standard User Interface, the mobile user uses a standard telephone key pad, with digits ‘0’ to ‘9’, plus ‘*’ and ‘#’ keys, to dial another mobile’s number. Each mobile is assigned a 7 digital number, and one or several group numbers. At the receiver, SCT3258 match the “Called Number” of the incoming call to its own individual number and group numbers. It reports a call

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match to the MCU if there is a match. Otherwise, it reports a not-match.

Dialing with wild characters, abbreviated dialing and mask dialing are supported.

## 2.9.1 Message Used for Standard User Interface

### 2.9.1.1 Enable Standard User Interface

To enable the Standard User Interface, the user uses message with field ADDRESSING_MODE, and set the mode to Configured Address Mode. The user can also set a mask for mask dialing with the same field. By default, mask dialing is disabled (mask length is 0).

To enable layer 3 processing, so that call match is performed at the receiver, the user should set the process mode to 3 with PROCESS_MODE field.

The follow message enables the Standard User Interface and enables layer 3 processing

84 A9 61 00 04 00 1A 03 28 01

### 2.9.1.2 Set Own Numbers

The user can set its own number with DPMR_OWN_ID_BCD field. For example, the following message set the own number to 1234567.

84 A9 61 00 05 00 53 12 34 56 70

The user can also set one or several group number with GROUP_ID field. For example, the following message set the first group number to 1122334. GROUP_ID is 1.

84 A9 61 00 05 00 29 11 22 33 41

### 2.9.1.3 Set Dialed Numbers

#### 2.9.1.3.1 Full 7 digit Number

The user can pass the dialed number to SCT3258 with DPMR_CALLED_ID_BCD field. For example, the following message set the dialed number to 1234567.

84 A9 61 00 05 00 52 12 34 56 70

#### 2.9.1.3.2 Wild Number Dialing

The user can also uses wild character ‘*’ to dial a group of numbers, by replacing ‘*’ with ‘A’. For example, the following message set the dialed number to 123456*.

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84 A9 61 00 05 00 52 12 34 56 A0

With this dialed number, called station with number ‘1234560’ to ‘1234569’ will respond.

#### 2.9.1.3.3 Abbreviated Dialing

The user does not always have to dial the full number of the called station. By using abbreviated dialing, the user can only dial the digits of the called station number that is different than this own number. For example, if caller’s own number is 1234567, and the called stations number is 1234568, the user only has to dial a single digit ‘8’. The message sent to SCT3258 is thus:

84 A9 61 00 05 00 52 8F FF FF F0.

Note that the un-dialed digits in DPMR_CALLED_ID_BCD is filled with ‘F’.

Upon receiving this message, SCT3258 forms the complete Called ID by extracting the un-dialed from its Own ID. In this example, the full Called ID, 1234568 will be sent to the far end.

#### 2.9.1.3.4 Masked Dialing

In some situation, the user is limited by the number of digits he/she can dial. To do this, the MCU sends a mask number (corresponding to the number of digits that is masked out for the user to dial) with ADDRESSING_MODE field.

For example, the following message limit the number of digital user can dial to 3.
84 A9 61 00 02 00 28 09.

If caller’s own number is 1234567, and the user dials ‘789’, the message sent to SCT3258 is thus:

84 A9 61 00 05 00 52 78 9F FF F0.

Upon receiving this message, SCT3258 forms the complete Called ID by extracting the un-dialed from its Own ID. In this example, the full Called ID, 1234789 will be sent to the far end.

If the number of digits user dial exceeds the maximum number of digits to dial, the MSB is discarded. In the previous examples, if user dials ‘99789’, the message sent to SCT3258 is:

84 A9 61 00 05 00 52 99 78 9F F0.

SCT3258 will remove ‘99’ and form the complete digit as ’1234789’.

### 2.9.1.4 Report Call Match

If layer 3 processing is enabled, the receiver matches the called number of incoming call to its own individual number or group number, and report the matching status with CALL_MATCH

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fields.

CALL_MATCH is reported as long as there is an incoming call, whether or not the call is matched or not.

CALL_MATCH also report the status of Color Code match.

For group call, CALL_MATCH also report the group number that matches the call. For group call with wild characters, the group number is 0.

For examples, the following reported message from SCT3258:

84 A9 61 00 02 00 27 00

tells the MCU that an individual call is matched, and Color Code is also matched;

84 A9 61 00 02 00 27 01

tells the MCU that an individual call is not matched, but Color Code is matched;

84 A9 61 00 02 00 27 02
indicates that a group call with wild characters is received, and Color Code is also matched;

84 A9 61 00 02 00 27 1A
indicates that a group call with group number 1 is received, but Color Code is not matched;

It is possible that a valid call may have a mis-match on the Color Code, as Color code, unlike the Called ID, is not FEC protected. When the MCU receives a call with un-matched Color Code, it may poll SCT3250 for Color Code in subsequent voice frames through reading DPMR_CC after waiting 160 milli-seconds to see if it is indeed a valid call.

The Called ID and Own ID information is embedded in the header frame (80 milliseconds) and in 4 CCH frames (320 milliseconds). For normal calls with valid header frame, call match is done within 80 milliseconds. However, with late arrival calls, SCT3258 has to extract call information over 4 error free CCH frames. Adding the time to synchronize FS2, the minimum time to report a call match is over 500 milliseconds. In case of high BER (bit error rate), the SYNC time can be longer.

## 2.10DMR Number and Dialing Plan

SCT3258 supports the Numberling Plan described in ANNEX C of ETSI TS 102 361-2.

With the Standard User Interface, the mobile user uses a standard telephone key pad, with digits

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‘0’ to ‘9’, plus ‘*’ and ‘#’ keys, to dial another mobile’s number. Each mobile is assigned a 7 digital number, and one or several group numbers. At the receiver, SCT3258 match the “Called Number” of the incoming call to its own individual number and group numbers. It reports a call match to the MCU if there is a match. Otherwise, it reports a not-match.

## 2.10.1 Message Used for DMR Dialing Plan

### 2.10.1.1 Set Own Numbers

The user can set its own number with DMR_OWN_ID_BCD field. For example, the following message set the own number to 1234567.

84 A9 61 00 05 00 7E 01 23 45 67

The user can also set one or several group number with GROUP_ID field. For example, the following message set the first group number to 1122334.

84 A9 61 00 05 00 29 11 22 33 41

### 2.10.1.2 Set Dialed Numbers

#### 2.10.1.2.1 Full 8 digit Number

The user can pass the dialed number to SCT3258 with DMR_CALLED_ID_BCD field. For example, the following message set the dialed number to 1234567.

84 A9 61 00 05 00 7D 01 23 45 67

#### 2.10.1.2.2.All Call Dialing

The user can use All Call Dialing. See the derails in the table.

| BCD ID | Remark |
|---|---|
| 84 A9 61 00 05 00 7D 00 aa aa(00******) | All Talkgroup ID0 |
| 84 A9 61 00 05 00 7D 01 aa aa(01******) | All Talkgroup ID1 |
| etc. | etc. |
| 84 A9 61 00 05 00 7D 09 aa aa(09******) | All Talkgroup ID9 |
| 84 A9 61 00 05 00 7D 0a aa aa(0*******) | All Talkgroup |

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#### 2.10.1.3 Report Call Match

In Easy Mode, the receiver matches the called number of incoming call to its own individual number or group number, and report the matching status with CALL_MATCH fields.

CALL_MATCH is reported as long as there is an incoming call, whether or not the call is matched or not.

CALL_MATCH also report the status of Color Code match.

For group call, CALL_MATCH also report the group number that matches the call.

For examples, the following reported message from SCT3258:

84 A9 61 00 02 00 27 00

tells the MCU that an individual call is matched, and Color Code is also matched;

84 A9 61 00 02 00 27 01

tells the MCU that an individual call is not matched, but Color Code is matched;

84 A9 61 00 02 00 27 02
indicates that a group call is received, and Color Code is also matched;

84 A9 61 00 02 00 27 1A
indicates that a group call with group number 1 is received, but Color Code is not matched;

It is possible that a valid call may have a mis-match on the Color Code, as Color code, unlike the Called ID, is not FEC protected. When the MCU receives a call with un-matched Color Code, it may poll SCT3250 for Color Code in subsequent frames.

The Called ID and Own ID information is embedded in the header frame (30 milliseconds) and in 6 voice frames (360 milliseconds). For normal calls with valid header frame, call match is done within 30 milliseconds. However, with late arrival calls, SCT3258 has to extract call information over 6 error free voice frames, or 360 milliseconds. In case of high BER (bit error rate), the SYNC time can be longer.

## 2.11Power Saving Mode

The host can put SCT3258 into low power modes with command CHIP_LOWPOWER (field type 0x15). The low power modes are listed below.

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0: Normal
1: Reserved
2: Reserved
3: Power down certain peripheral of SCT3258 including serial port, DMA and timers.
4: SCT3258 enters IDLE mode
5: SCT3258 enters Sleep mode
6: SCT3258 enters Halt mode

If SCT3258 is in power saving mode lower or equal to 4, it can be waken up with command CHIP_LOWPOWER with field value 0 (normal power mode). SCT3258 resume all states before it enters the lower power mode.

If SCT3258 enters power saving mode 5, it can be waken up with INT0. SCT3258 resume all states before it enters the lower power mode.

If SCT3258 enters power saving mode 6, it can be waken up with NMI. Once NMI is received by SCT3258, it restarts the SCT3258. All information sent to SCT3258 is lost and need to be reinitialized. The difference of reset with NMI and reset with RESET chip is that firmware would not need to be reloaded with NMI.

The same command can be used to power down the external vocoder and codec by setting bit 5 and bit 5 of field value.

## 2.12 Codec Selection

SCT3258 has native support for the following two codecs:

TI TLV320AIC3204
Wolfson WM8758B

TLV320AIC3204 and WM8758B are stereo codecs. With a stereo codec, the left ADC is connected to Microphone input, the right ADC is connected to the FM demod port. The right DAC is connected to the VCTCXO port, while the left DAC is connected to the audio line port in the RX mode and the VCO port in the TX mode.

The codec can be selected with CODEC_SELECT command (field ID 0x2D).

Other codecs can be used with SCT3258 if it supports I2C interface and conforms to the serial format of SCT3258 high speed serial port.

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## 2.13 Codec Configurations

The host can send I2C command to codec through SCT3258. The commands are sent to SCT3258 via HPI interface, with bit 6 of TYPE field set to 1. Upon receiving such commands, SCT3258 extracts information from the packet after the TYPE fields, and adds I2C address of the codec, and sends the information to the codec via I2C interface.

To configure codec, the host first sends the I2C address of the codec to SCT3258 with command SET_I2C_ADDRESS. It then can send codec configuration command directly to the codec by setting the packet type to be 0x40.

For example, to send command to WM8758B to reset it, the user can use the following commands:
Command ID: SET_I2C_ADDRESS
Send Command: 84 A9 61 00 02 03 2E 34
Receive Command:84 A9 61 00 02 03 2E 00
Command ID: CODEC_COMMAND
Send Command: 84 A9 61 00 02 40 00 00
Receive Command:84 A9 61 00 02 40 17 00

When SCT3258 receives the above command from the host, it extract “00 00” from the command, and 0x34 to the beginning, and sends “0x34, 0, 0” to the codec via I2C interface.

## 2.14 Debug Mode

### 2.14.1 Setting Up Two Point Modulation

SCT3258 has two outputs in TX mode connecting to VCO and VCTCXO port of the RF unit to facilitate 2 point modulation. The user should refer to “SCT3258 Application Notes – Analog Interface Setup” for details on how to adjust two point modulation.

### 2.14.2 BER Test

To check the link quality, SCT3258 can be configured to do BER test.

The user should refer to “SCT3258 Application Notes – Analog Interface Setup” for details on how to configure for BER test..

### 2.14.3 Loop Back

To verify the different part of SCT3258 hardware and software, a set of loop backs can be enabled.

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The following loop back modes are especially useful in verifying external codec and vocoder connection of SCT3258.

### 2.14.3.1 Audio Codec Loop Back and Modem Codec Loop Back

With audio codec loop back, the audio signal from the microphone goes through the ADC in the TX path and loops back to the DAC in the RX path, and sent out to the headphone output.

With modem codec loop back, the modem signal from the DEMOD_Q port goes through the ADC in the RX path and loops back to the DAC in the TX path, and sent out to the MOD_Q port.

The user can apply signal generated at the input and monitor the output with a scope. The correct operation of audio codec loop back and modem codec loop indicate the correct connection of external codec and the analog audio and modem path.

To set up for audio codec loop back and modem codec loop back, the user should follow these steps:

Send command to set SCT3258 in audio codec loop back mode:
ALOOP_CODEC_LOOP: 61 00 04 00 18 03 11 00

The user should then be able to hear his voice from the headphone, and see the signal from a scope.

### 2.14.3.2 Audio Vocoder Loop Back

With audio vocoder loop back, the audio signal from the microphone goes through the ADC, the decimator, the voice encoder in the TX path and loops back to the voice decoder, the interpolator and the DAC in the RX path, and sent out to the headphone output.

Correct vocoder loop back indicate correct operation of the external vocoder (e.g. AMBE3000).

To set up for audio vocoder loop back, the user should follow these steps:

Send command to set SCT3258 in audio vocoder loop back mode:
ALOOP_VOCODER_LOOP: 61 00 04 00 18 03 03 00

The user should then be able to hear his voice from the headphone.

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